

Sheth Laherchand Uttamchand Jhaveri (L.U.J.) College of Arts & Sir Mathuradas Vissanji (M.V.) College of Science & Commerce

Sub : Data Visualization with Power BI and Tableau

Aim : Implementing Advanced Visualization Techniques and Customizations.

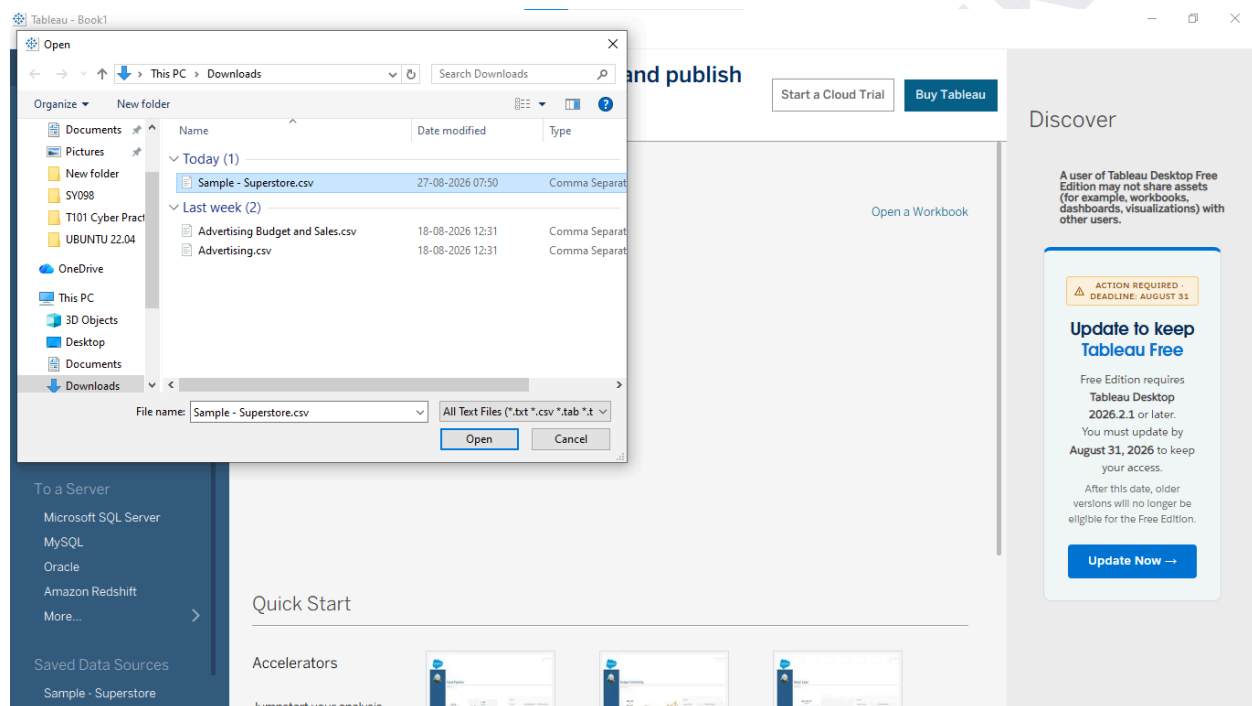
A: Create Slope Charts and Lollipop Charts

1. Create a Slope Chart (Year-over-Year Progression)

Objective: Build a two-point line chart comparing total sales for each Sub-Category between two specific baseline years (2014 vs 2017).

Step 1: Set Up the Base Worksheet

1. Open Tableau Desktop connected to Superstore.csv



2. Create a new worksheet and rename it: Slope Chart.

3. Drag Order Date to the Columns shelf. (It will appear as YEAR(Order Date)).

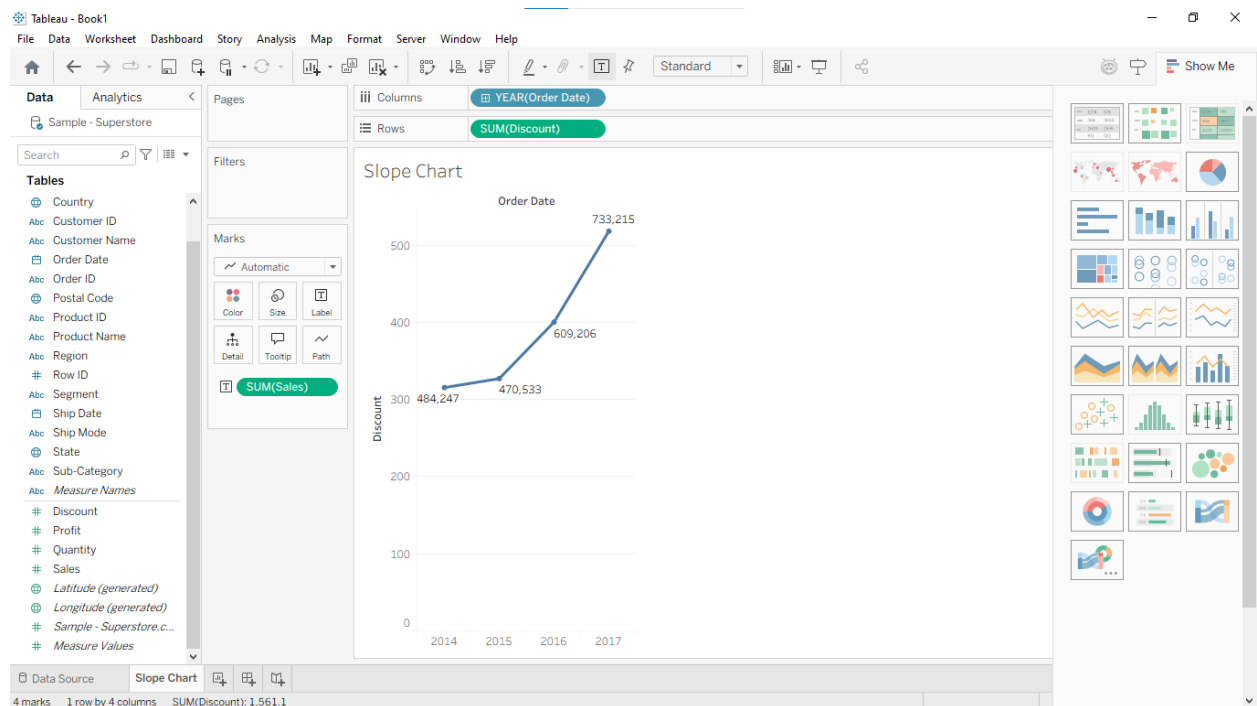
4. Drag Sales to the Rows shelf (appears as SUM(Sales)).

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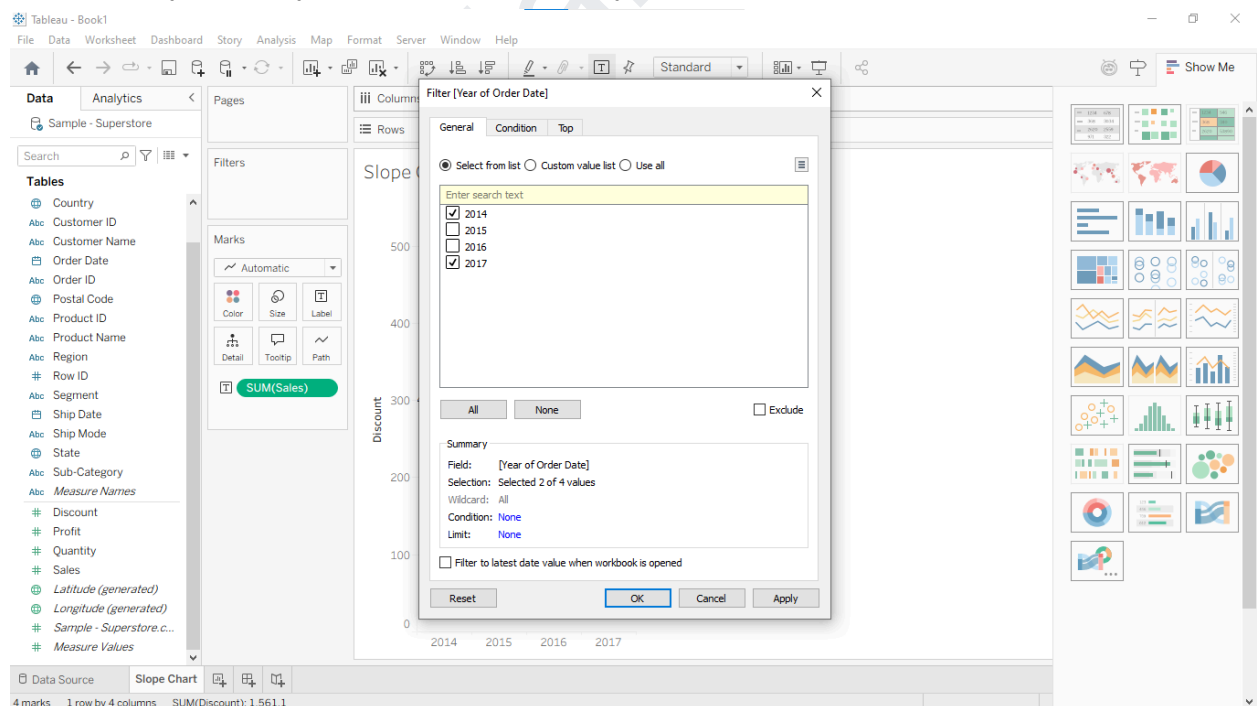
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Step 2: Filter to Baseline & Comparison Years

1. Hold Ctrl (or Cmd on Mac), drag Order Date to the Filters shelf, select Years, and click Next.
2. Check only the first year (2014) and the last year (2017). Click OK.



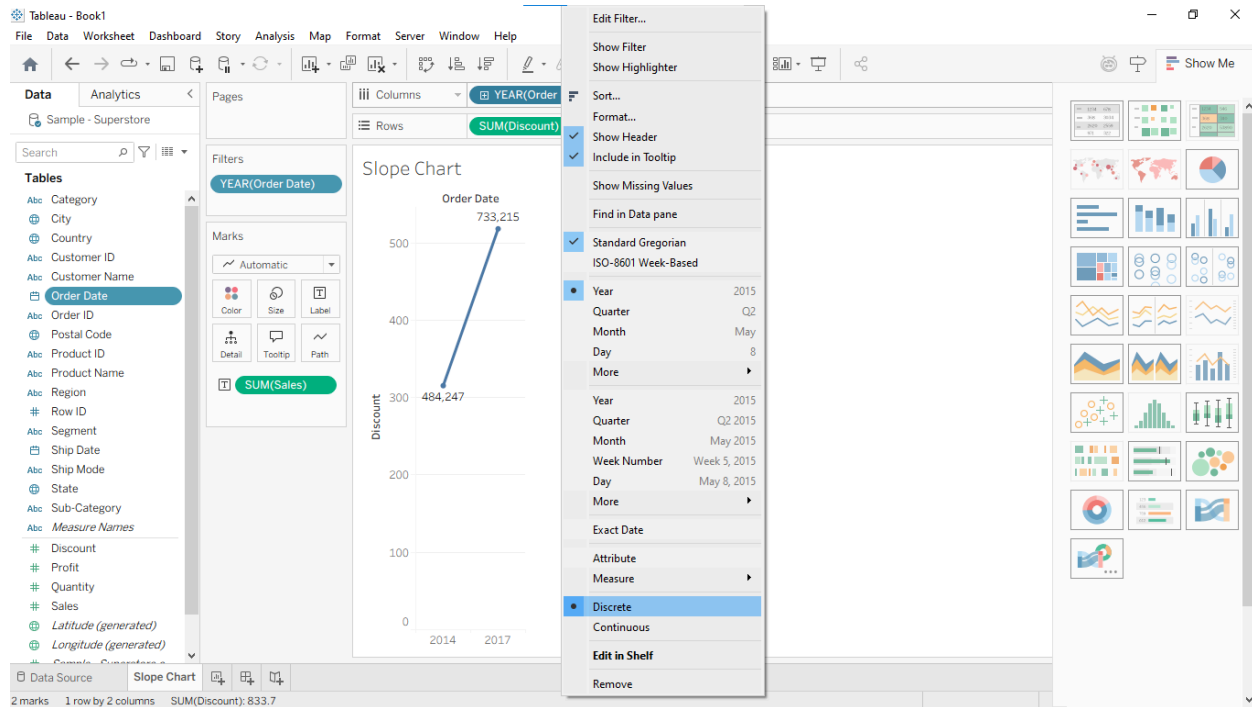
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3. Right-click on the YEAR(Order Date) pill on the Columns shelf --> select Discrete(blue pill) so it acts as two distinct category pillars.

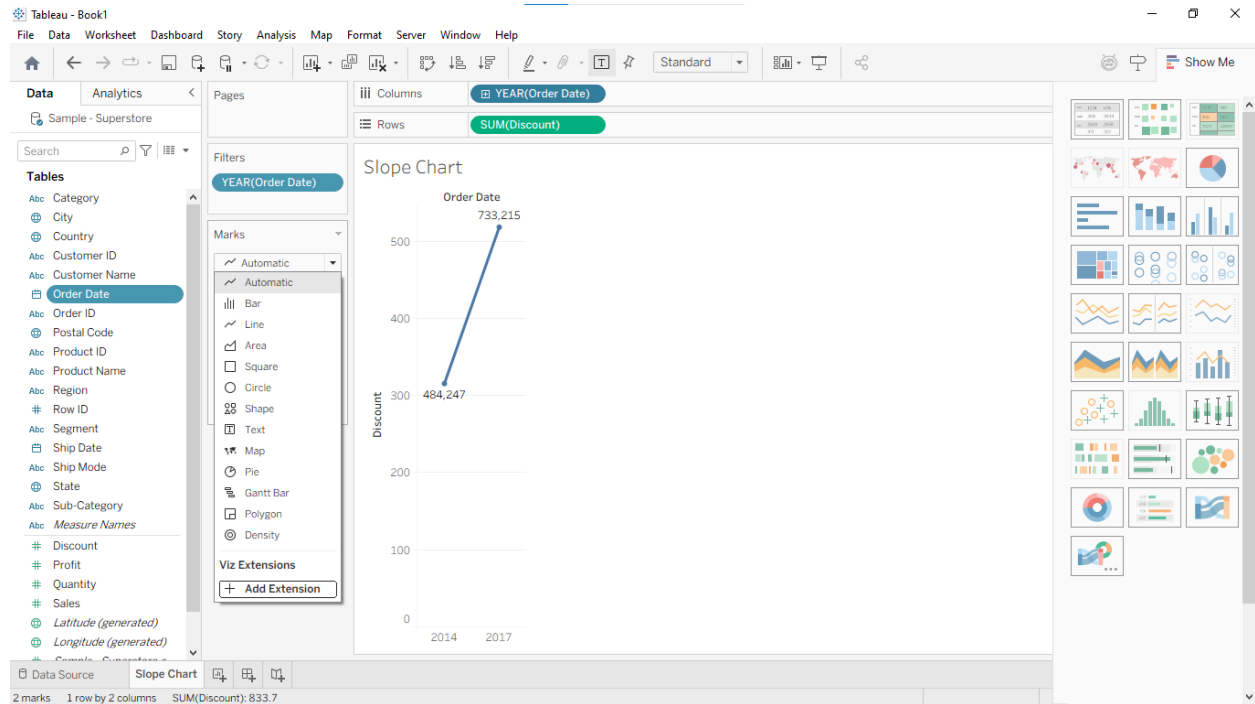


Step 3: Disaggregate by Sub-Category

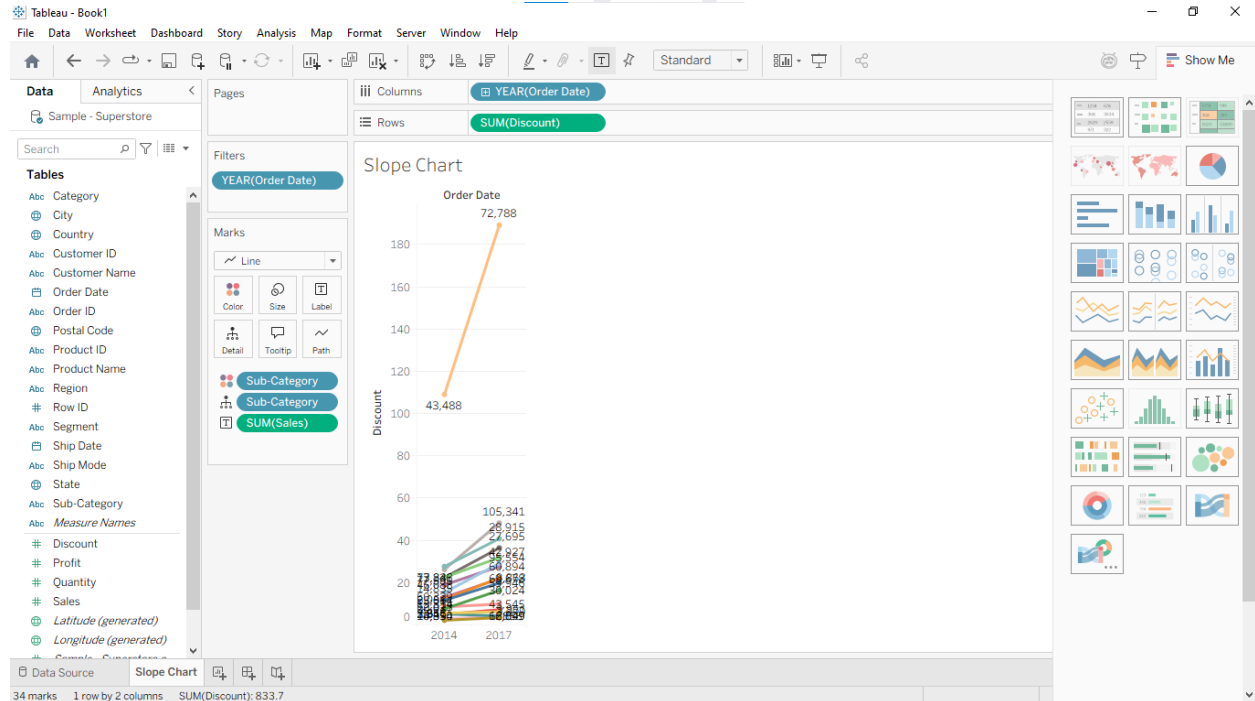
1. On the Marks card, change the mark type dropdown from Automatic to Line.

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2. Drag Sub-Category onto the Detail tile (and onto the Color tile) on the Marks card.



3. You will now see 17 distinct sloping lines connecting 2014 to 2017.

Step 4: Format Labels and Endpoints

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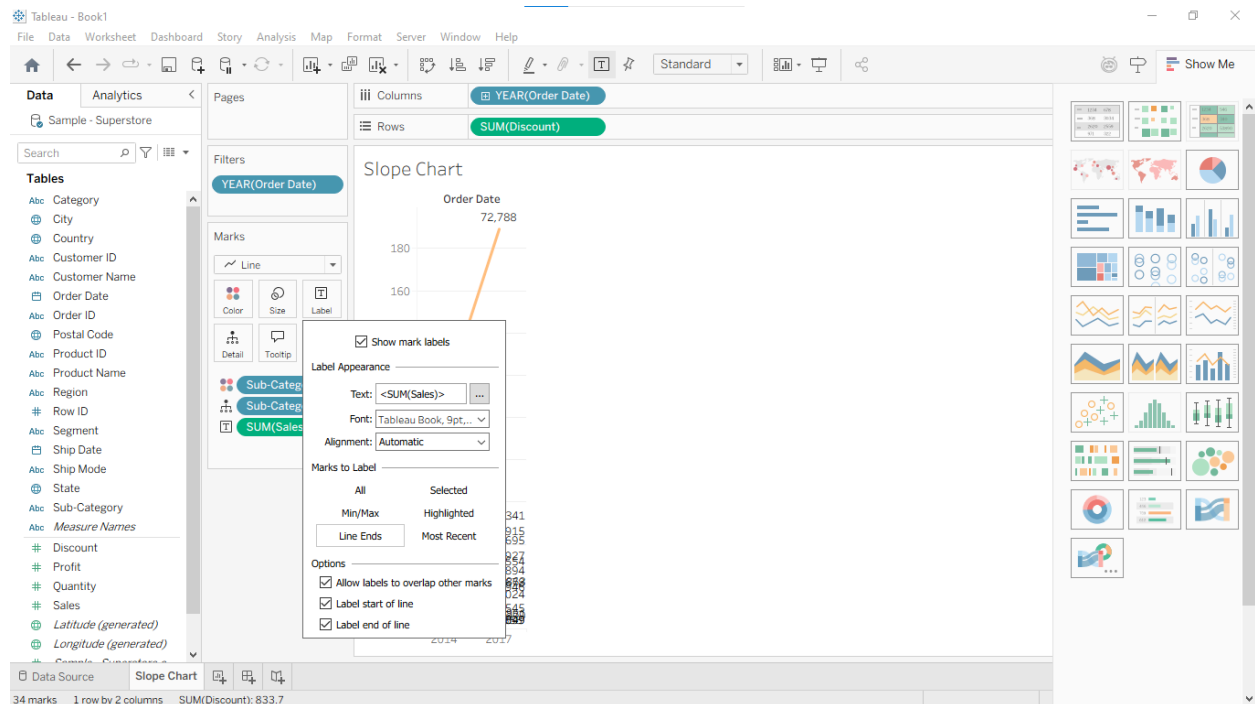
1. On the Marks card, click the Label tile:

- o Check Show mark labels.

- o Under Marks to Label,

choose Line Ends

- o Check Label start of line and Label end of line



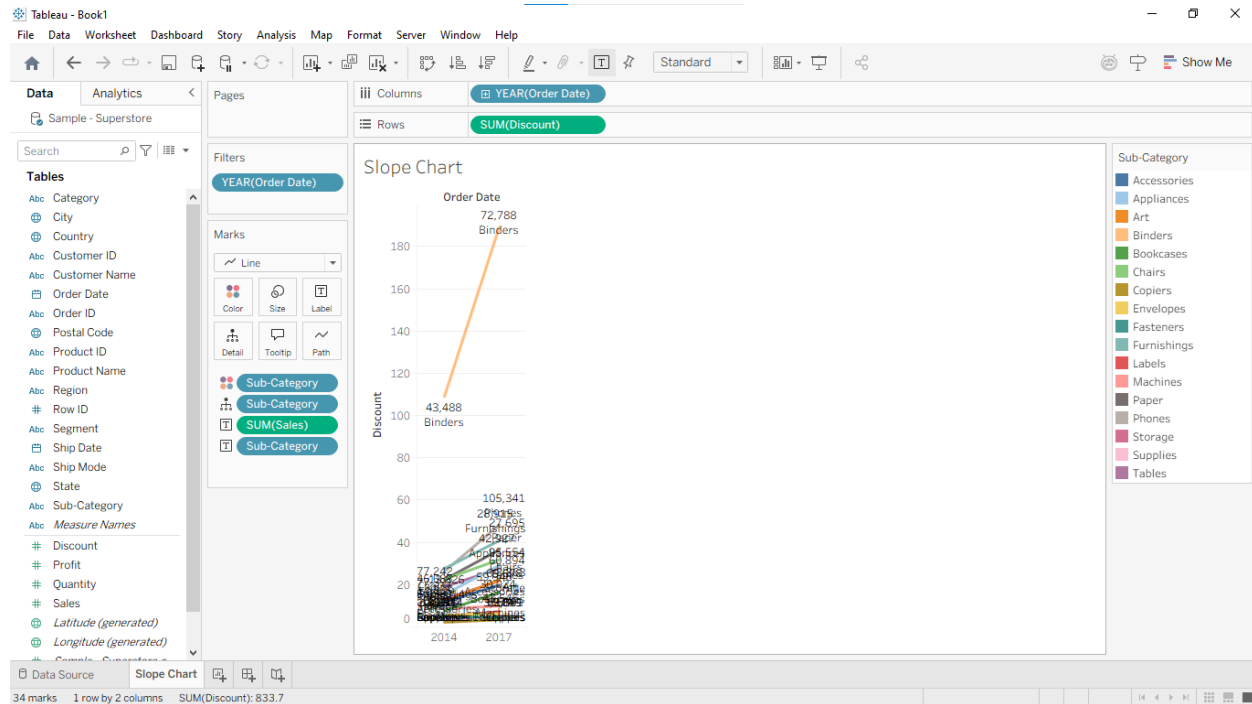
2. Drag Sub-Category onto Label on the Marks card to display the product name directly at the line endpoints

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2. Create a Lollipop Chart (Sales with Profit Highlighting)

Step 1: Set Up the Base Worksheet

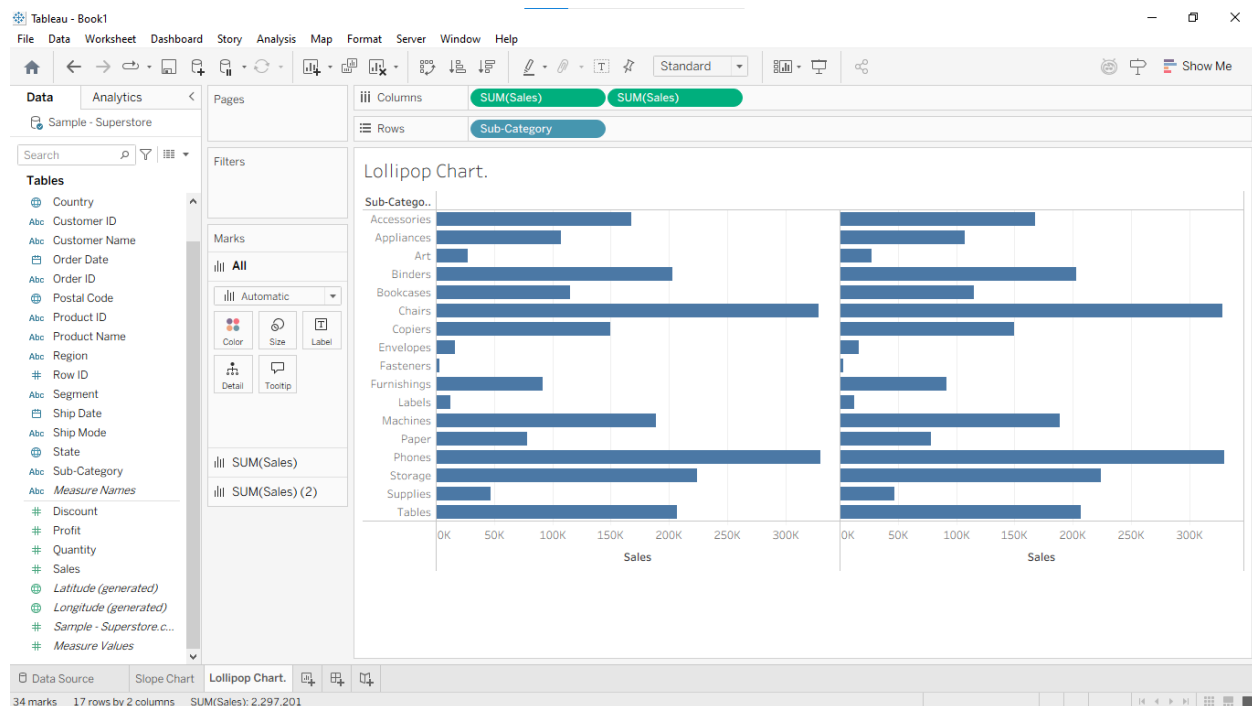
1. Create a new worksheet and rename it: **Lollipop Chart**.
2. Drag Sub-Category onto the Rows shelf.
3. Drag Sales onto the Columns shelf.
4. Drag Sales onto the Columns shelf a second time (creating two adjacent SUM(Sales) pills on Columns).

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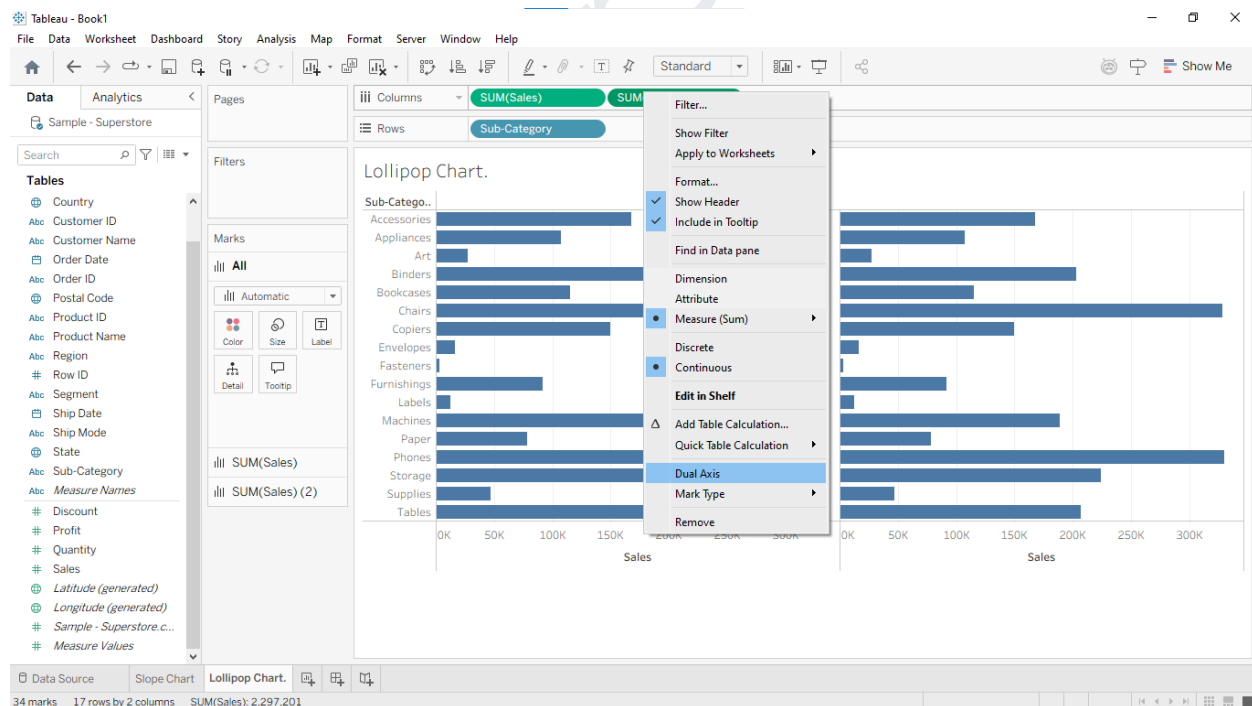
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Step 2: Convert to Dual Axis

1. Right-click the second SUM(Sales) pill on the Columns shelf --> select Dual Axis.



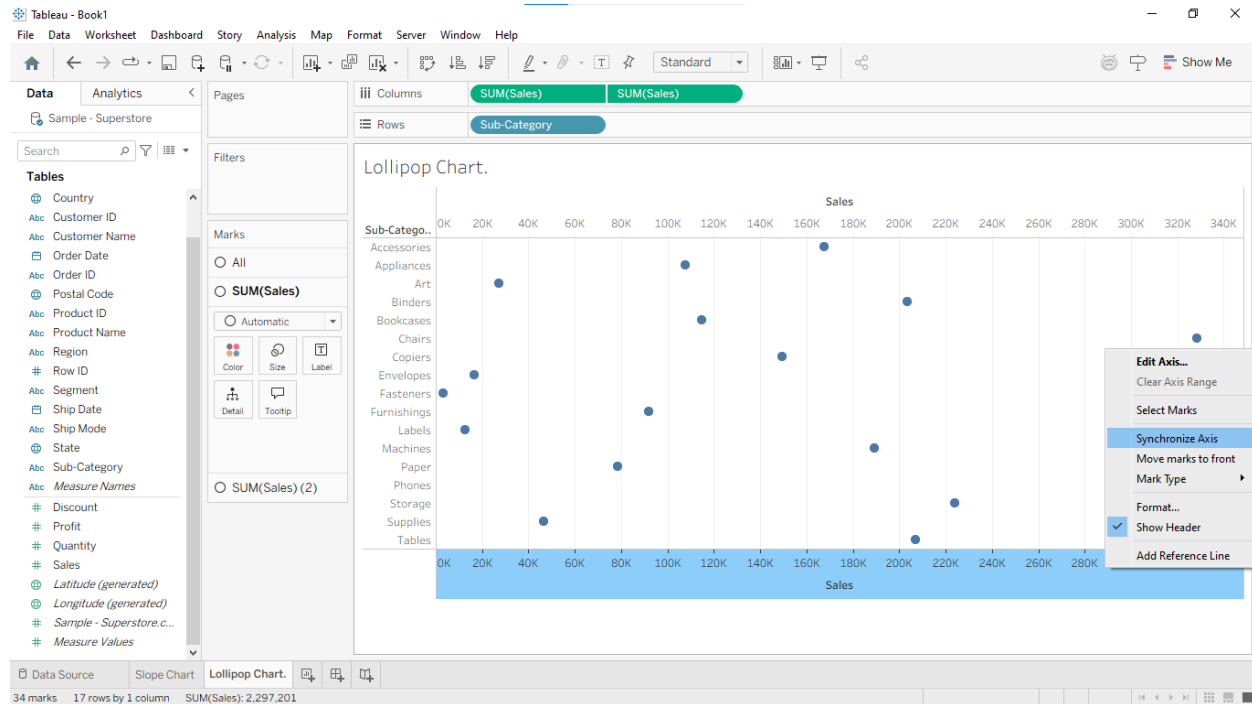
2. Right-click the top numerical axis in the visualization view --> select Synchronize Axis (crucial to ensure both axes align).

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Step 3: Configure the Marks Cards

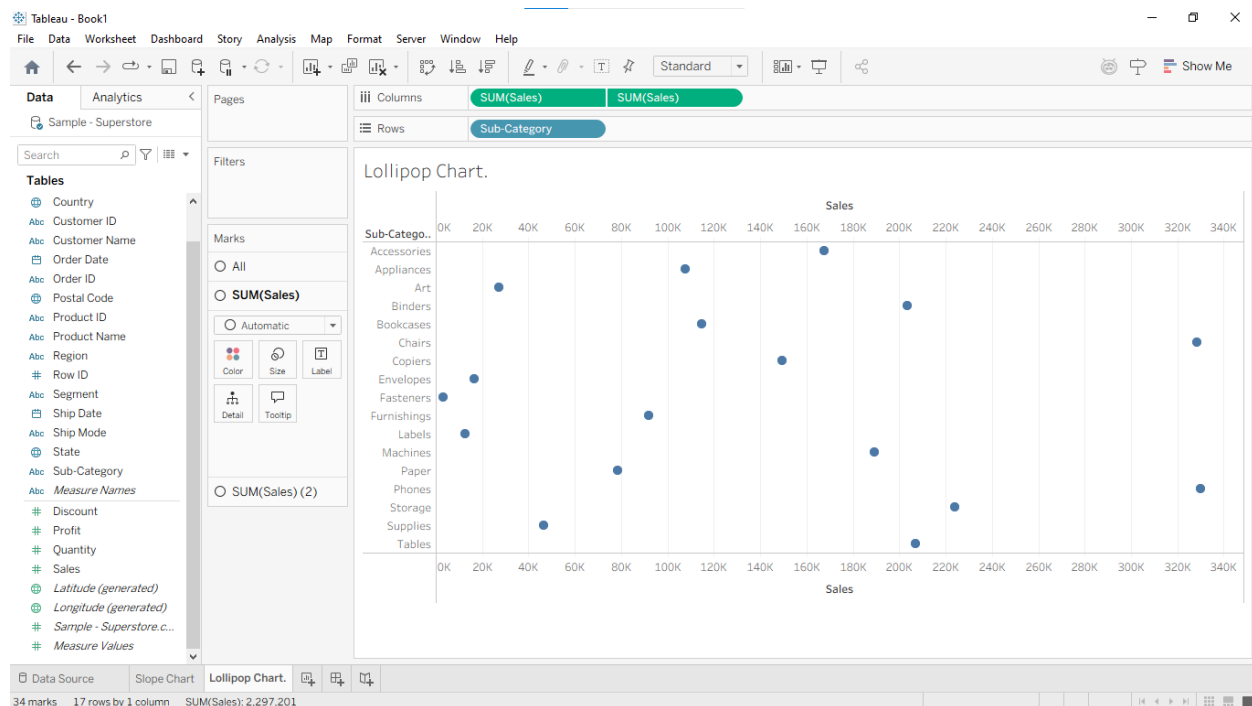
In the Marks pane on the left, you will now see three cards: All, SUM(Sales) (first mark), and SUM(Sales) (2) (second mark).

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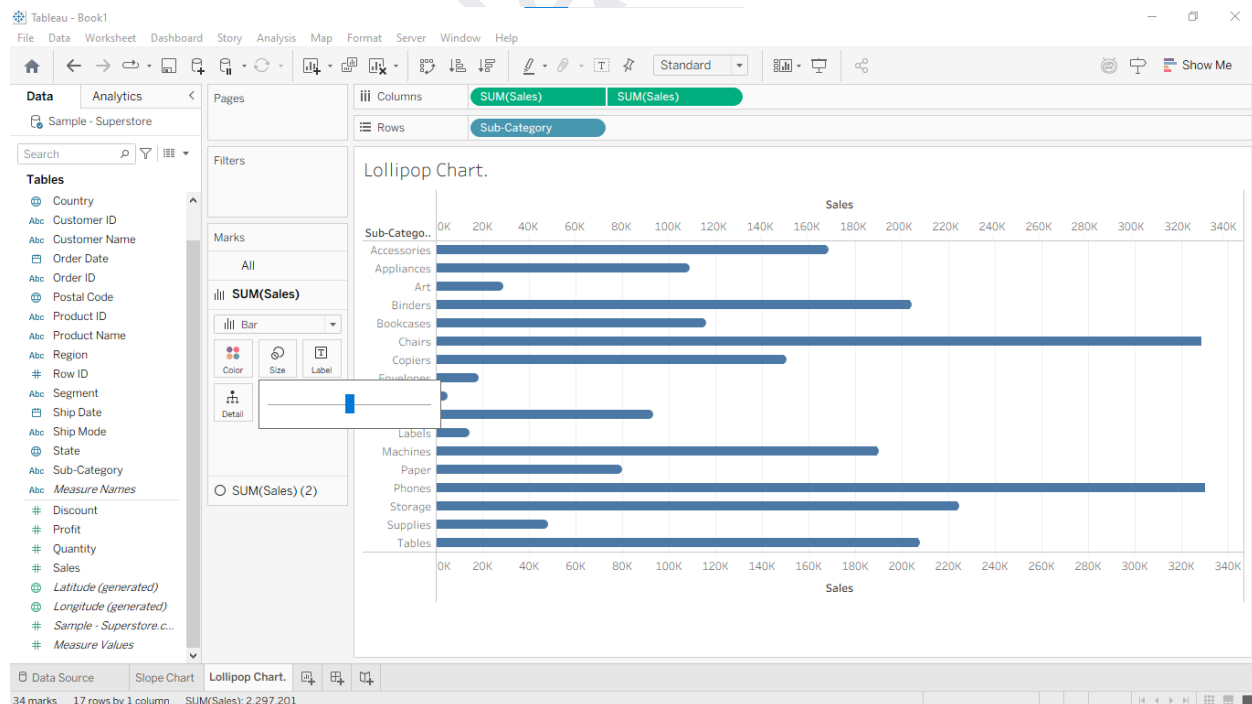
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1. Configure the Stem (Bar):

- o Click on the first SUM(Sales) mark card.
- o Change the mark type dropdown to Bar.
- o Click the Size tile --> pull the slider down until the bars become thin, elegant stems.



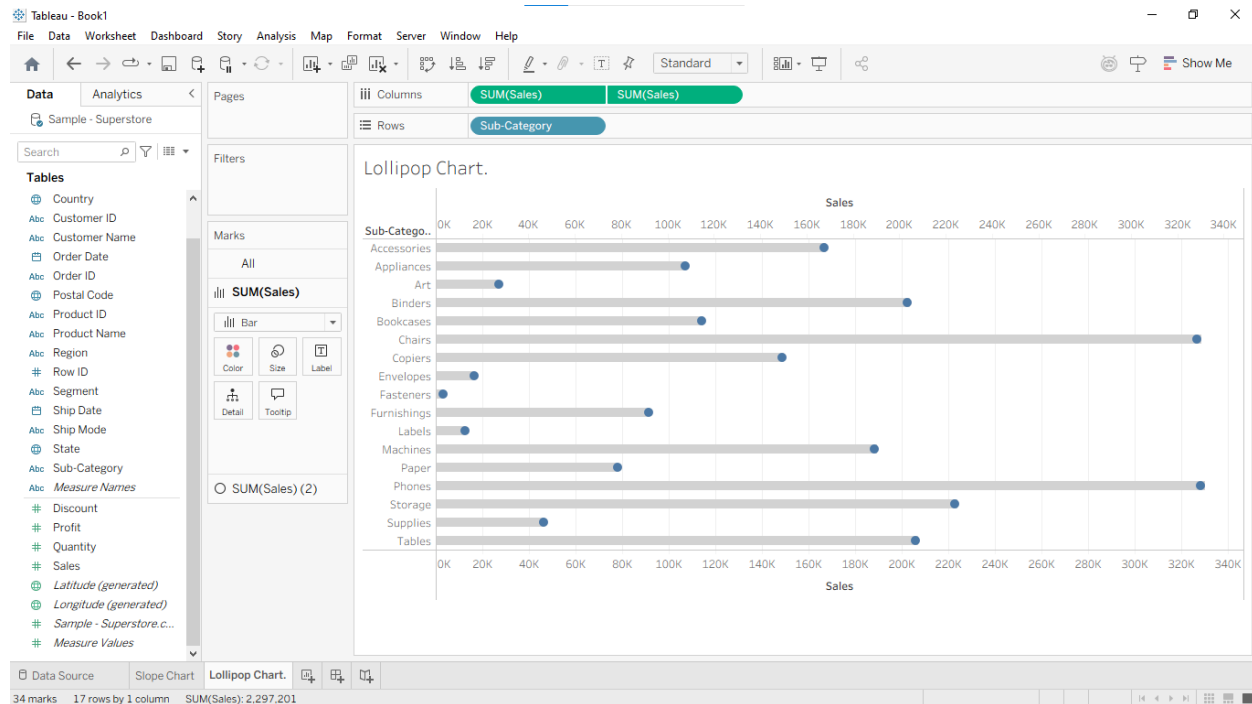
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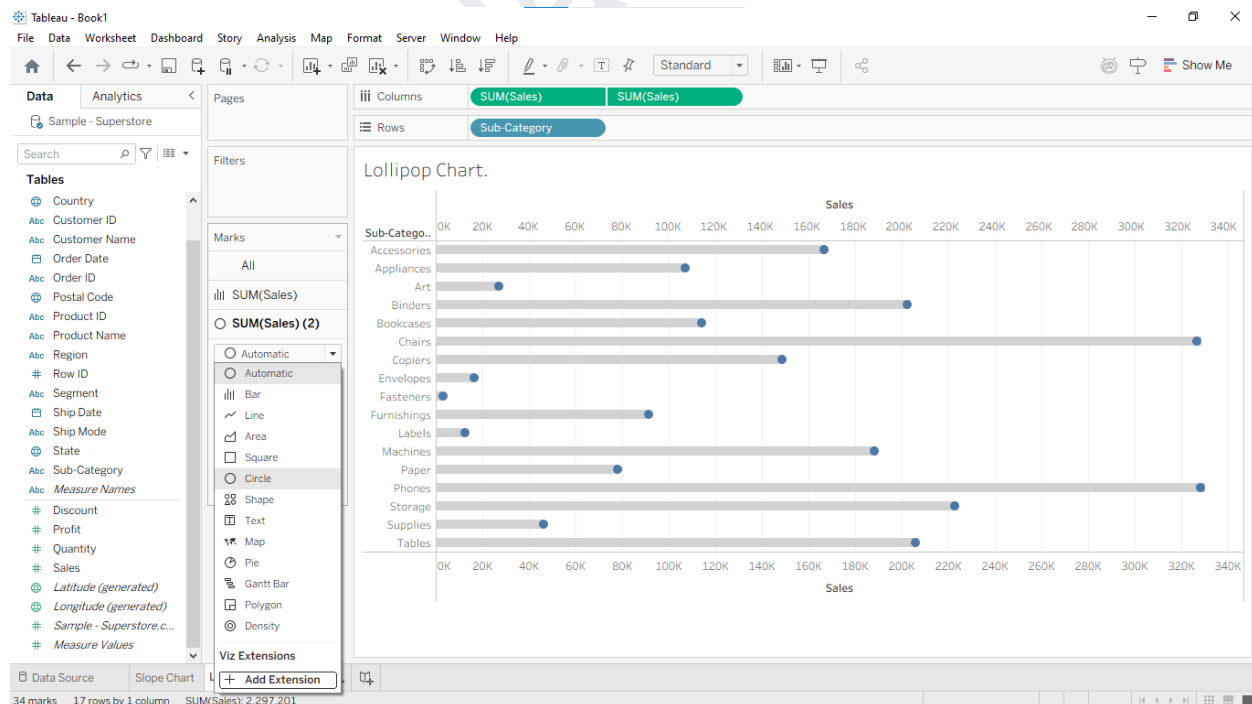
o Click Color --> set the bar color to a neutral gray (e.g., #B0B0B0).



2. Configure the Head (Circle):

o Click on the second SUM(Sales) (2) mark card.

o Change the mark type dropdown to Circle.



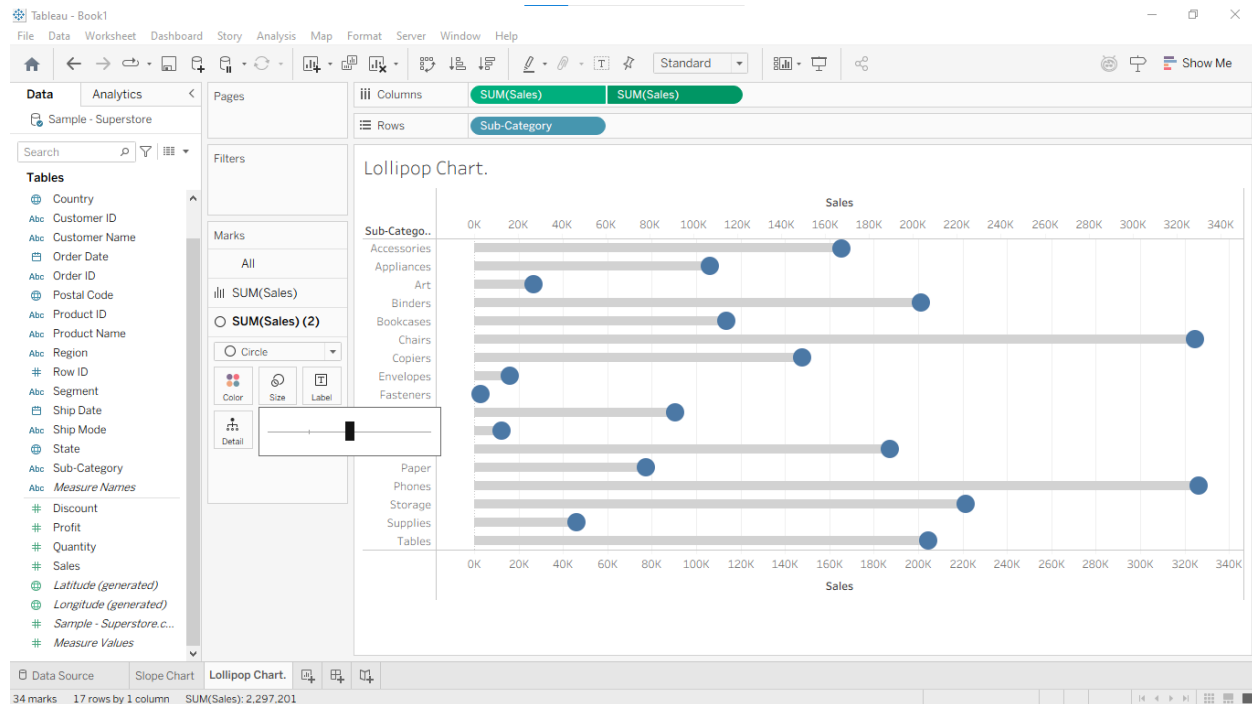
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o Click the Size tile --> increase the circle size so it sits neatly at the tip of each bar stem.



o Drag Profit onto the Color tile of this second marks card.

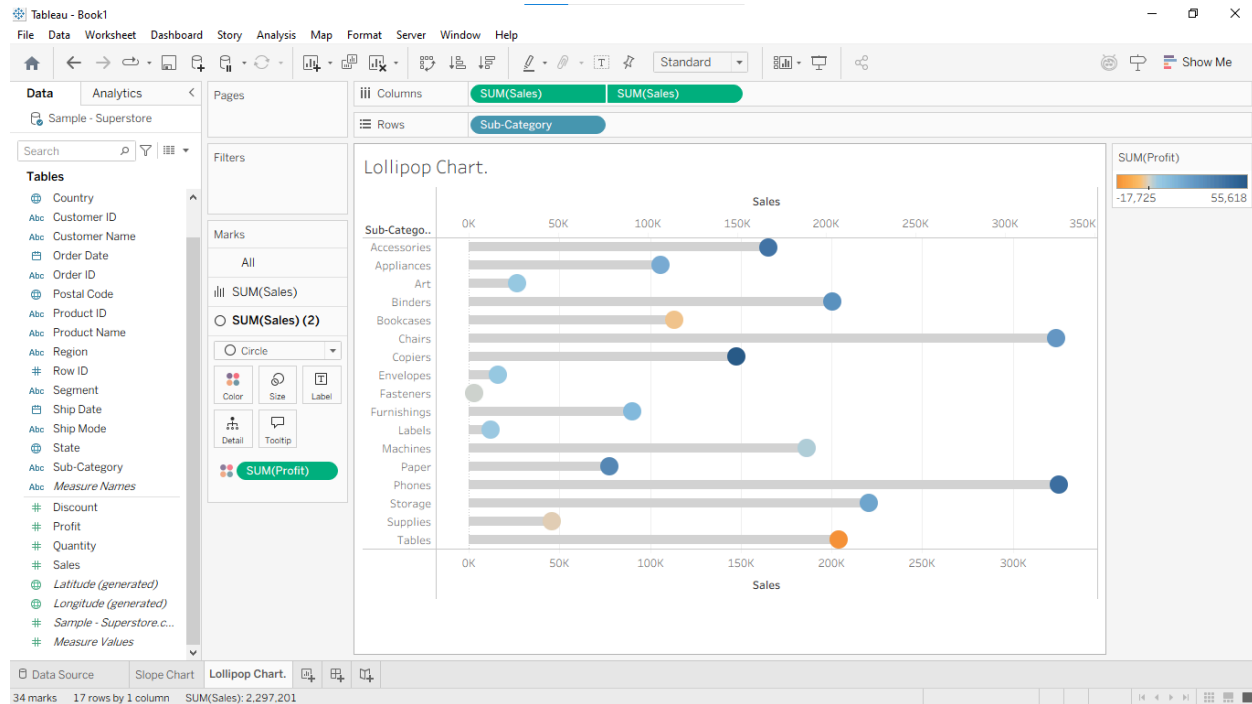
o (Observation): Sub-categories like Tables immediately show negative profit (red/orange) while having high sales.

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Step 4: Clean Up Axis Formatting

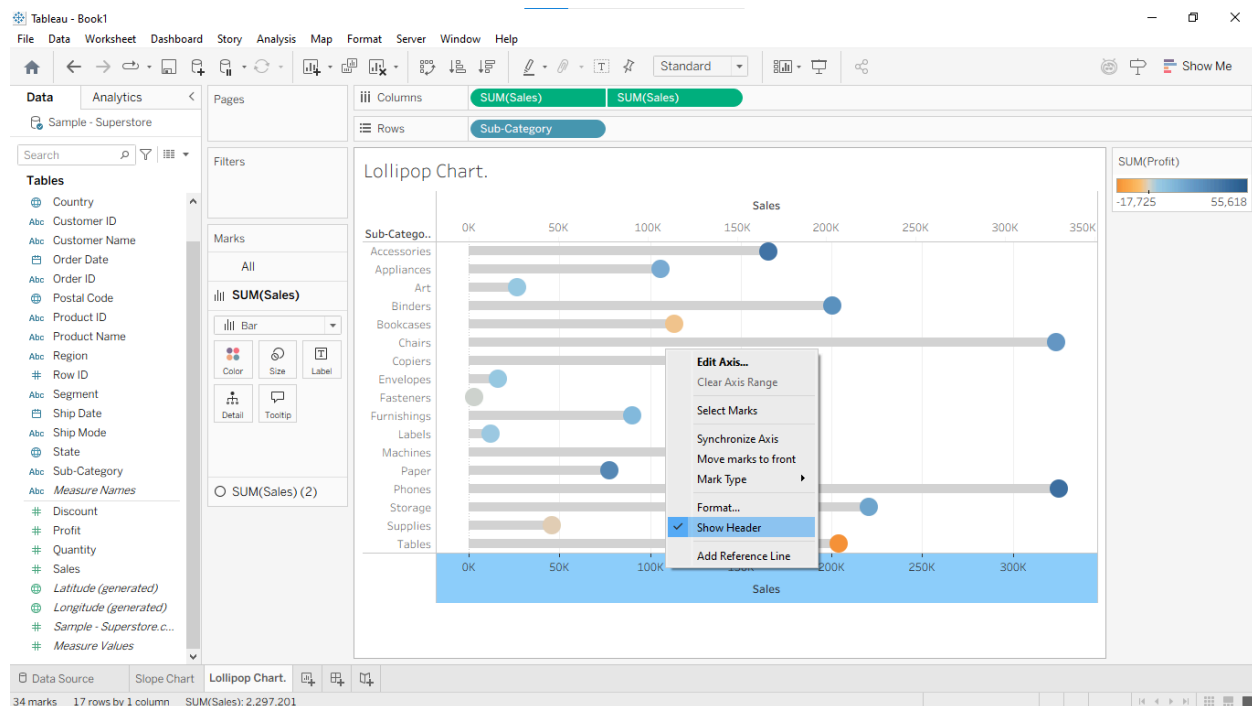
1. Right-click the top header axis --> uncheck Show Header

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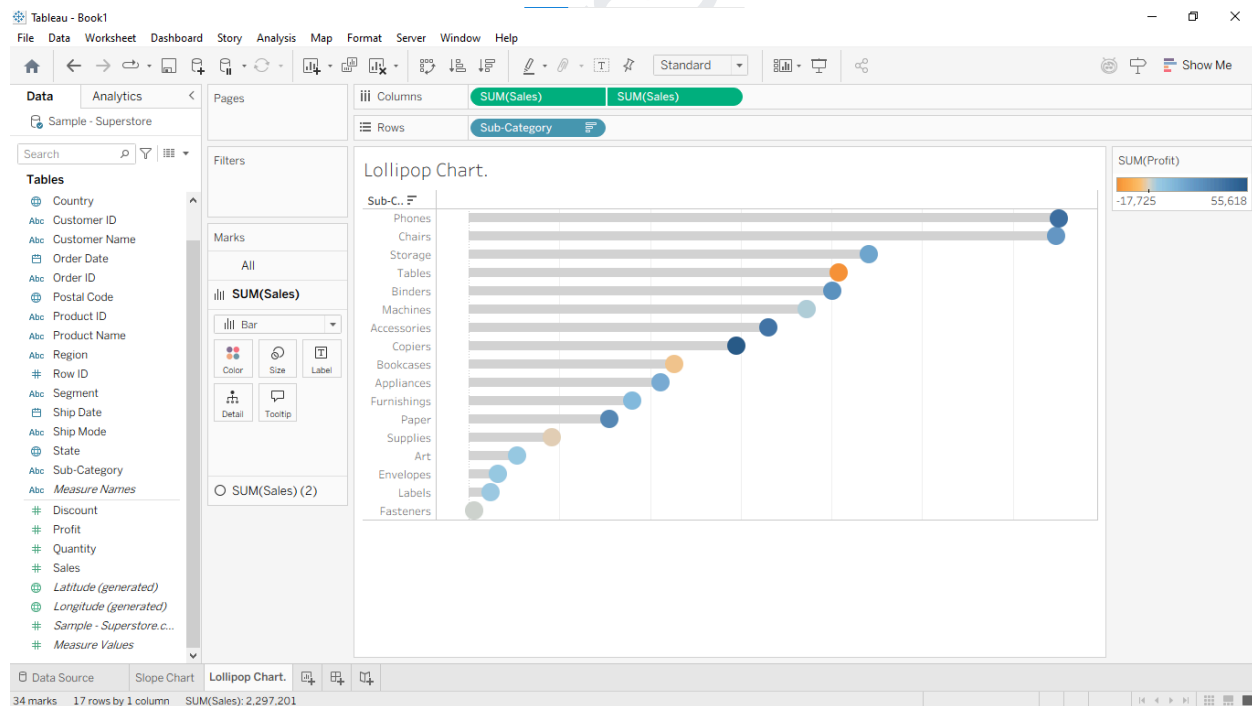
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2. Click the Sort Descending button in the top toolbar to sort by total sales.



B: Design Waterfall Charts

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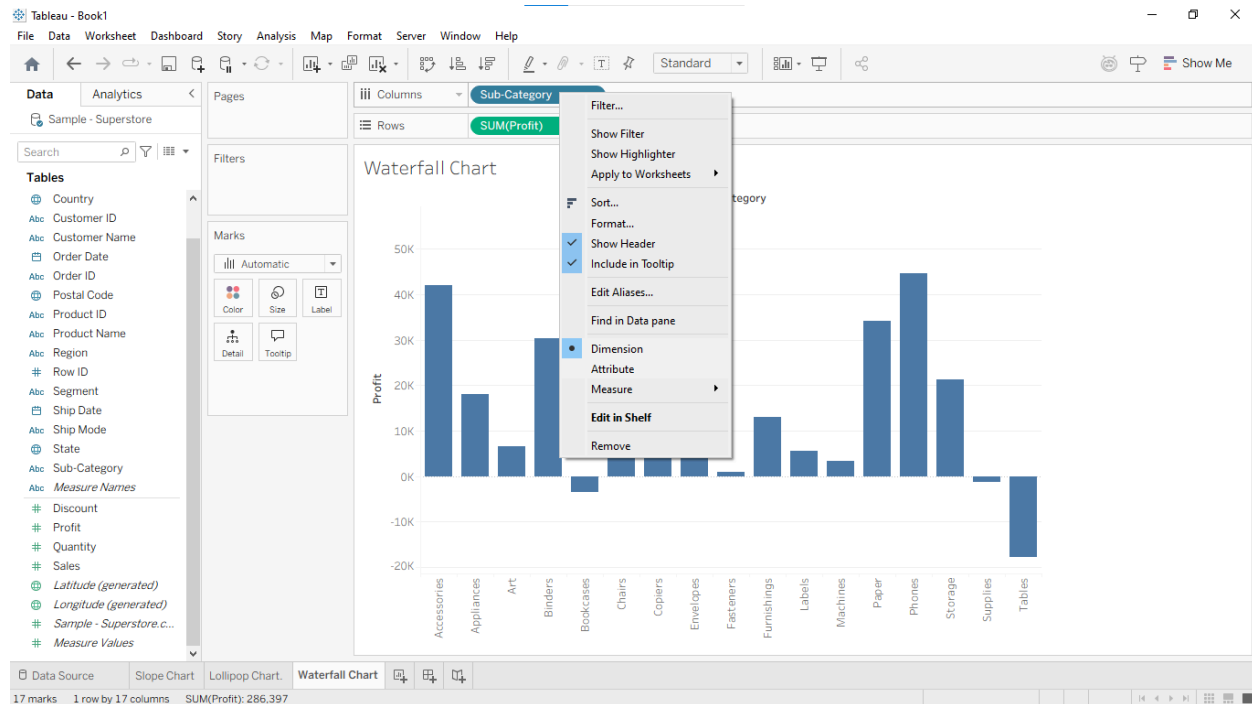
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Step 1: Set Up the Base Worksheet

1. In Tableau Desktop, create a new worksheet and rename it: Waterfall Chart.
2. Drag Sub-Category onto the Columns shelf.
3. Drag Profit onto the Rows shelf (appears as SUM(Profit)).
4. Sort the view: Click the Sort Descending button on the top toolbar so the highest profit sub-categories appear first.

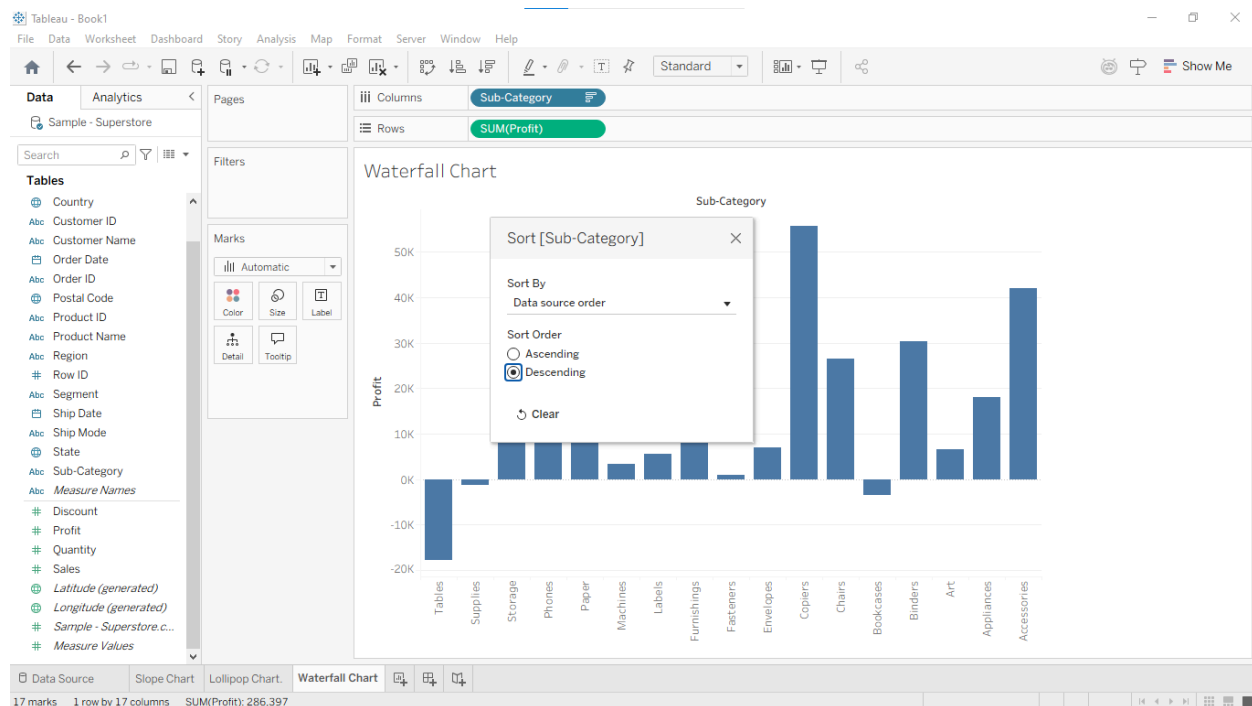


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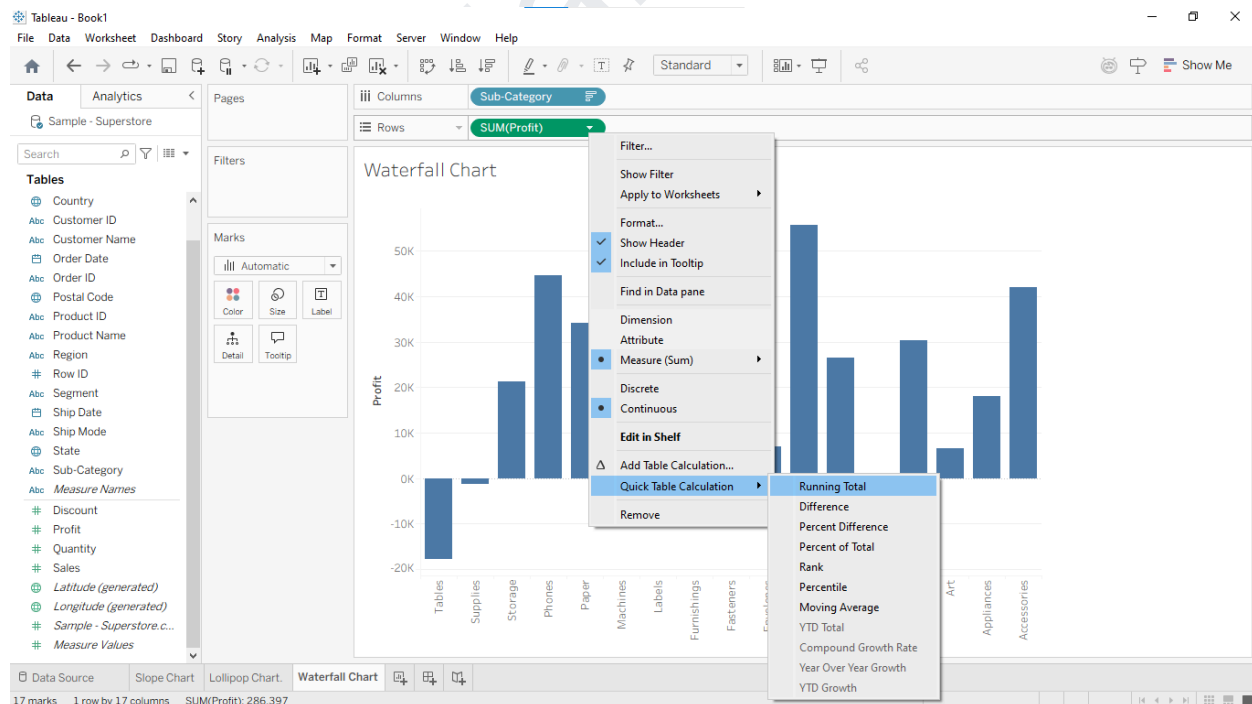
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Step 2: Apply Running Total Table Calculation

1. Right-click the SUM(Profit) pill on the Rows shelf.
2. Select Quick Table Calculation --> Running Total.
3. Notice that the line/bars now accumulate upward across the columns from left to right.

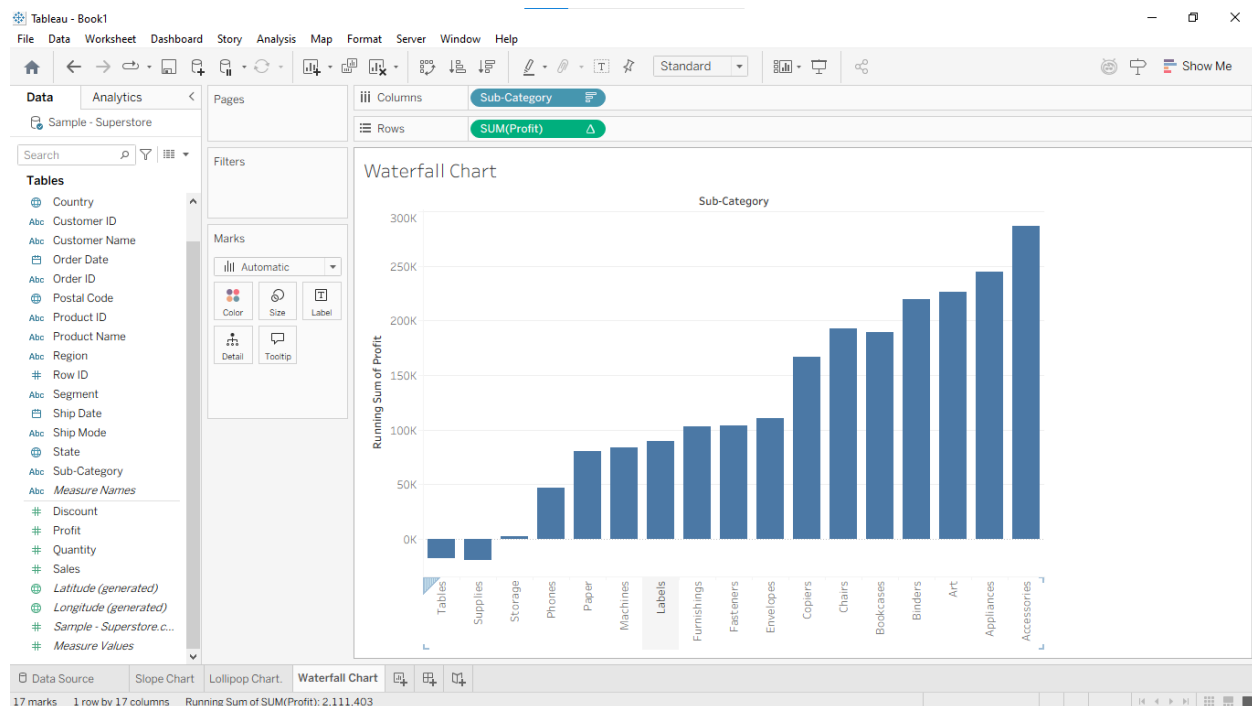


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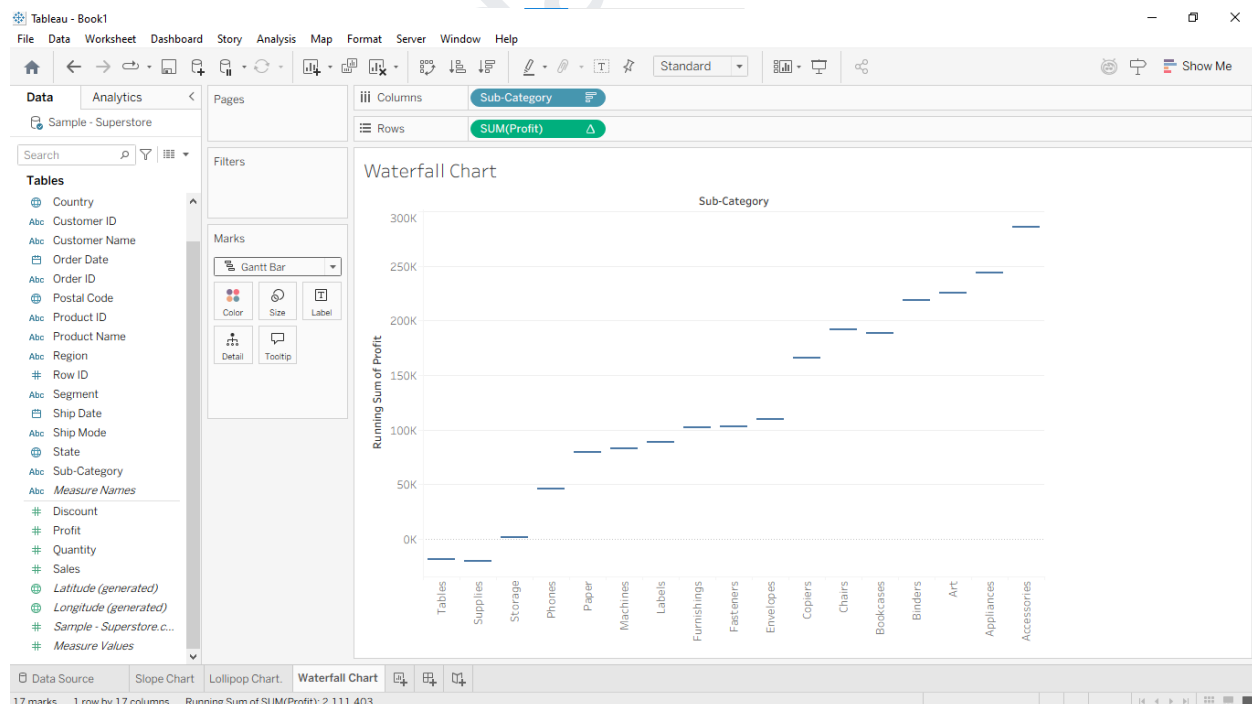
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Step 3: Switch Mark Type to Gantt Bar

1. On the Marks card, change the dropdown selector from Automatic (or Bar) to Gantt Bar.
2. You will see thin horizontal baseline markers representing the running total cumulative level at each sub-category step.



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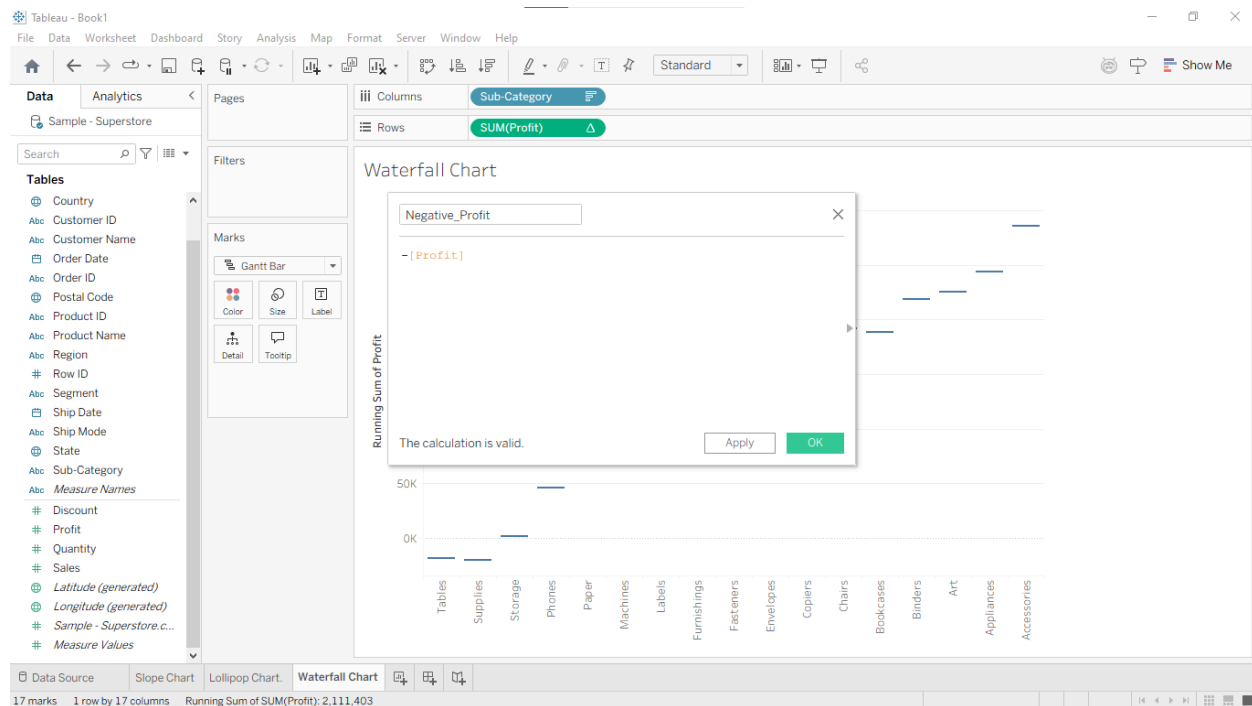
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Step 4: Create the Negative Profit Sizing Calculation

1. In the Data Pane on the left, right-click an empty area --> select Create Calculated Field....
 2. Set the field details: o Field Name: Negative_Profit o Tableau Formula: Code snippet $-\text{[Profit]}$
- o Click OK



Step 5: Size and Color the Waterfall Blocks

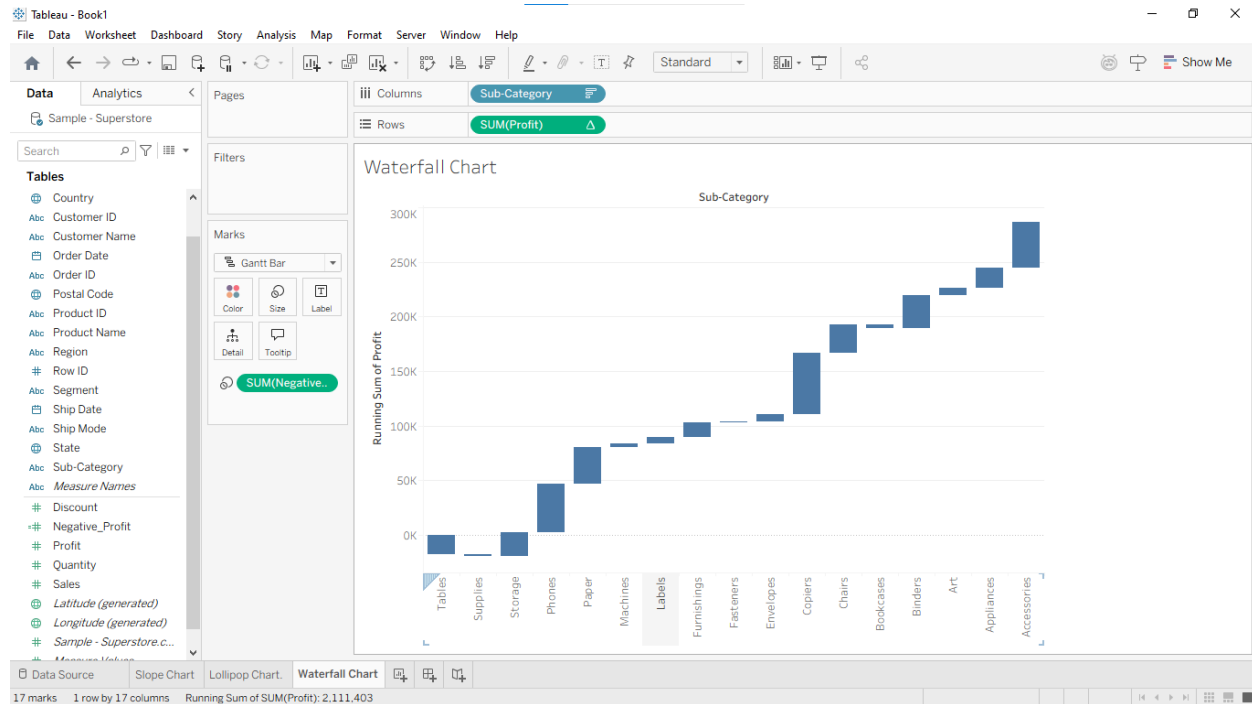
1. Drag Negative_Profit onto the Size tile on the Marks card. o The floating Gantt bars will now expand vertically, bridging each step to the next

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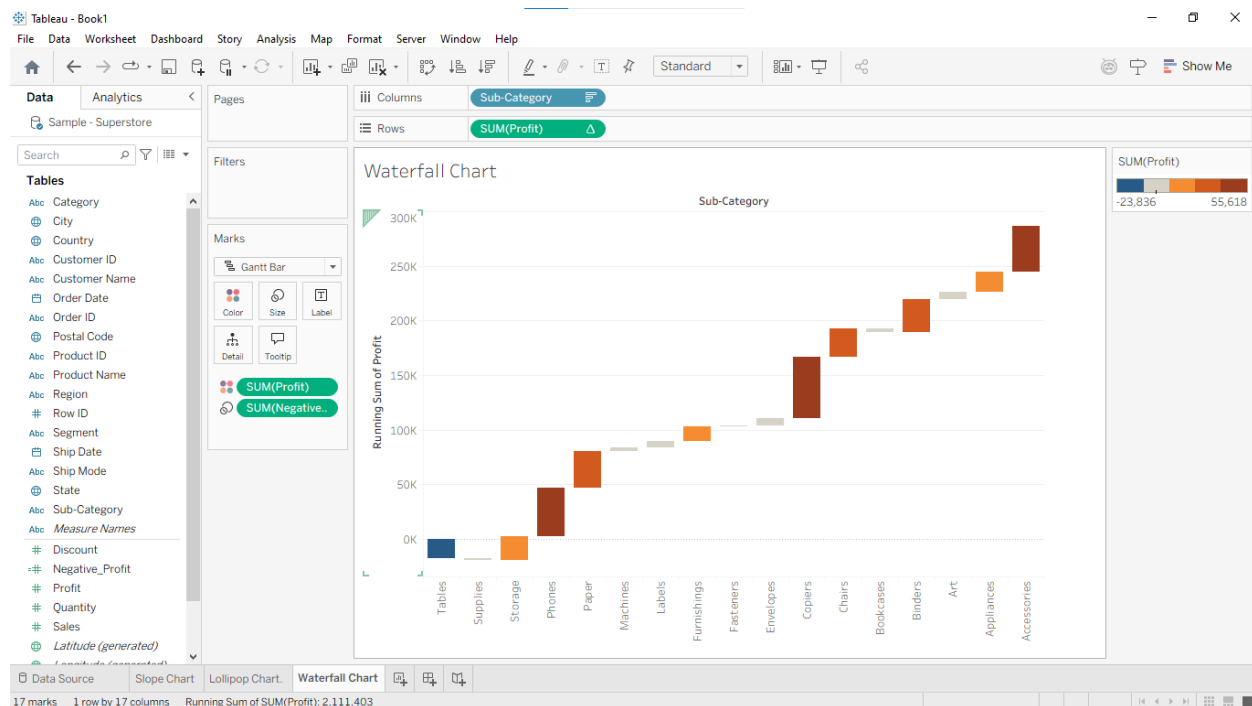
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2. Drag Profit (original measure) onto the Color tile on the Marks card.
3. Click the Color tile --> select Edit Colors...:
 - o Change Palette to Red-Green Diverging (or Red-Blue).
 - o Check Use Full Color Range or set Center at 0.
 - o Click OK.

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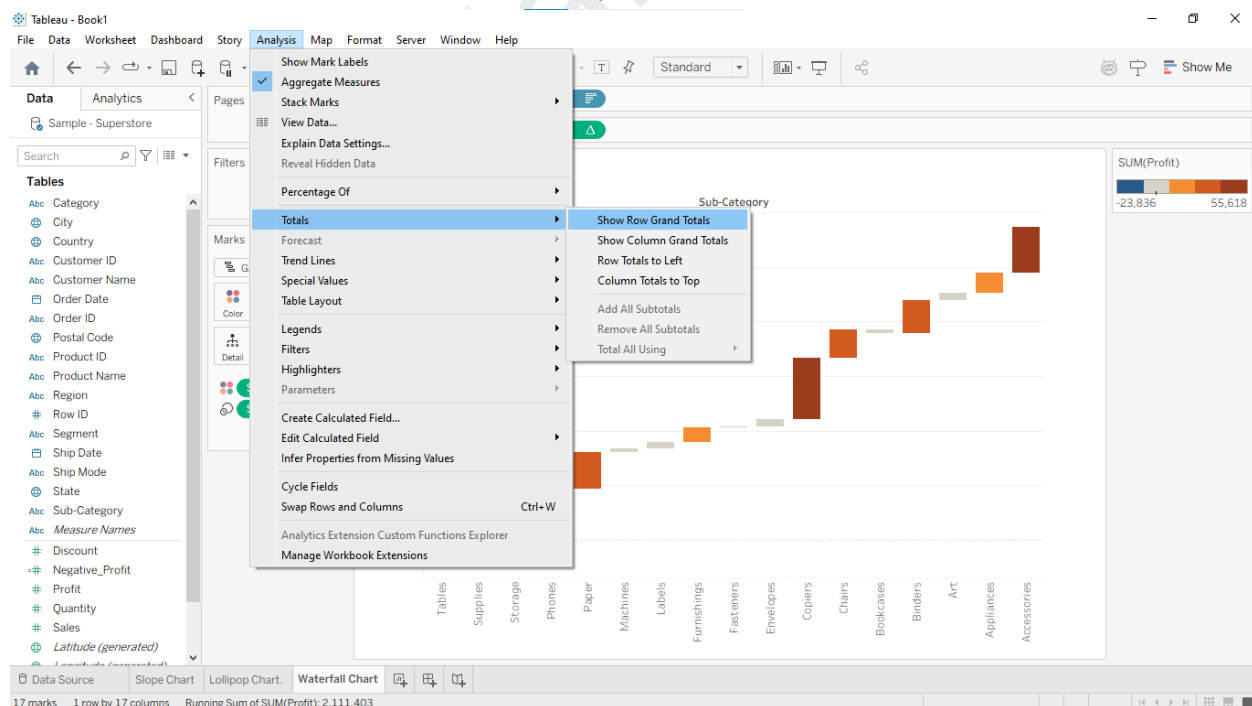
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Step 6: Add Grand Totals to Close the Waterfall

1. Go to the top application menu: select Analysis --> Totals --> Show Row Grand

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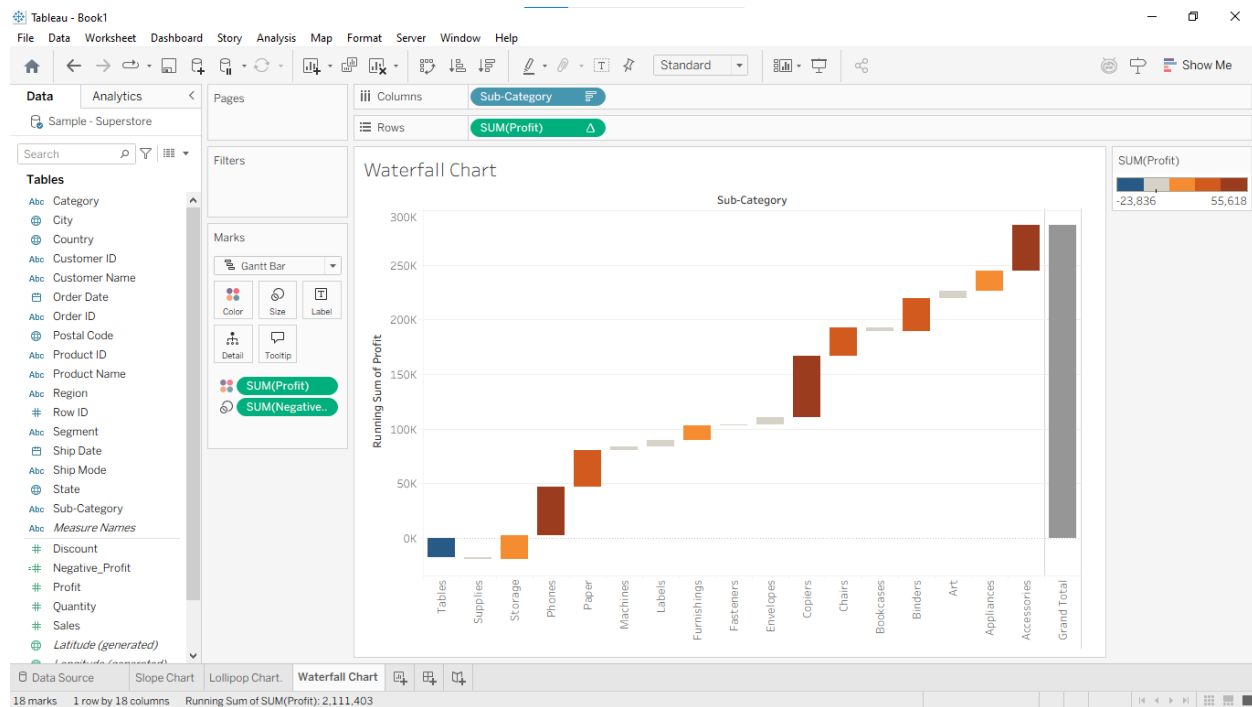
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2. A final anchor pillar will appear at the right end showing the final cumulative profit (\$286,397.02)



C: Build Sparklines for Trend Analysis

Step 1: Set Up the Base Worksheet and Table Dimensions

1. In Tableau Desktop, create a new worksheet and rename it: Sparklines View.
2. Drag Category onto the Rows shelf.
3. Drag Sub-Category onto the Rows shelf (to the right of Category).

Step 2: Add Continuous Time-Series for Trend Shape

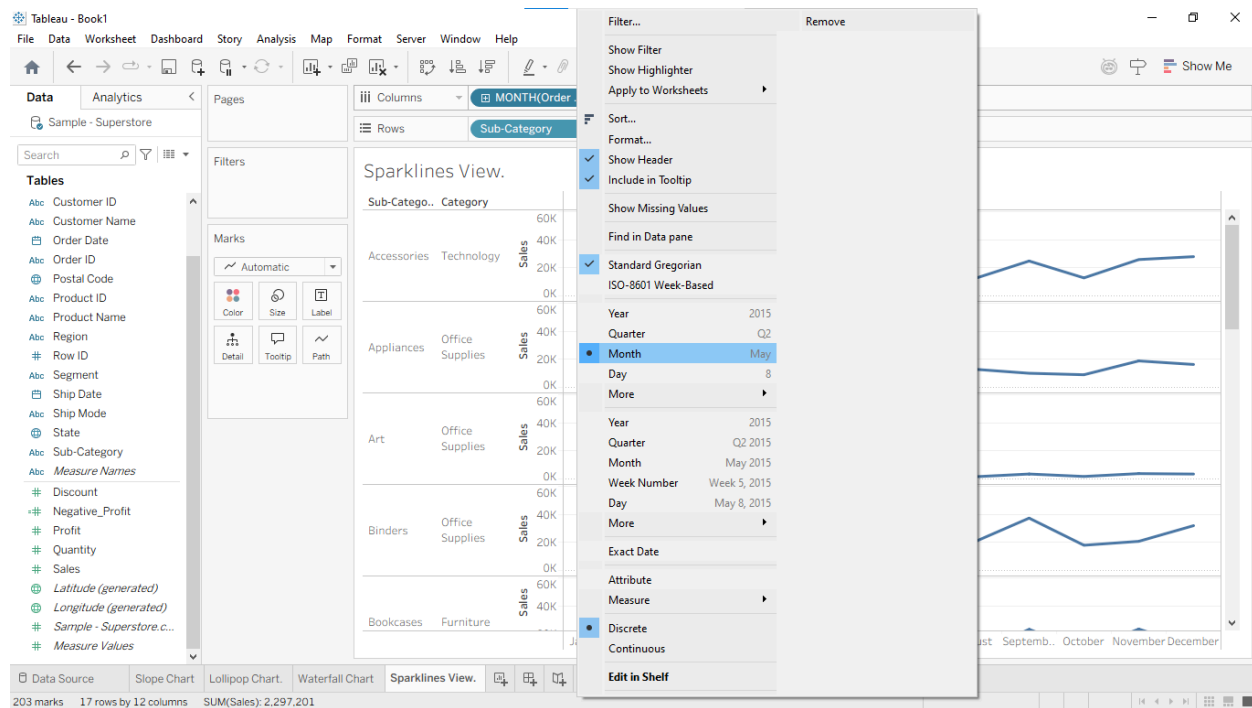
1. Drag Order Date onto the Columns shelf.
2. Right-click the Order Date pill on Columns --> choose the continuous Month option (the second Month entry in the dropdown list, showing Month May 2015, green pill MONTH(Order Date)).

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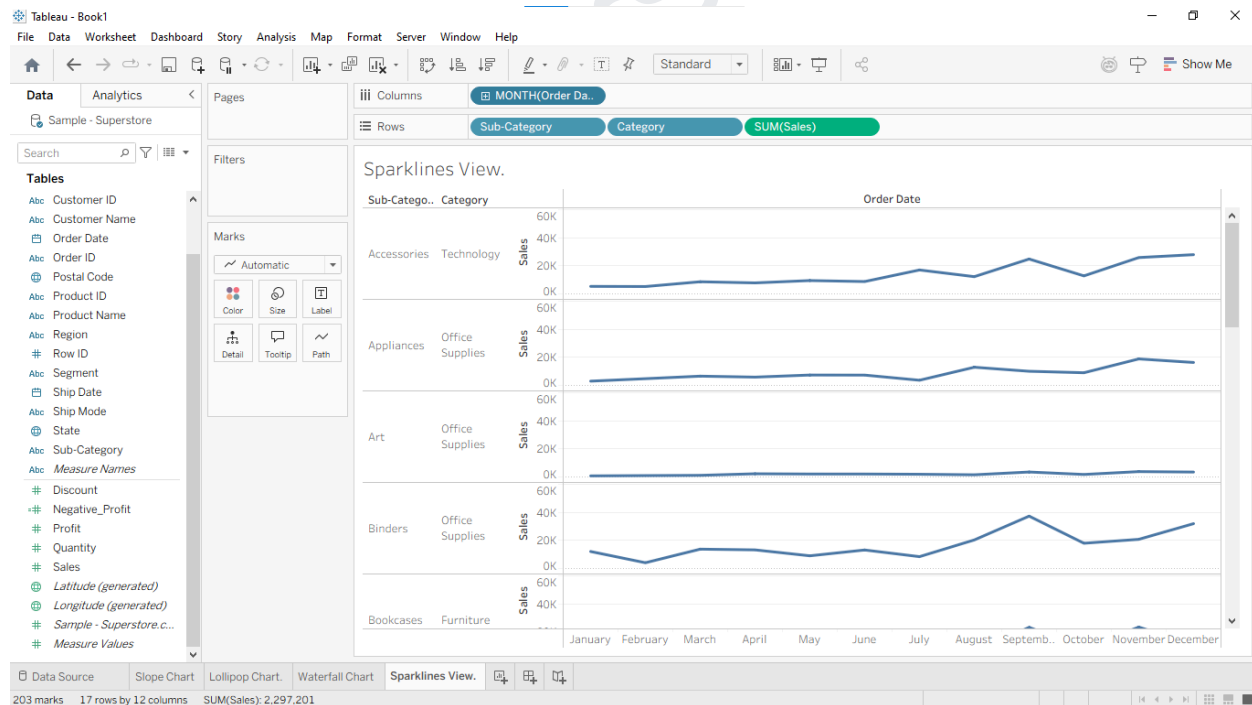
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3. Drag Sales onto the Rows shelf (appears as SUM(Sales)).



4. Notice that Tableau renders 17 individual line charts broken down by Sub-Category.

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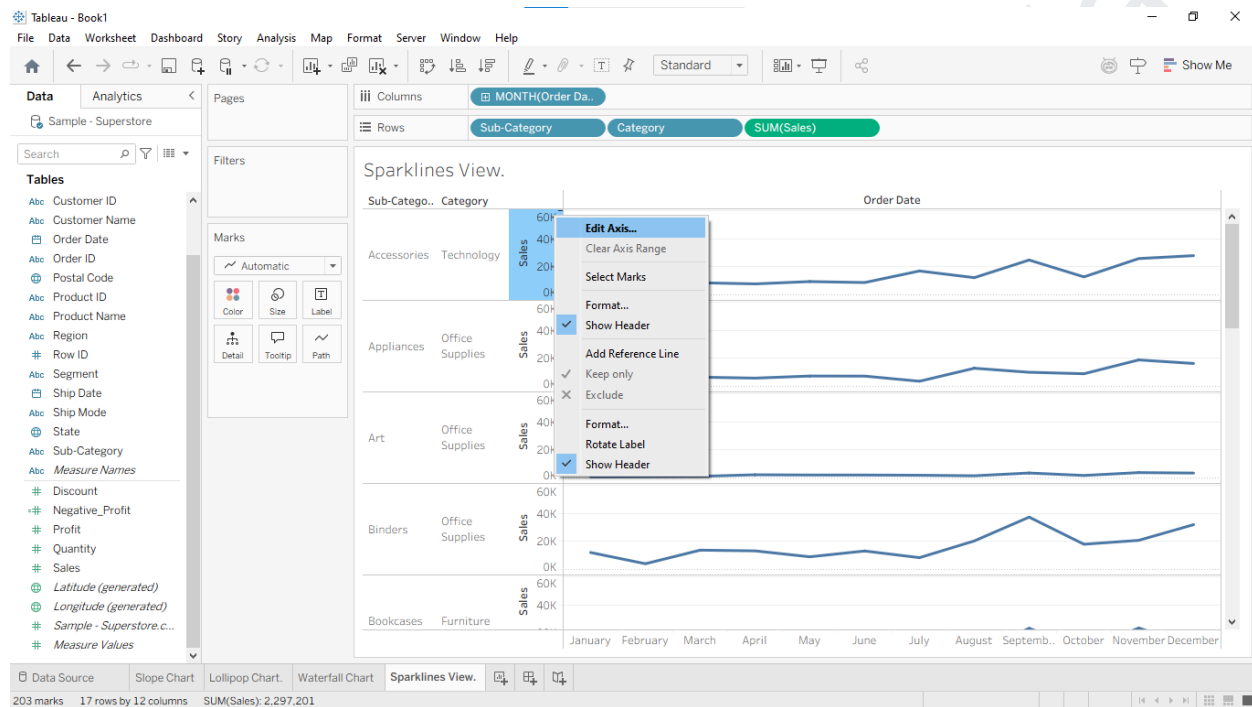
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Step 3: Add Total Sales Metric Pill alongside the Sparkline

1. Drag Sales onto the Rows shelf again (to the left of SUM(Sales) or on Columns to create a side-by-side KPI cell). o Alternative Clean Method: Drag Sales onto the Tooltip / Label or create a discrete total pill.

2. To create the classic tabular layout: o Leave MONTH(Order Date) on Columns. o Right-click the vertical Y-axis on any sparkline row --> select Edit Axis....



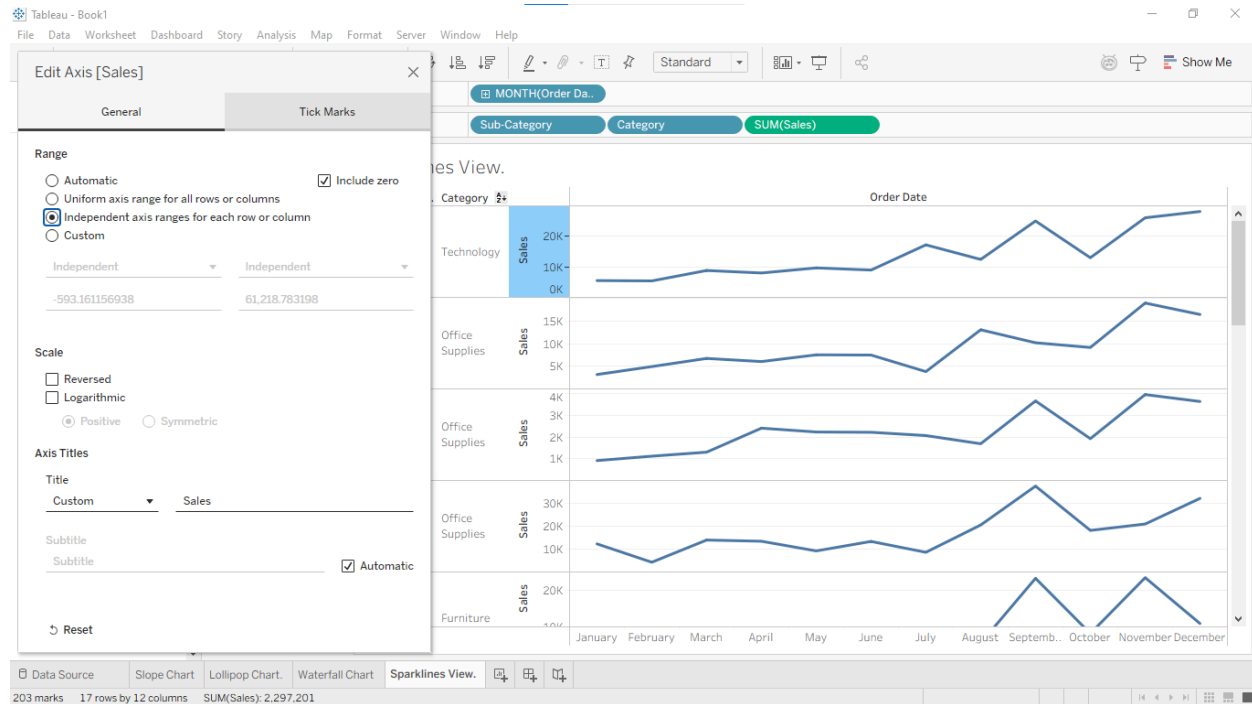
o Under the Range settings, choose Independent axis ranges for each row or column (this allows each sub-category to show its distinct trend profile without being compressed by larger categories)

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o Close the Edit Axis dialog.

Step 4: Strip Away Visual Clutter (Clean Sparkline Design)

1. Hide Axes:

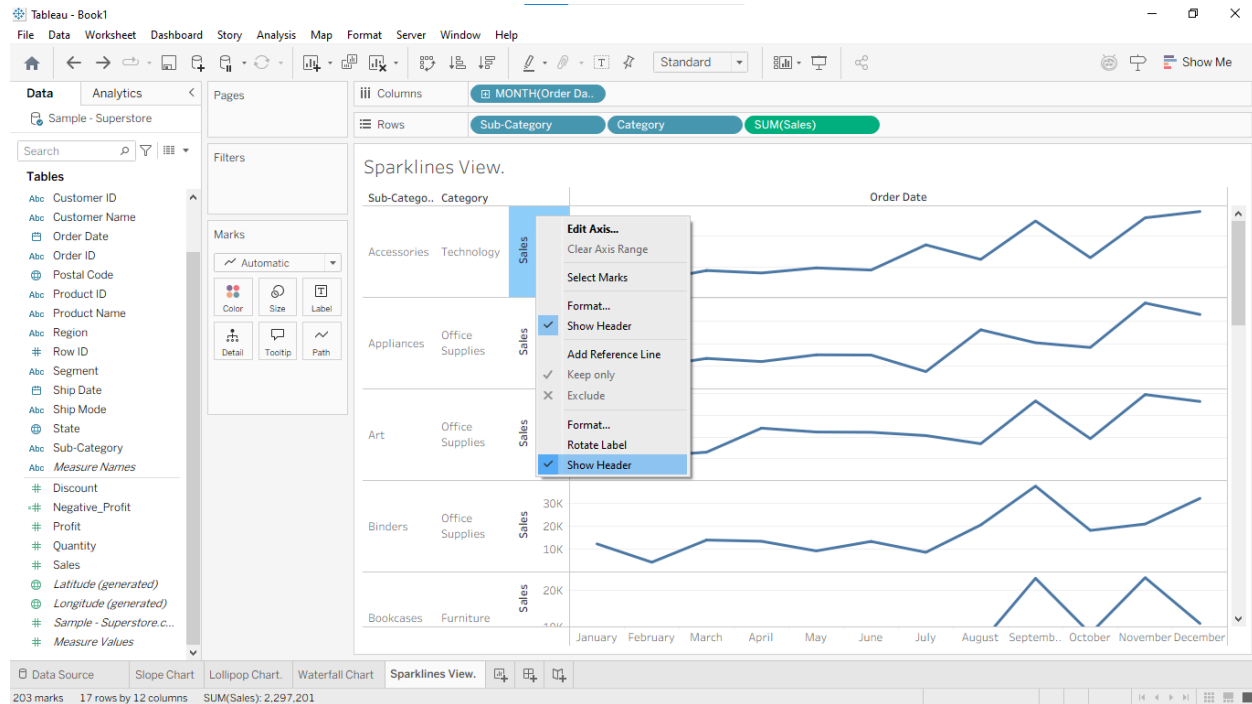
o Right-click the vertical Y-axis (Sales) --> uncheck Show Header.

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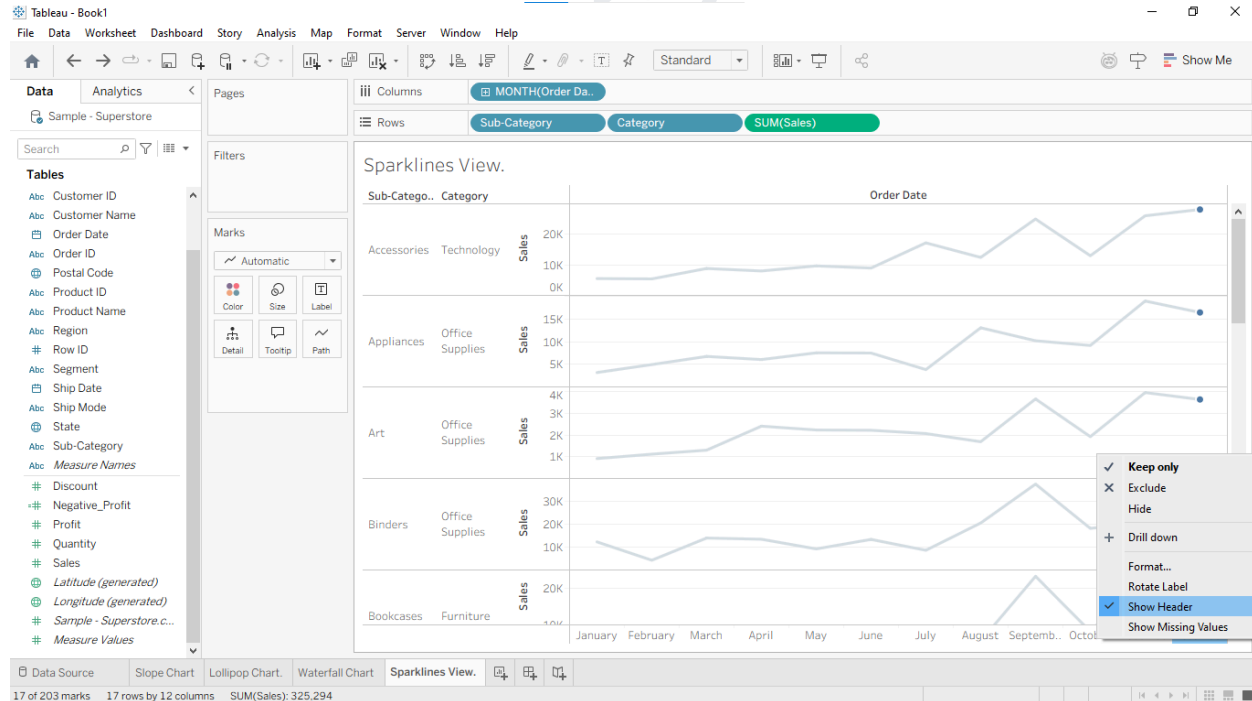
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o Right-click the bottom horizontal X-axis (Order Date) --> uncheck Show Header

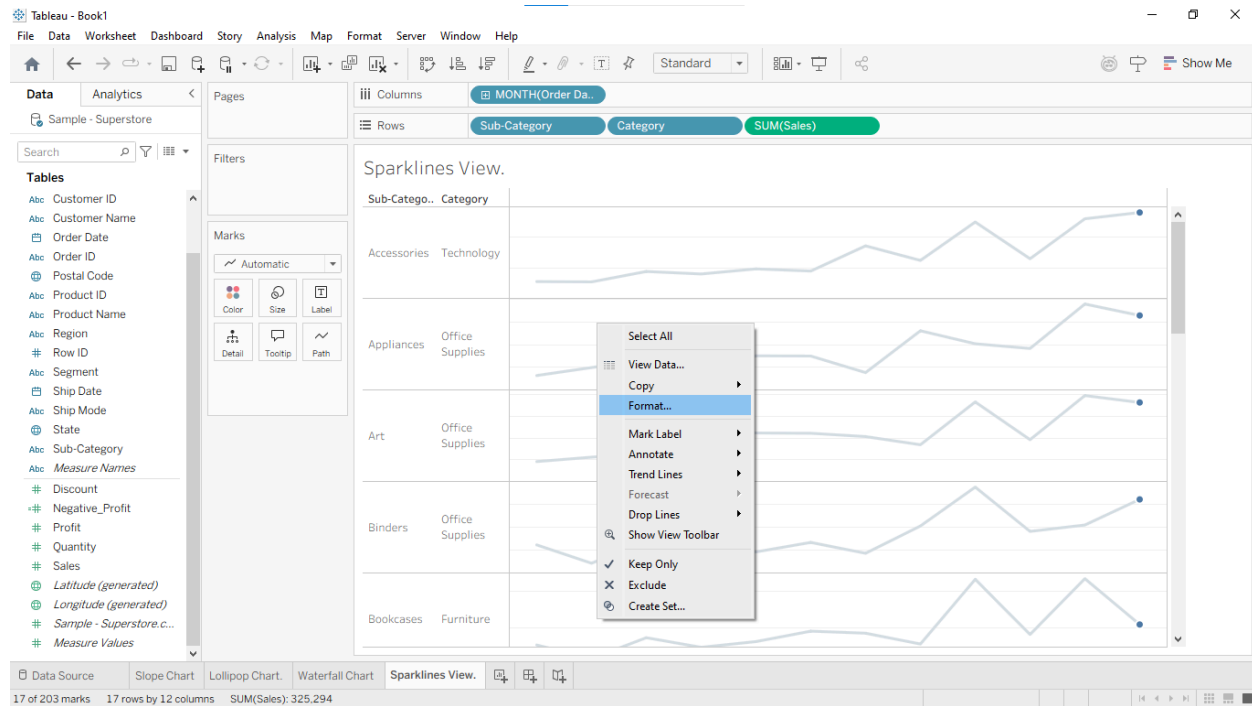


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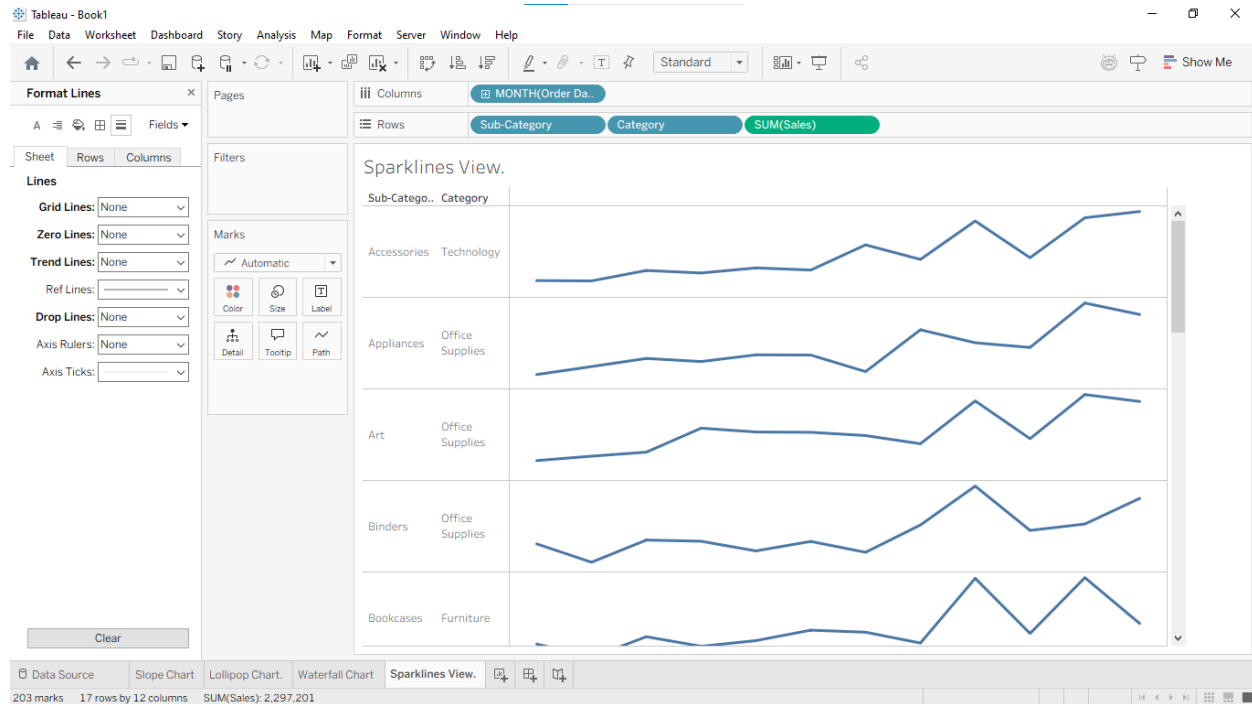
2. Remove Gridlines & Dividers: o Right-click anywhere in the chart canvas --> select Format....



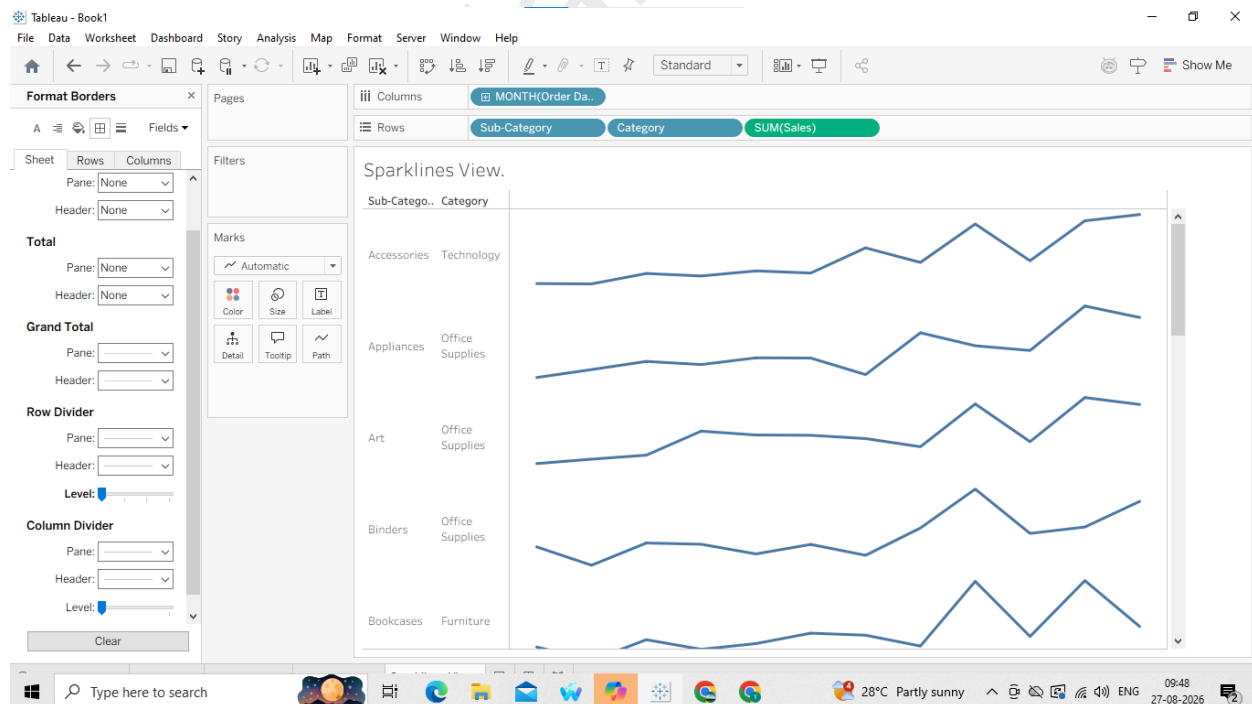
- o In the Format pane on the left, Click the Lines icon (paintbrush with lines):
- Set Grid Lines to None.
- Set Zero Lines to None.
- Set Trend Lines and Drop Lines to None.

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- o Click the Borders icon (square outline):
- Set Row Divider to a light thin line.
- Set Column Divider to None.



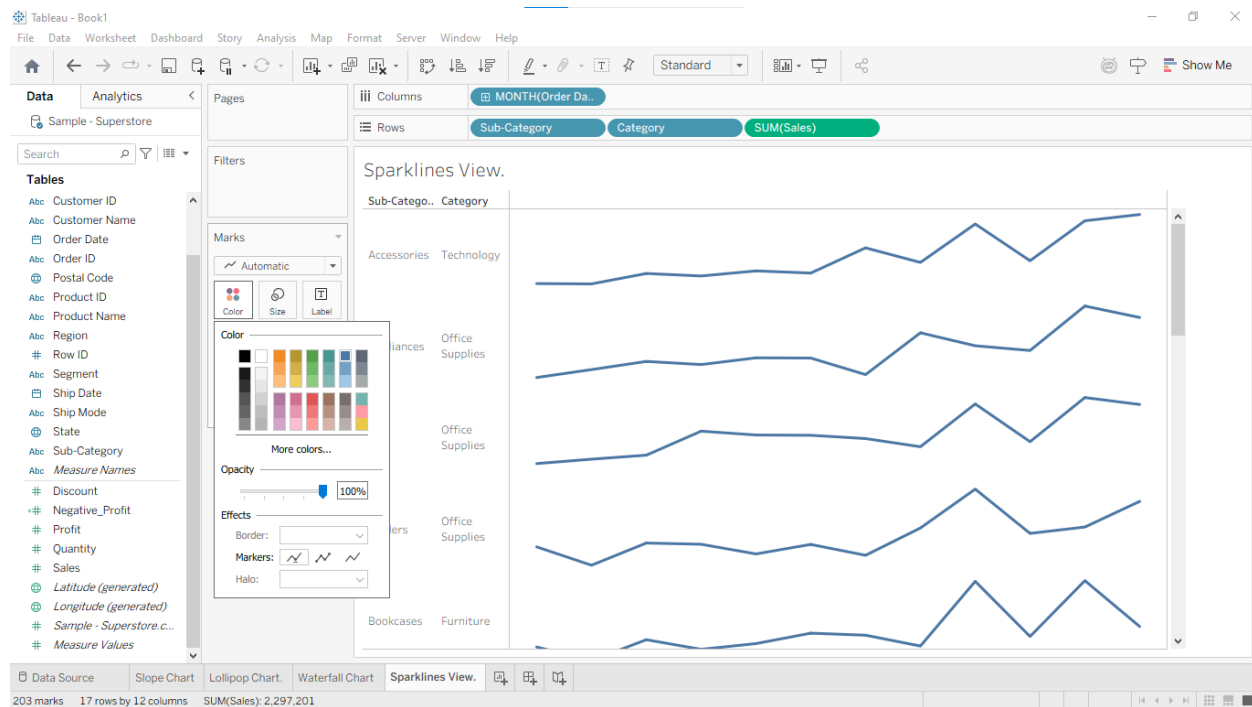
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Step 5: Highlight Minimum and Maximum Milestone Endpoints

1. On the Marks card, click the Color tile --> set the line color to dark slate gray or deep navy.



2. On the Marks card, click the Label tile:

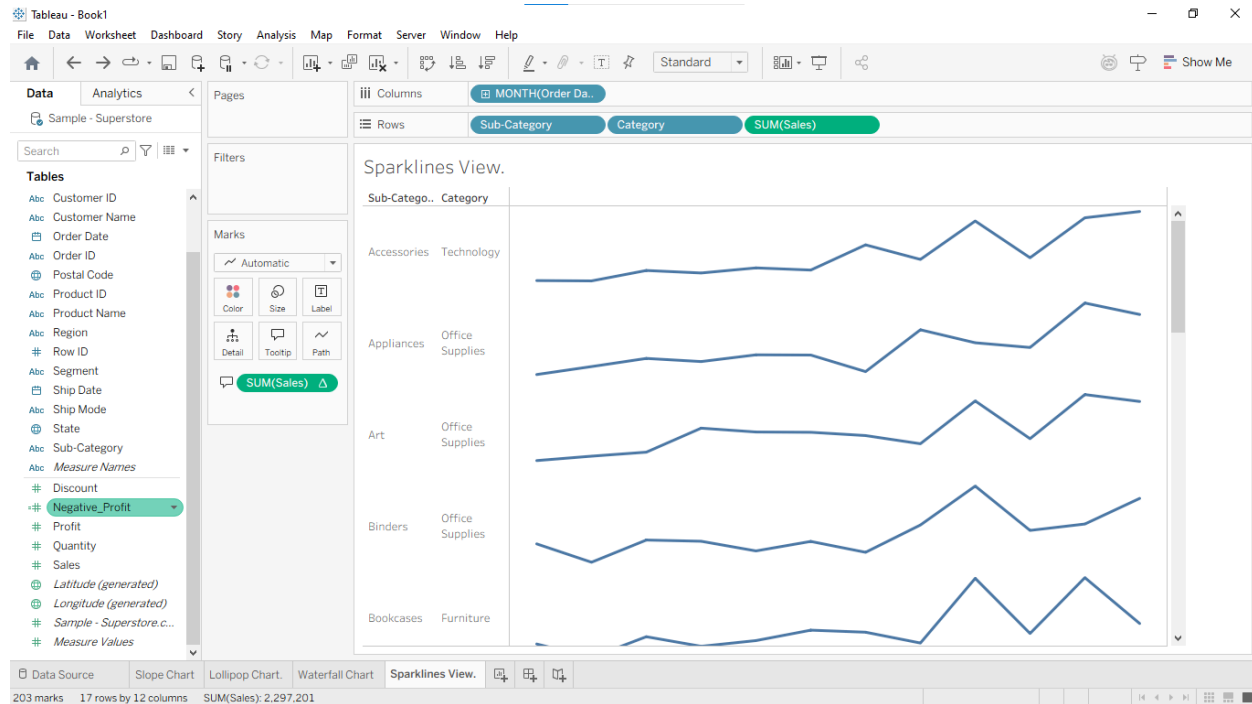
- o Check Show mark labels.
- o Under Marks to Label, select Min / Max.
- o Under Field, ensure SUM(Sales) is selected.
- o Under Options, check Label minimum value and Label maximum value

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3. Now, each sparkline cleanly displays its lowest and peak performance milestones without crowding intermediate points.

D: Implement Sheet Swapping for Dynamic Dashboards

Step 1: Create the Control Parameter

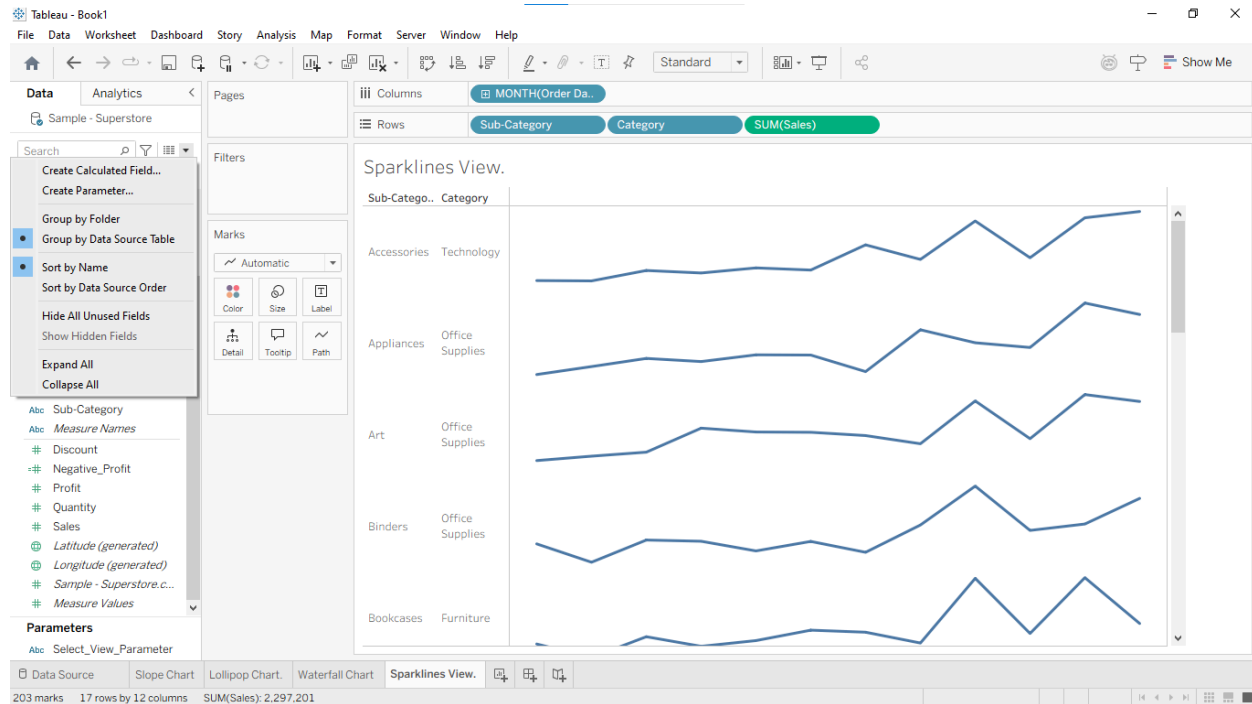
1. In the Data Pane on the left, click the small dropdown arrow next to the search bar (or right-click an empty area) --> select Create Parameter..

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2. Configure the parameter properties:

- o Name: Select_View_Parameter
- o Data type: String
- o Allowable values: Choose List
- o In the List of values table, type the two view names:

1. Category Sales (Bar)

2. Regional Profit (Map)

o Current value: Set to Category Sales (Bar)

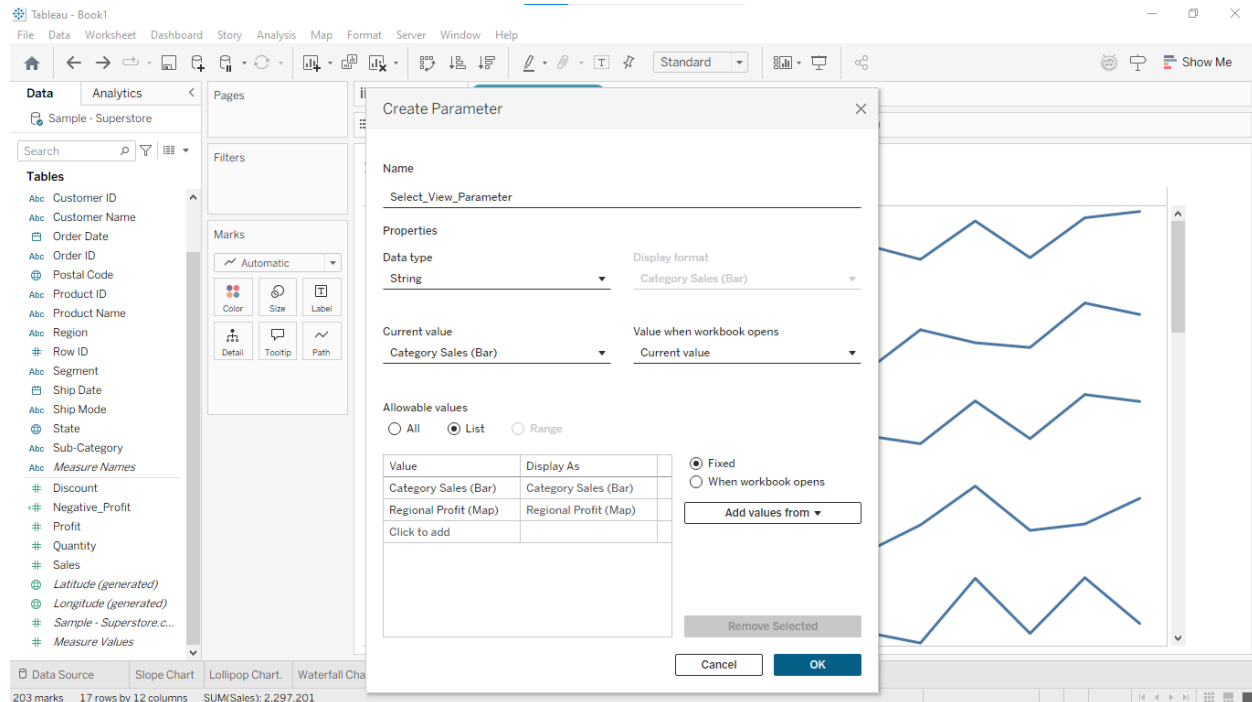
3. Click OK.

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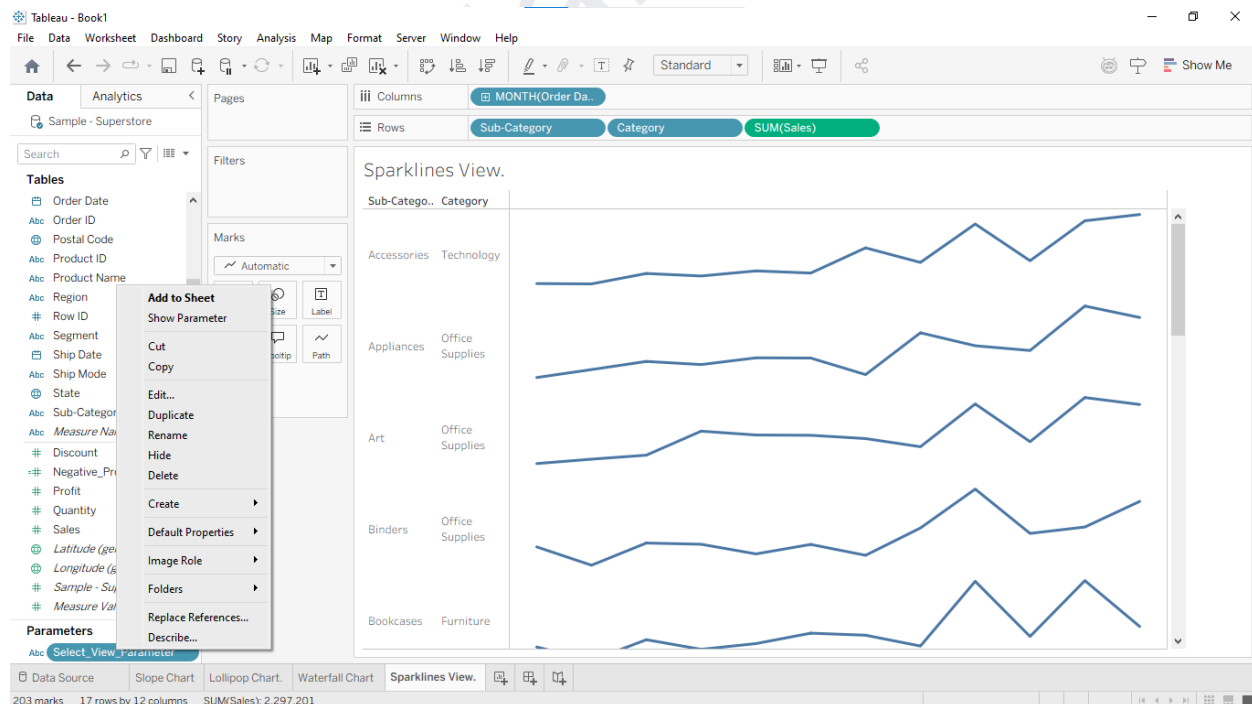
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4. In the Data Pane under Parameters, right-click `Select_View_Parameter` --> select Show Parameter.



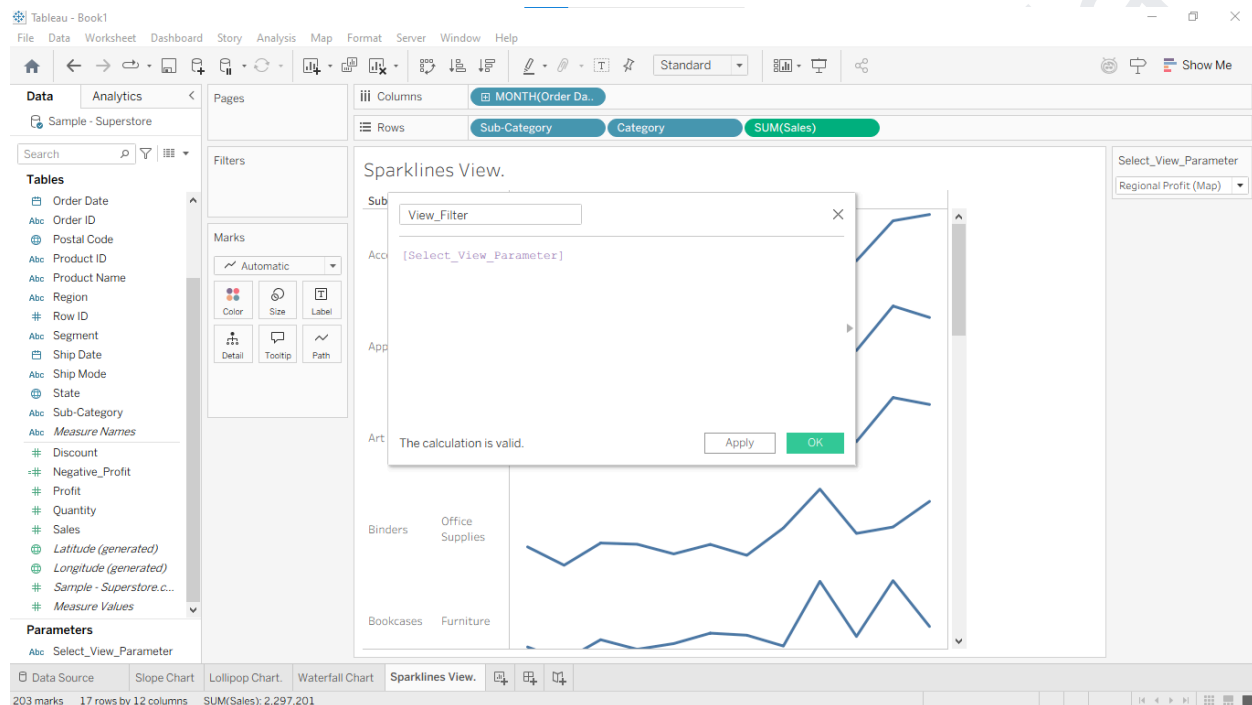
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Step 2: Create the Dynamic Filter Field

1. Right-click an empty space in the Data Pane --> select Create Calculated Field....
2. Configure the field:
 - o Field Name: View_Filter
 - o Tableau Formula: Code snippet [Select_View_Parameter]
 - o (Note: This formula simply outputs the exact string selected in the parameter).
3. Click OK



Step 3: Build Sheet 1 (View 1 - Category Sales)

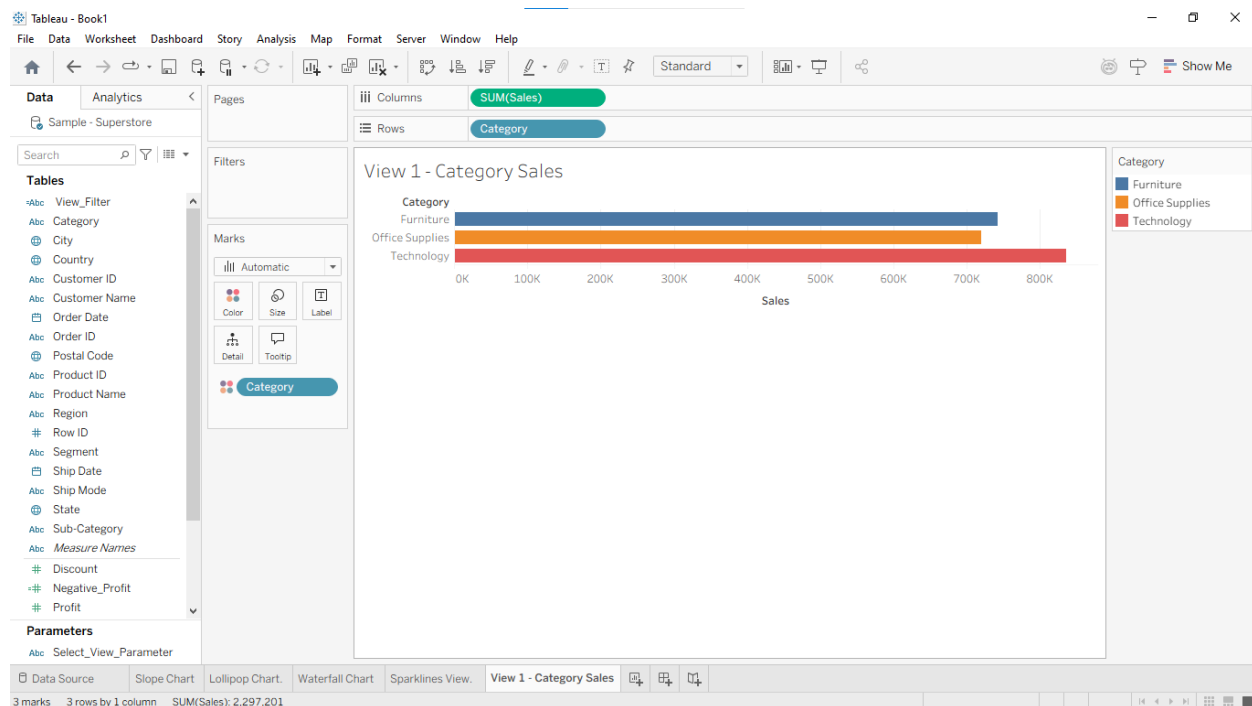
1. Create a new worksheet and name it: View 1 - Category Sales.
2. Build the chart:
 - o Drag Category to the Rows shelf.
 - o Drag Sales to the Columns shelf.
 - o Drag Category to Color on the Marks card.
 - o Sort bars descending

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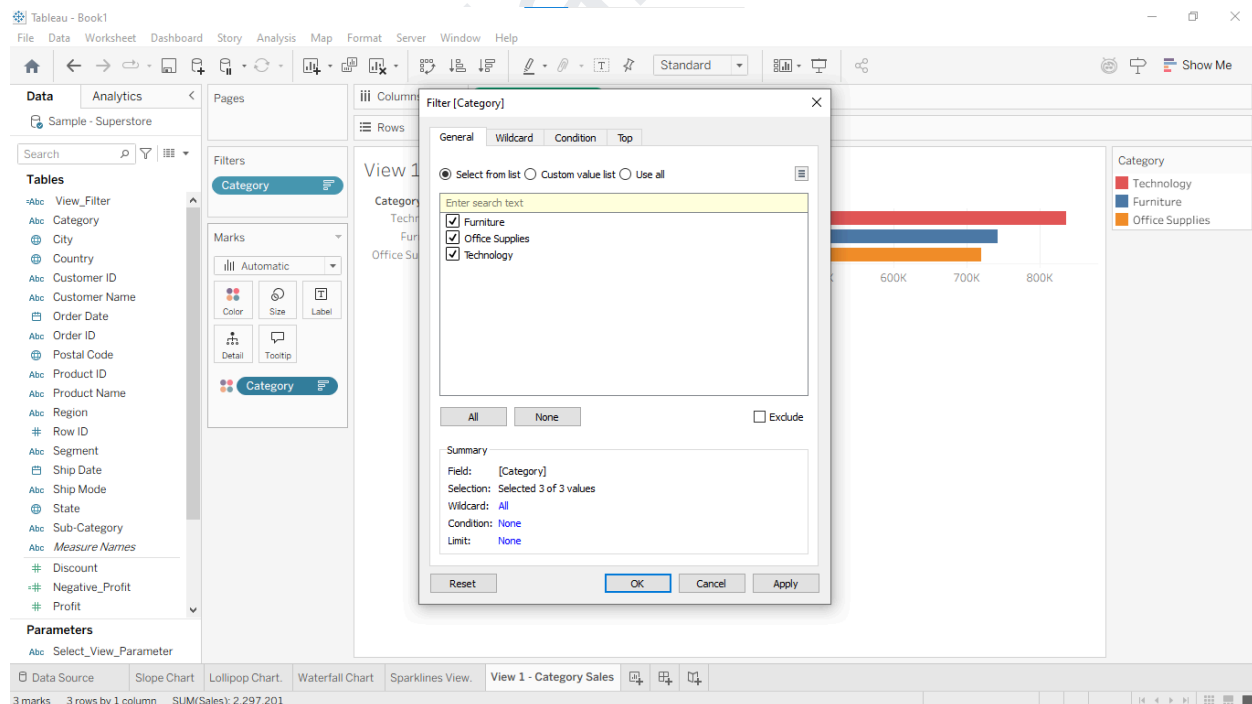
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3. Apply the dynamic filter:

- o Ensure your parameter is set to Category Sales (Bar).
- o Drag the calculated field View_Filter to the Filters shelf.
- o Select Custom value list (or use the list) --> type or check Category Sales (Bar) --> click OK.



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Step 4: Build Sheet 2 (View 2 - Regional Profit)

1. Create another new worksheet and name it: View

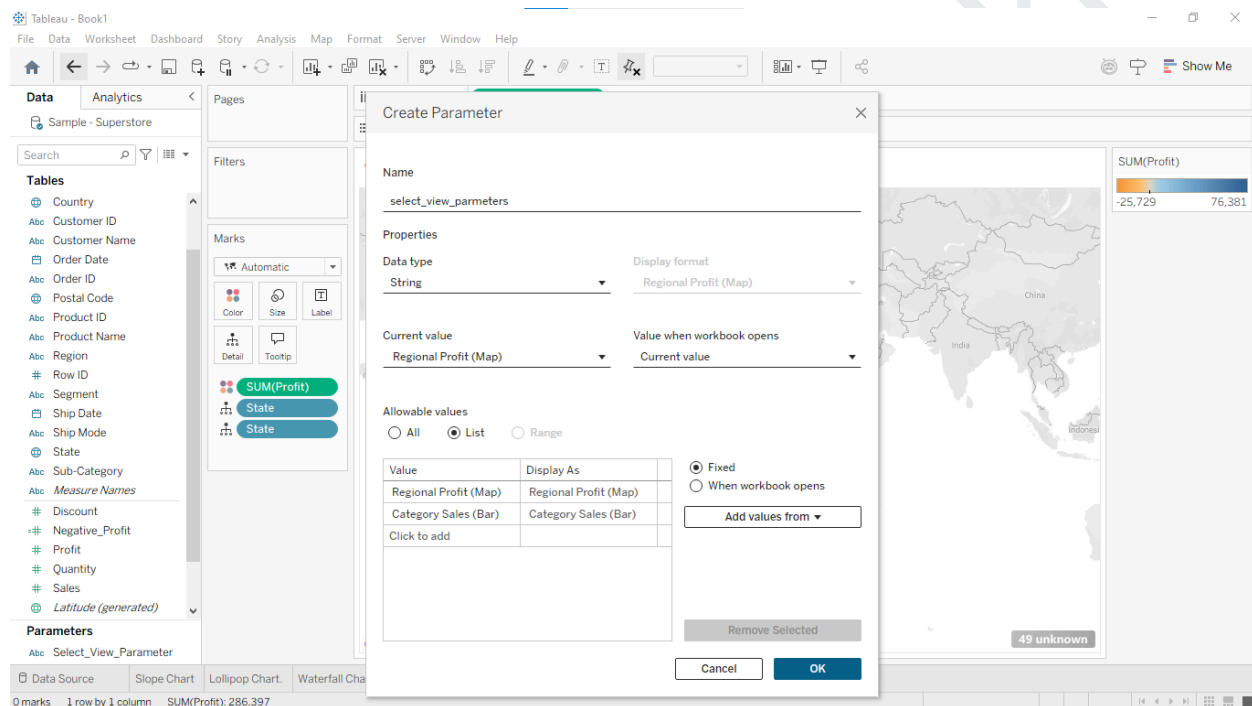
2 - Regional Profit. 2. Build the chart:

o Drag State to the canvas (or double-click to generate latitude/longitude map marks).

o Drag Profit to the Color tile on the Marks card.

o (Alternative bar view if offline: Drag Region to Rows, Profit to Columns).

3. Switch the Parameter to View 2: o In the parameter control, switch the selection to Regional Profit (Map).



4. Apply the dynamic filter

Drag View_Filter to the Filters shelf.

o Check or type Regional Profit (Map) --> click OK

Step 5: Construct the Dynamic Swapping Dashboard

1. Create a new Dashboard (click the New Dashboard icon at the bottom tab bar).

2. Under the Objects pane on the bottom-left: o Click and drag a Vertical Layout Container (or Horizontal) onto the blank dashboard canvas.

3. Drag View 1 - Category Sales into the layout container.

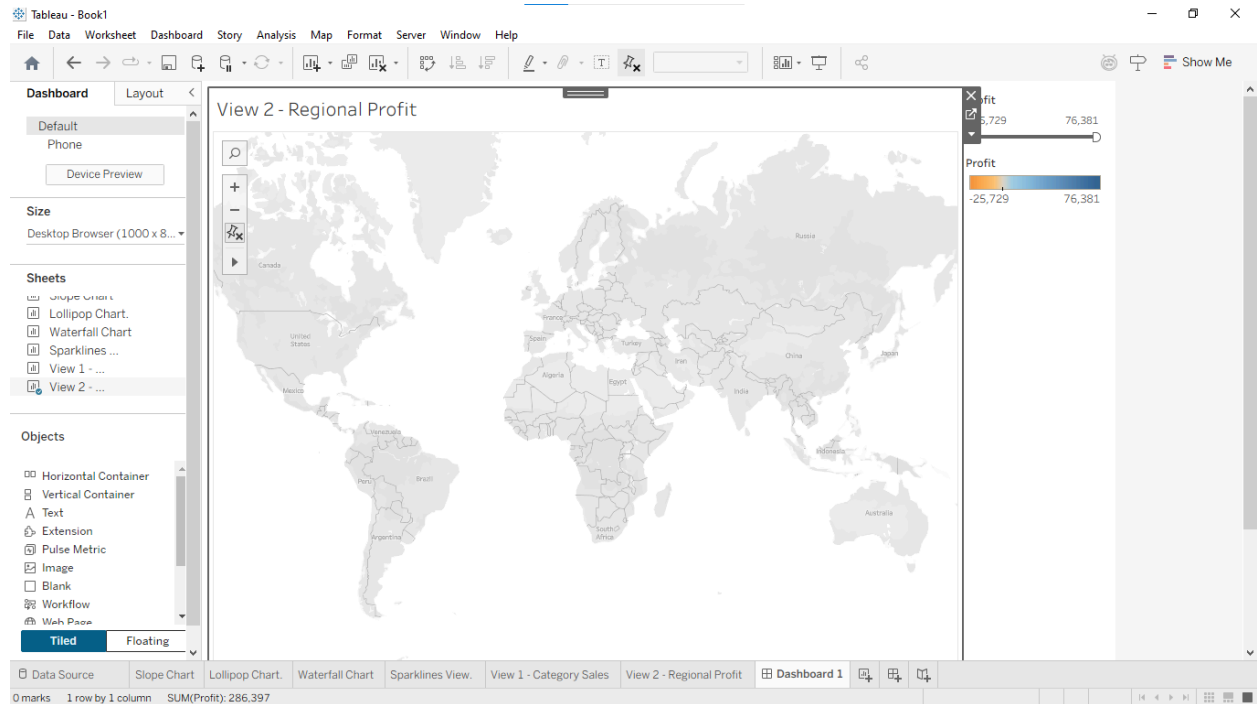
4. Drag View 2 - Regional Profit into the same layout container directly below or above the first sheet.

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5. Critical Step for Sheet Swapping (Hide Titles): Data Visualization with Power BI and Tableau

Dr. Mahendra K. o Right-click the title bar of View

1 - Category Sales on the dashboard --> select Hide Title.

o Right-click the title bar of View

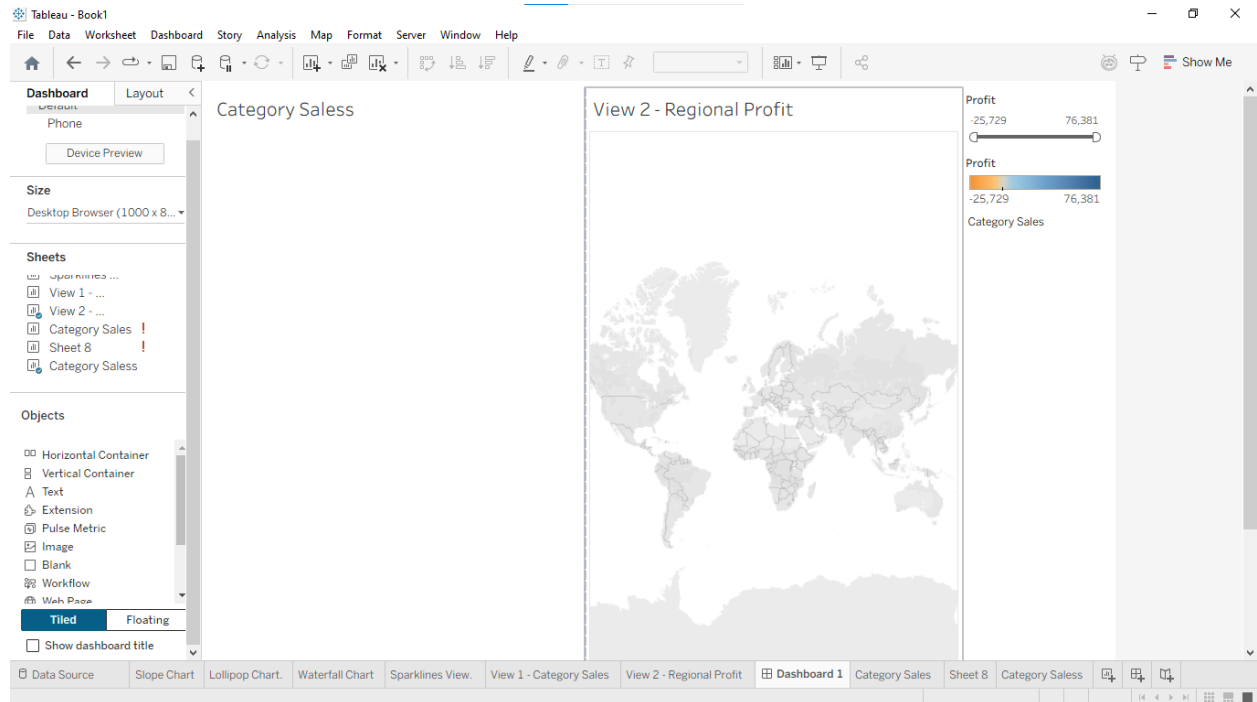
2 - Regional Profit on the dashboard --> select Hide Title. o (Why? If titles remain visible, Tableau will keep the blank title box pinned even when the sheet collapses)

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5. Critical Step for Sheet Swapping (Hide Titles): Data Visualization with Power BI and Tableau

Dr. Mahendra K. o Right-click the title bar of View

1 - Category Sales on the dashboard --> select Hide Title.

o Right-click the title bar of View 2 - Regional Profit on the dashboard --> select Hide Title. o

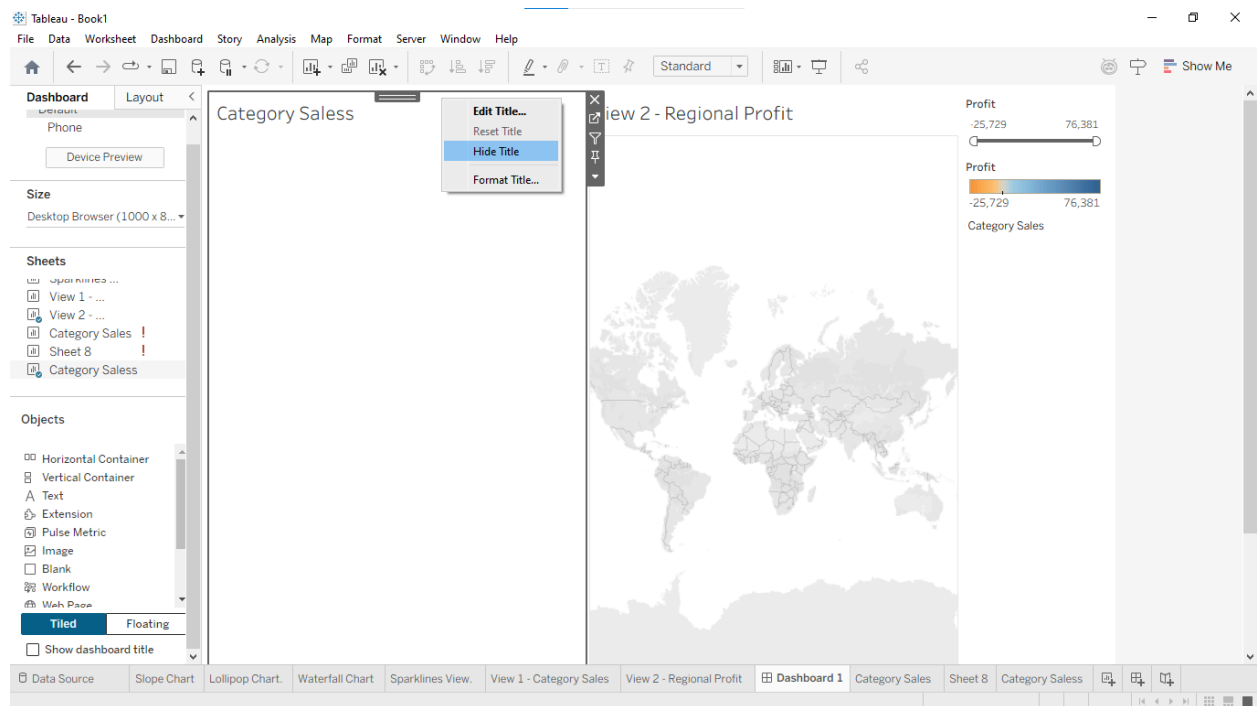
(Why? If titles remain visible, Tableau will keep the blank title box pinned even when the sheet collapses).

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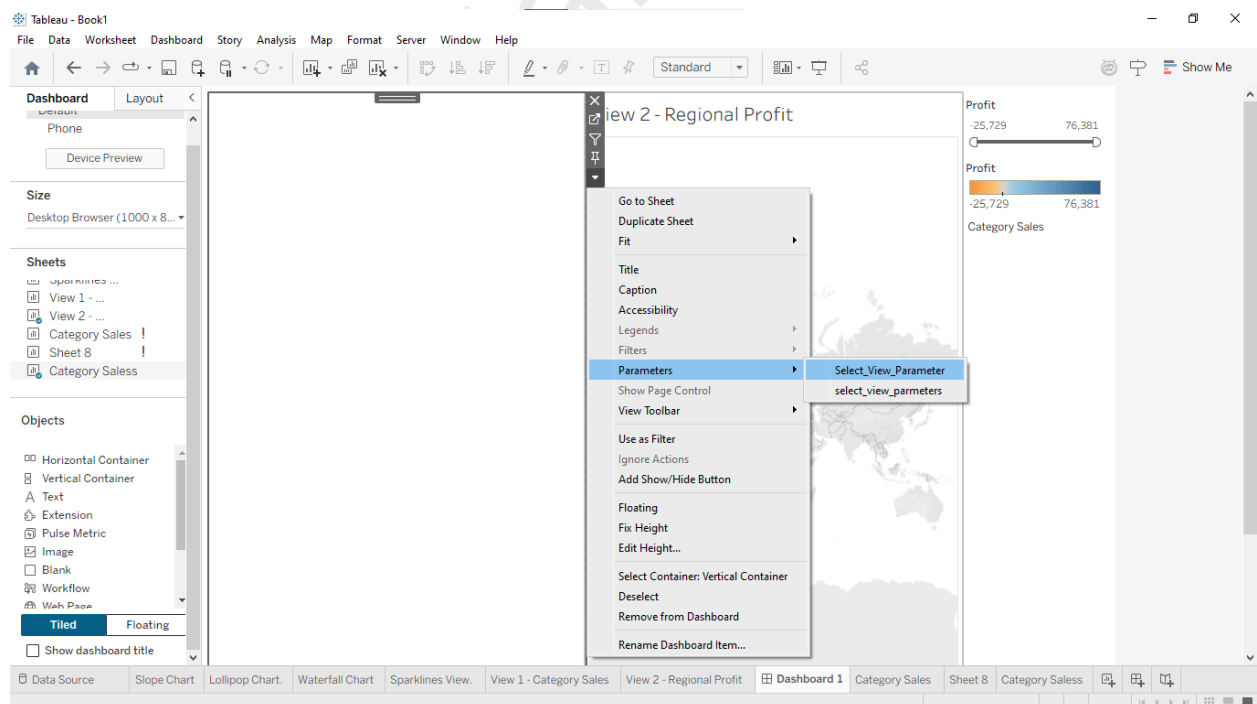
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6. Add the Parameter selector: o Click on the visible sheet --> click the small dropdown arrow on its container border --> Parameters --> Select_View_Parameter.



Step 6: Test the Dynamic Swapping

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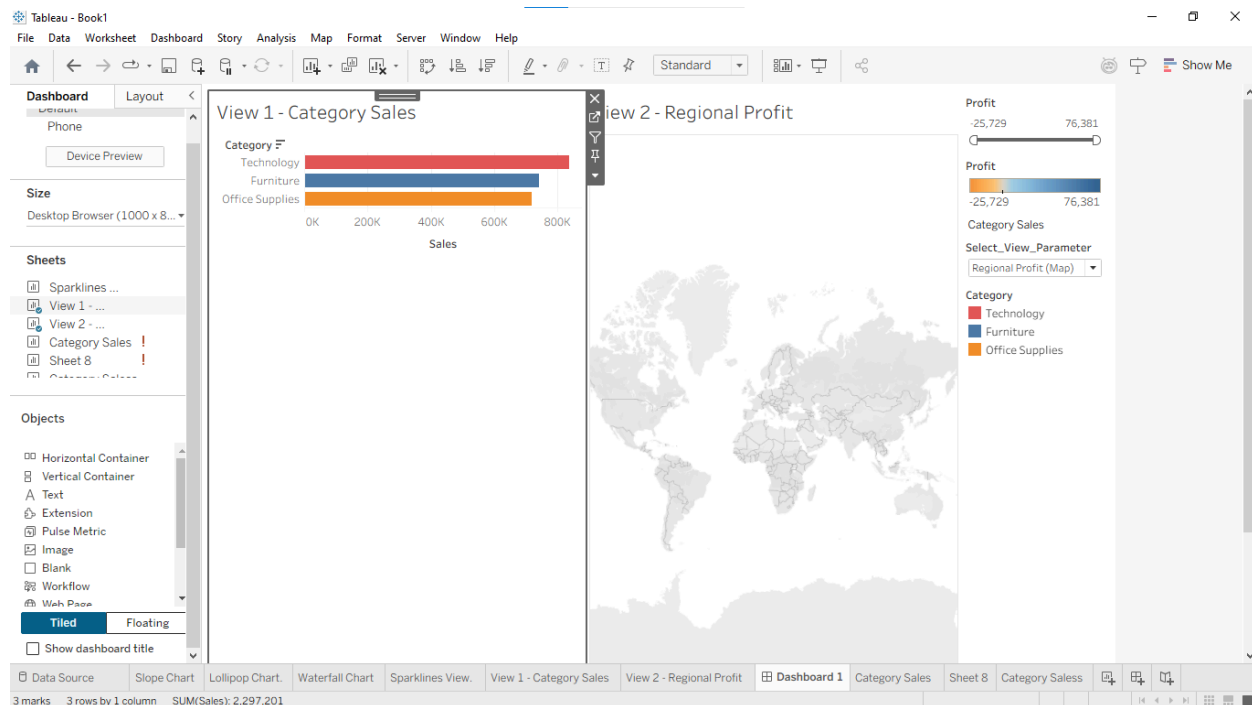
Roll no : T117

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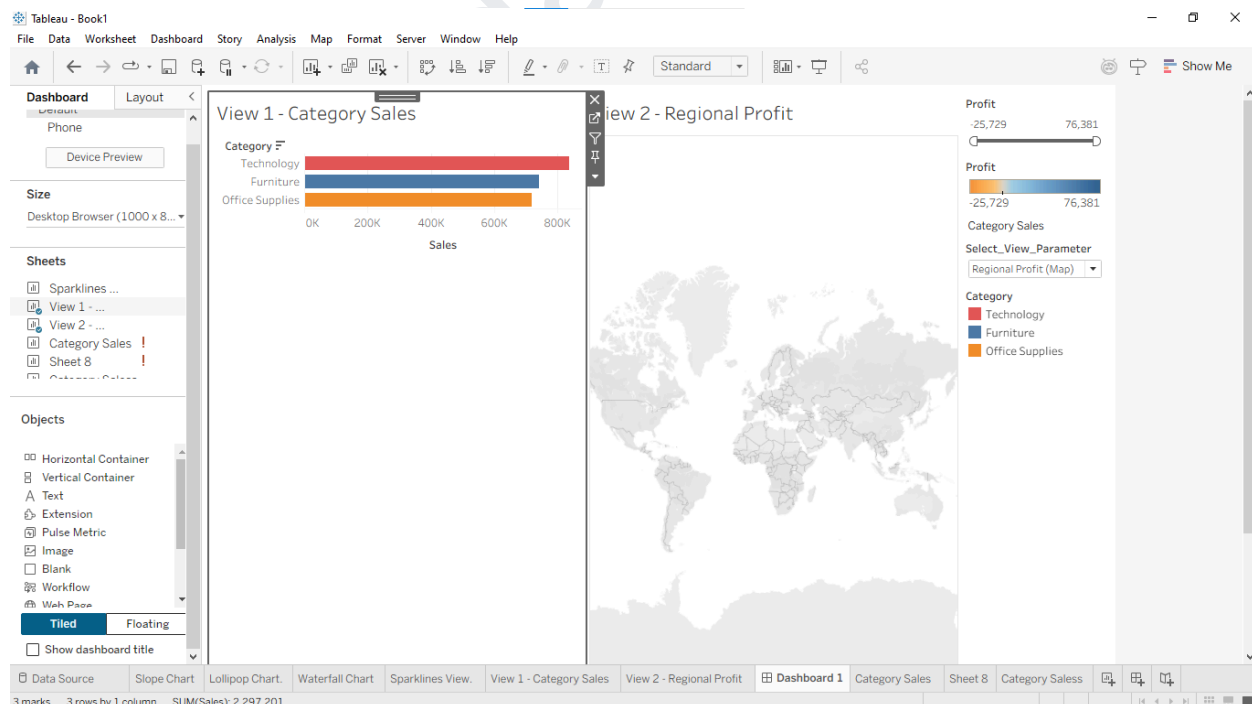
Sub : Data Visualization with Power BI and Tableau

In the parameter dropdown control: o Select Category Sales (Bar):

Sheet 1 expands to fill 100% of the container, while Sheet 2 collapses entirely to 0 height.



o Select Regional Profit (Map): Sheet 1 collapses completely, and Sheet 2 expands to occupy the container.



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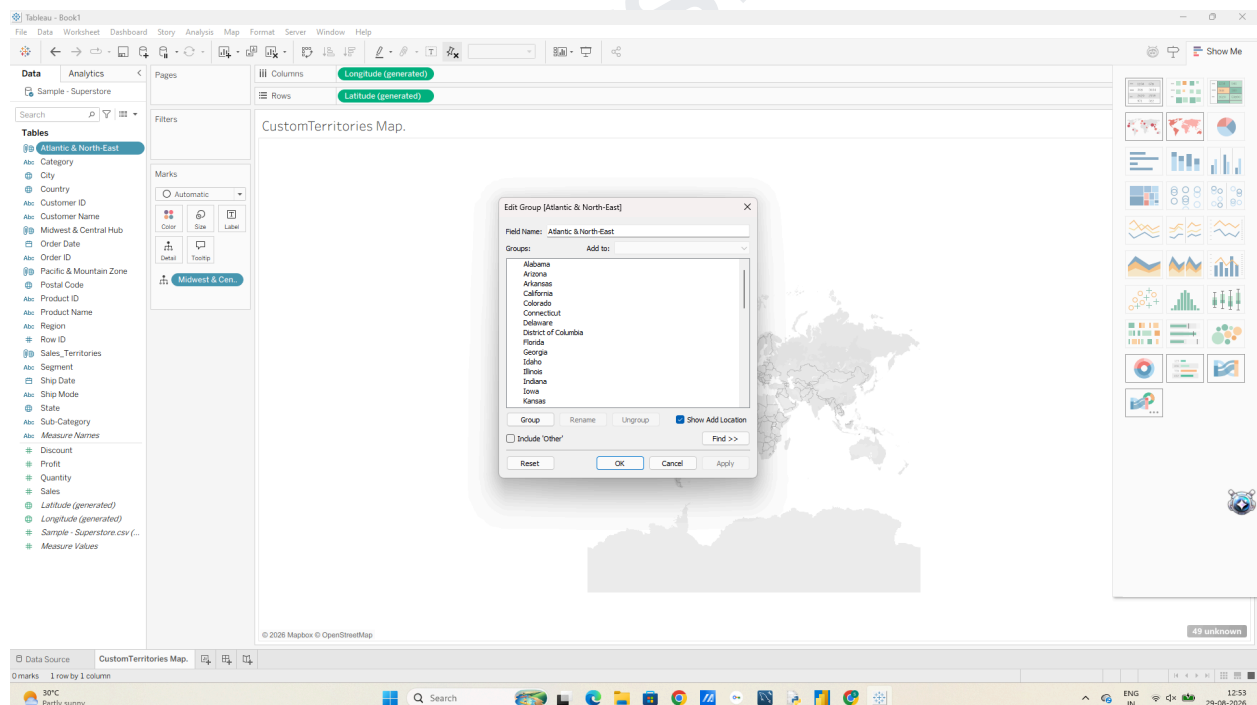
Sub : Data Visualization with Power BI and Tableau

Step 1: Create a Custom State Grouping (Territories)

1. In Tableau Desktop, navigate to a new worksheet and rename it: Custom Territories Map.
2. In the Data Pane on the left, locate the State field.
3. Right-click State --> select Create --> Group....
4. In the grouping dialogue box: o Select states belonging to your custom zones and click Group:

Group 1 (Pacific & Mountain Zone): Select California, Washington, Oregon, Nevada, Arizona, Utah, Idaho, Montana, Wyoming, Colorado, New Mexico --> Click Group --> Rename group to: Pacific & Mountain Zone.

Group 2 (Midwest & Central Hub): Select Texas, Illinois, Wisconsin, Nebraska, Michigan, Ohio, Indiana, Minnesota, Iowa, Missouri, Kansas, Oklahoma, North Dakota, South Dakota -->

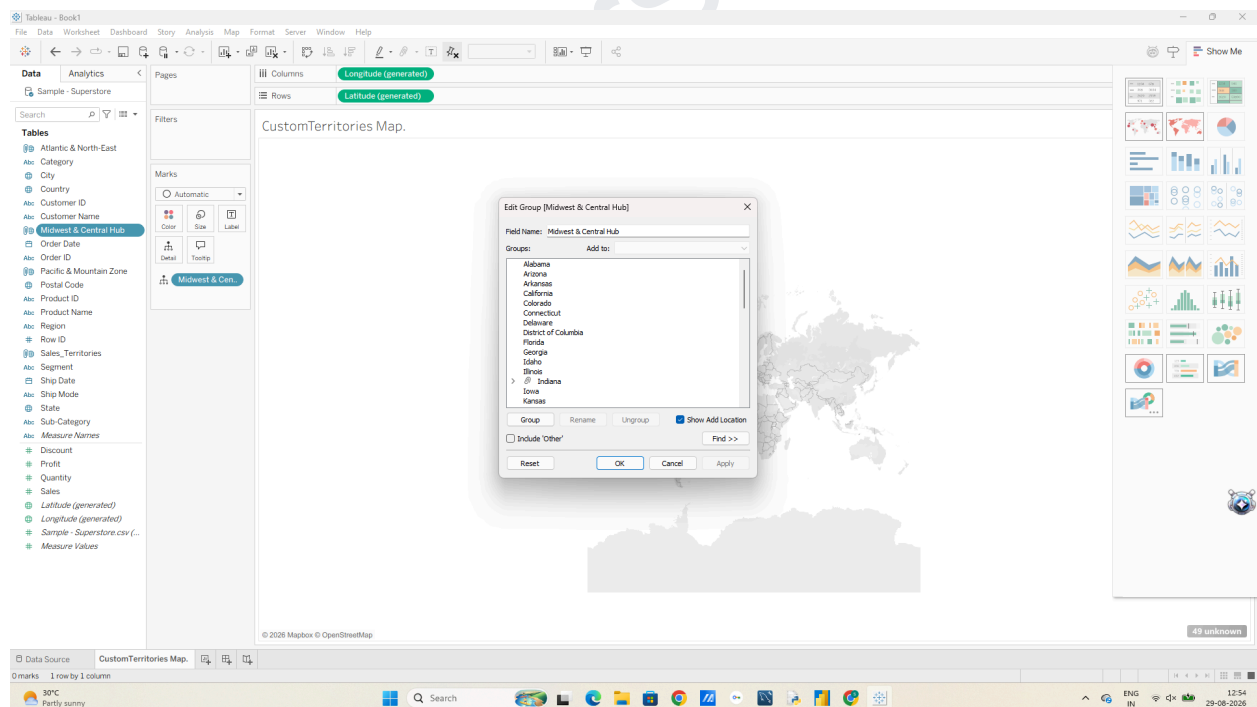
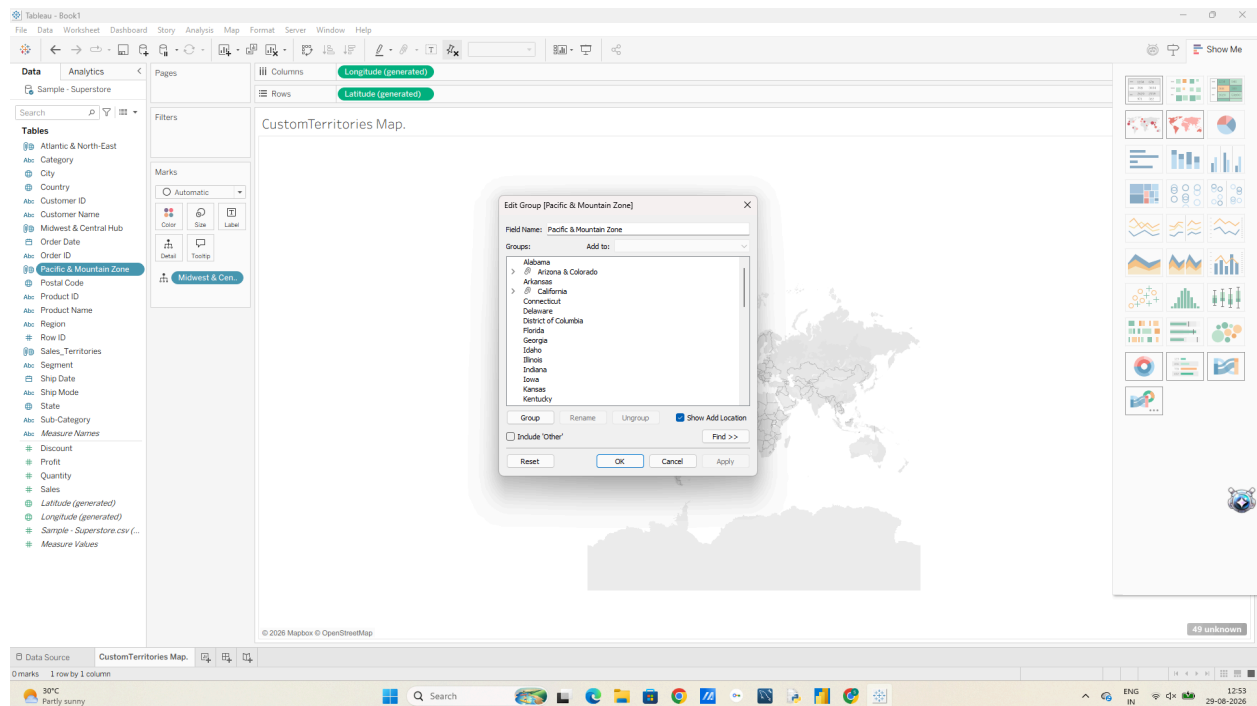


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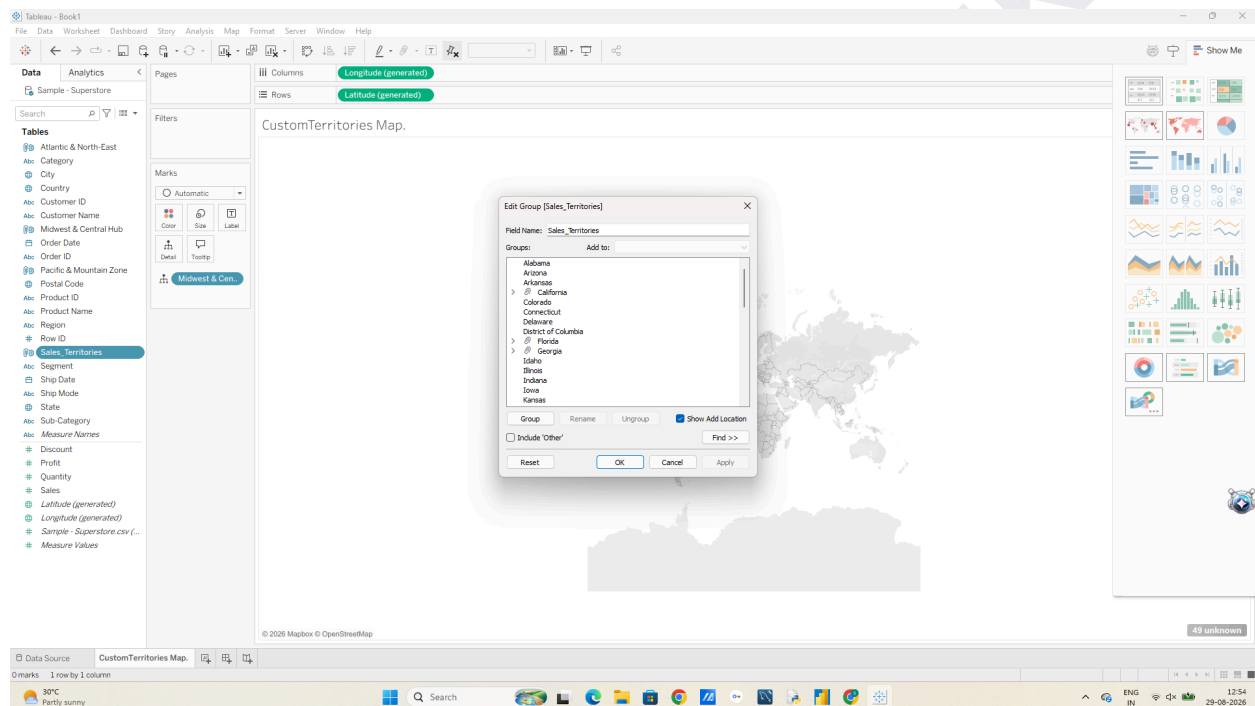
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Click Group --> Rename group to: Midwest & Central Hub. Group 3 (Atlantic & North-East): Select New York, Pennsylvania, Massachusetts, New Jersey, Delaware, Maryland, Connecticut, Rhode Island, Vermont, New Hampshire, Maine --> Click Group --> Rename group to: Atlantic & North-East.

Group 4 (Southern Enterprise Territory): Check the box Include 'Other' (or group remaining southern states: Florida, Georgia, North Carolina, Virginia, Tennessee, Kentucky, Alabama, Mississippi, Arkansas, Louisiana, South Carolina) --> Rename to: Southern Enterprise Territory.

o Change the Field Name at the top to: Sales_Territories.

o Click OK.



Step 2: Assign Geographic Role to the Custom Group

1. In the Data Pane, find your newly created field Sales_Territories (marked with a paperclip icon).
2. Right-click Sales_Territories --> select Geographic Role --> Create From... --> select State.
3. Notice that the icon next to Sales_Territories changes to a Globe icon (loplus), indicating it is now a valid spatial geometry dimension.
4. NOTE: Since we have made groups manually so we can completely skip this step.

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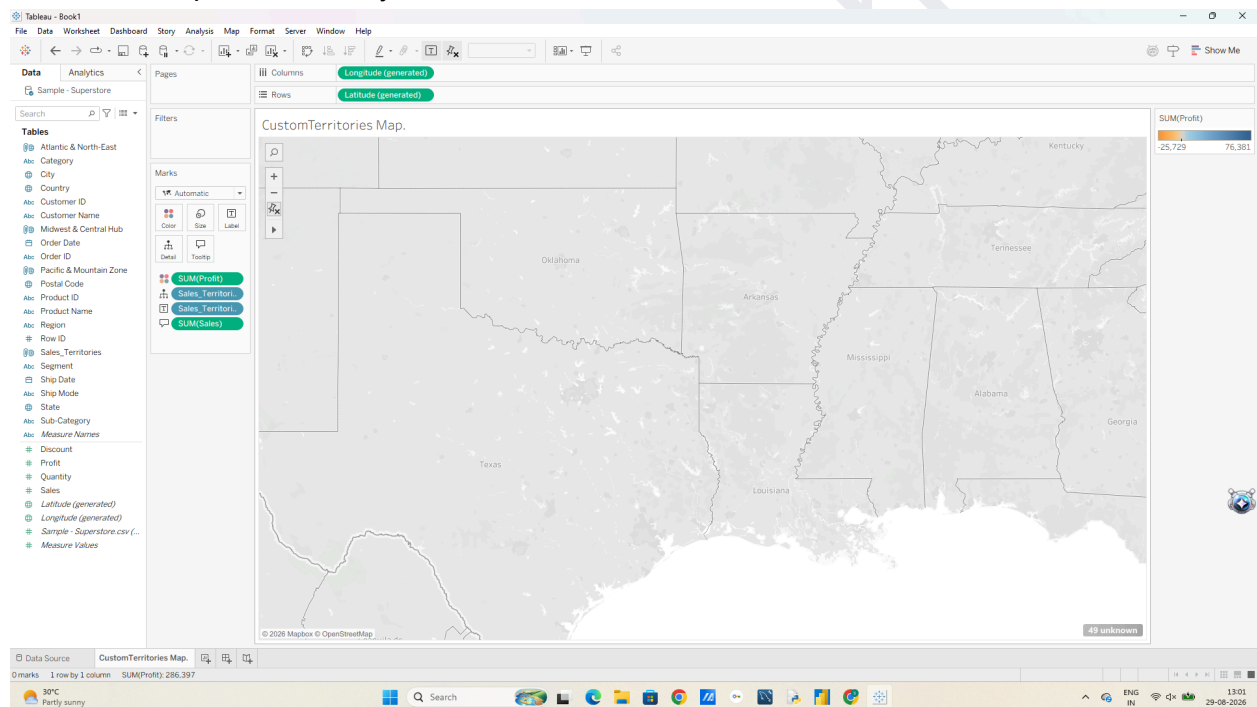
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Step 3: Build the Consolidated Territory Map

1. Double-click Sales_Territories in the Data Pane (or drag Latitude (generated) to Rows and Longitude (generated) to Columns, and drag Sales_Territories onto the Detail tile).
2. On the Marks card, change the mark type dropdown from Automatic to Map.
3. Observe the Spatial Fusion: o Tableau merges the internal borders of individual states, rendering 4 cohesive, contiguous regional polygon territories.
4. Drag Sales onto the Color tile on the Marks card.
5. Drag Profit onto the Tooltip tile.
6. Drag Sales_Territories onto the Label tile to display each territory's name centered within its consolidated spatial boundary.



Step 4: Compare State Granularity vs. Custom Territory

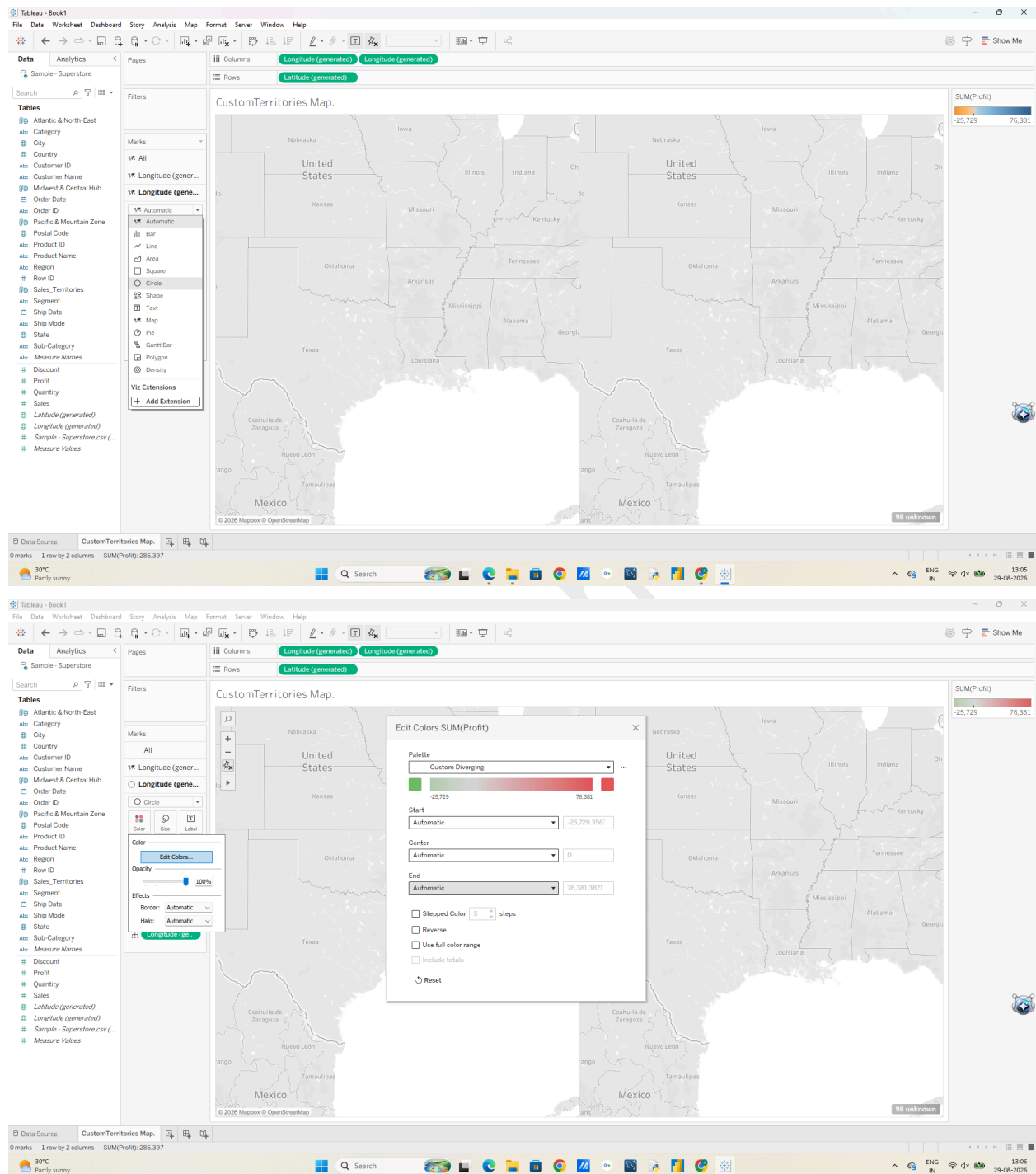
1. Hold Ctrl (or Cmd on Mac) and drag Longitude (generated) on the Columns shelf to create a second copy next to itself (Dual Axis Map).
2. On the second map Marks card: o Drag State onto Detail. o Change mark type to Circle. o Drag Profit onto Color (diverging Red-Green).

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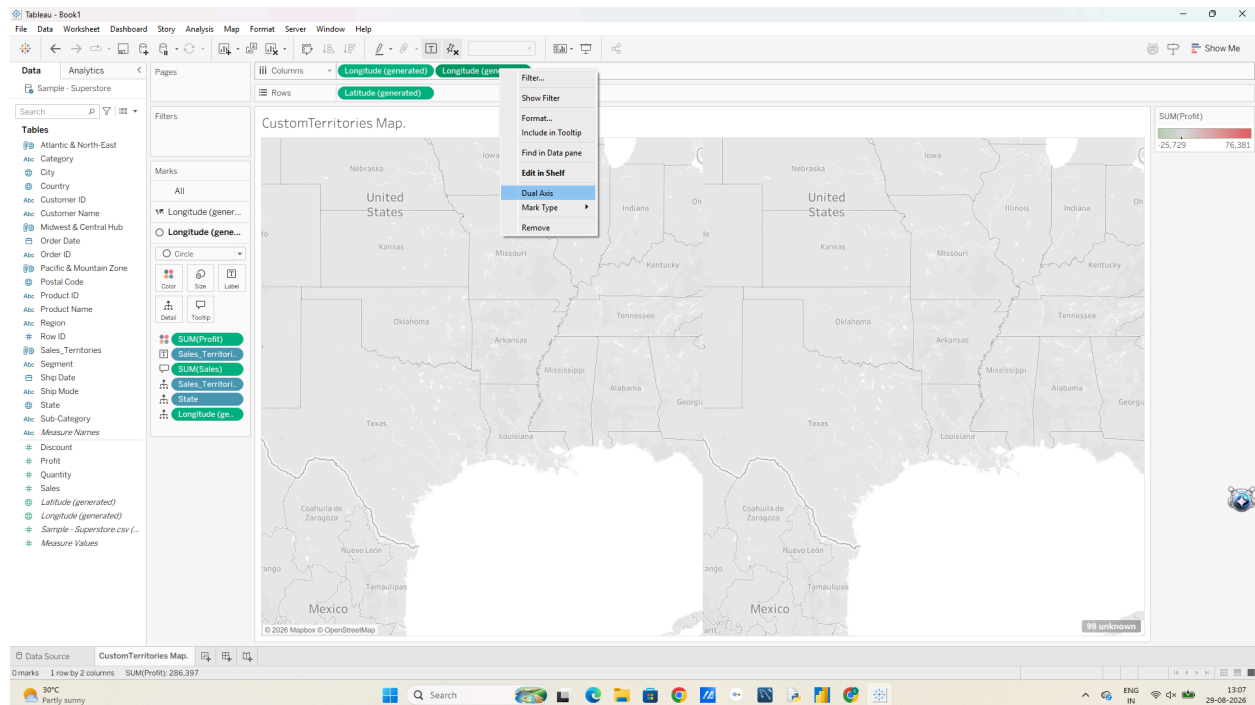


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3. Right-click the second Longitude (generated) pill on Columns --> select Dual Axis.



4. You now have a unified map showing custom territory polygon fills with individual state bubble indicators superimposed over them

F: Use Background Images in Dashboards

Step 1: Create the Coordinate Calculated Fields

1. In Tableau Desktop, create a new worksheet named: Store Floor Plan Heatmap.

2. Create the Store_X_Coordinate field:

o Right-click in the Data Pane --> Create Calculated Field....

Field Name: Store_X_Coordinate

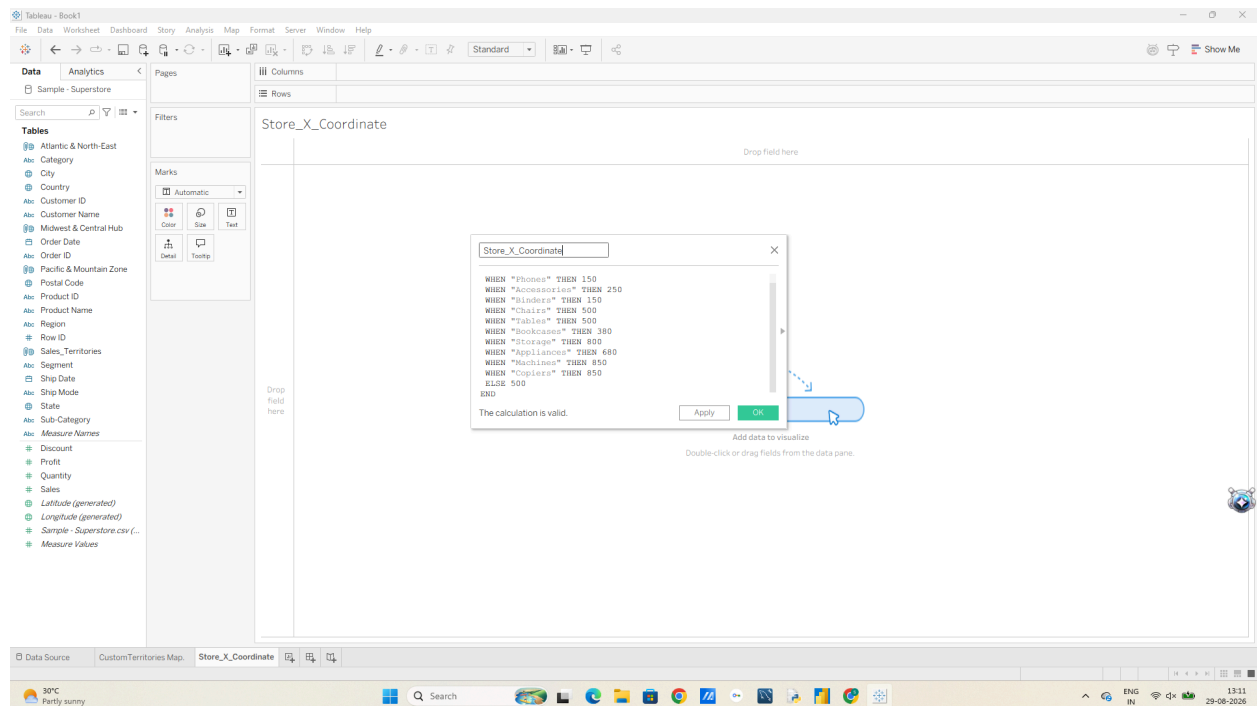
o Tableau Formula: Code snippet CASE [Sub-Category] WHEN "Phones" THEN 150 WHEN "Accessories" THEN 250 WHEN "Binders" THEN 150 WHEN "Chairs" THEN 500 WHEN "Tables" THEN 500 WHEN "Bookcases" THEN 380 WHEN "Storage" THEN 800 WHEN "Appliances" THEN 680 WHEN "Machines" THEN 850 WHEN "Copiers"

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3. Create the Store_Y_Coordinate field:

- o Right-click in the Data Pane --> Create Calculated Field....

- o Field Name: Store_Y_Coordinate o Tableau Formula: Code snippet CASE [Sub-Category]

WHEN "Phones" THEN 480 WHEN "Accessories" THEN 360 WHEN "Binders" THEN 200

WHEN "Chairs" THEN 480 WHEN "Tables" THEN 200 WHEN "Bookcases" THEN 100 WHEN

"Storage" THEN 480 WHEN "Appliances" THEN 100 WHEN "Machines" THEN 200 WHEN

"Copiers" THEN 360 ELSE 300 END

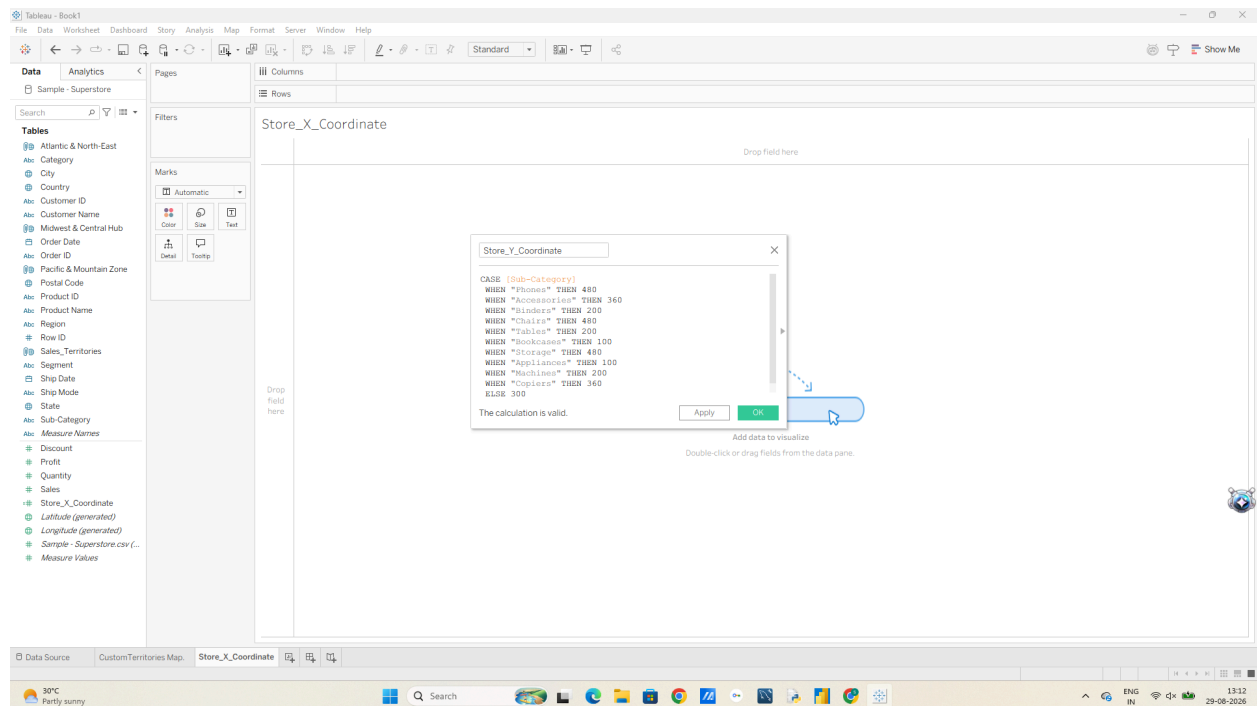
- o Click OK.

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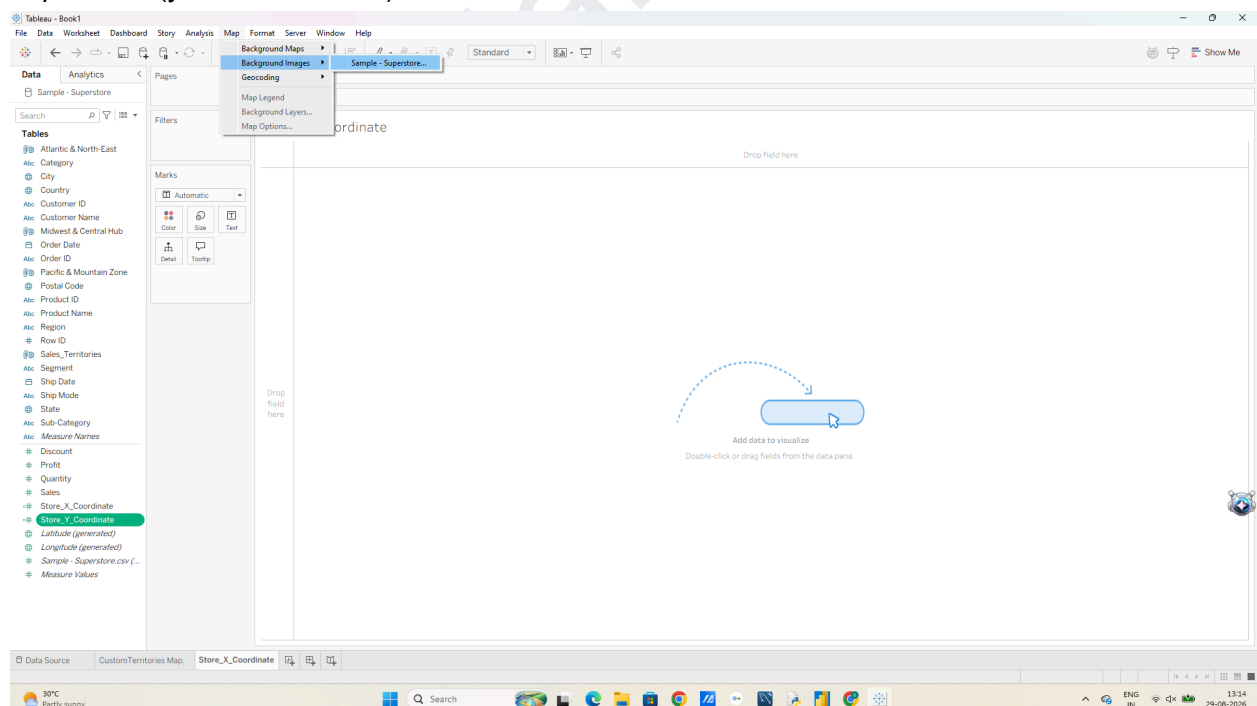
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Step 2: Configure the Custom Background Image in Tableau 1. Go to the top application menu: select Map --> Background Images --> select Superstore (your data source).



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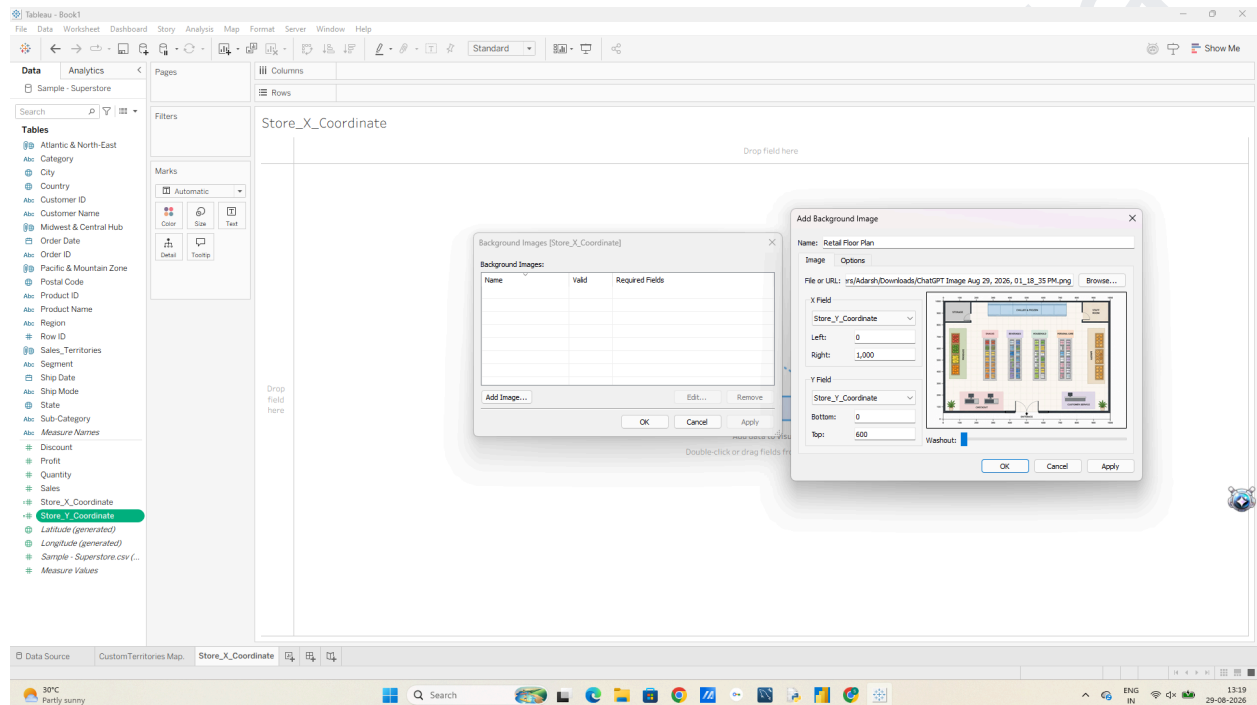
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2. In the Background Images dialog, click Add Image....

3. In the Add Background Image window: o Name: Retail Floor Plan o File or URL: Click Browse and select any image file of a floor plan / blueprint on your machine (or use a placeholder rectangle/grid image).

o X Field (Horizontal axis): Field: Select Store_X_Coordinate Left: 0 Right: 1000

o Y Field (Vertical axis): Field: Select Store_Y_Coordinate Bottom: 0 Top: 600



4. Click the Options tab: o Check Lock Aspect Ratio.

o Check Always Show Entire Image.

5. Click OK, then click OK again to close the window.

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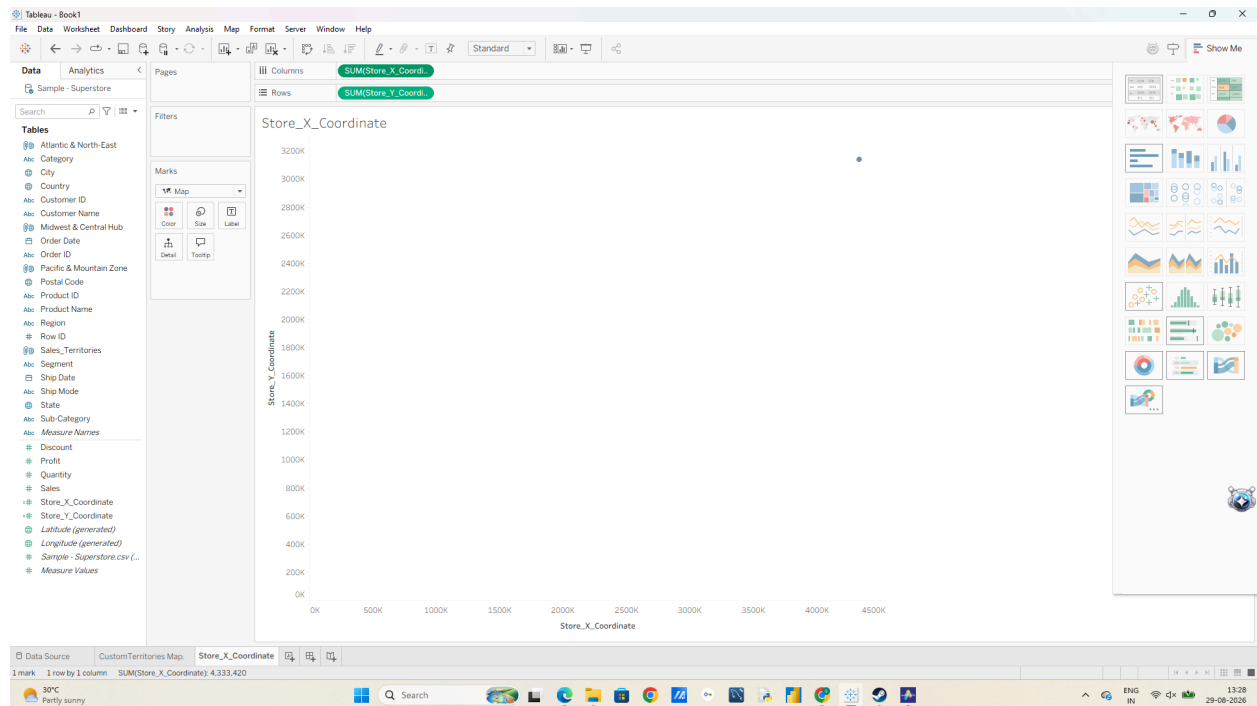
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Step 3: Build the Floor Plan Spatial Visualization

1. Drag Store_X_Coordinate onto the Columns shelf.
2. Drag Store_Y_Coordinate onto the Rows shelf.



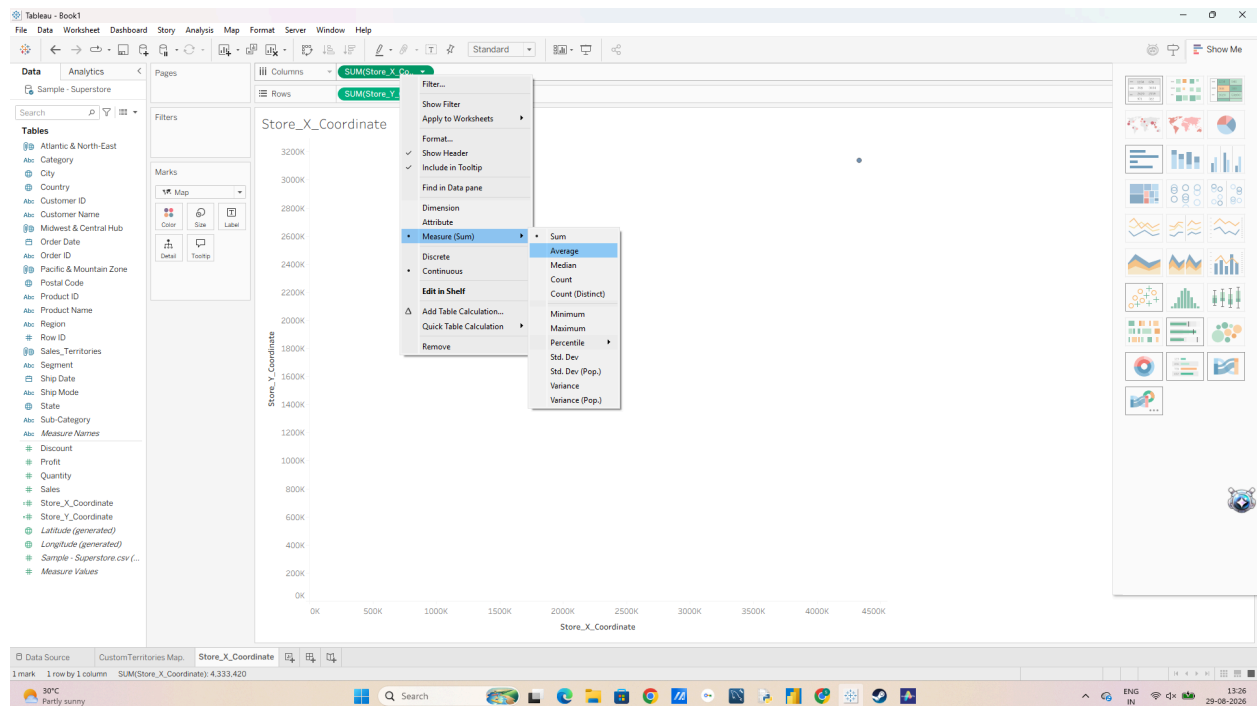
4. Ensure both coordinate pills are aggregated or treated properly: o Right-click SUM(Store_X_Coordinate) on Columns --> select Dimension(or keep as AVG).

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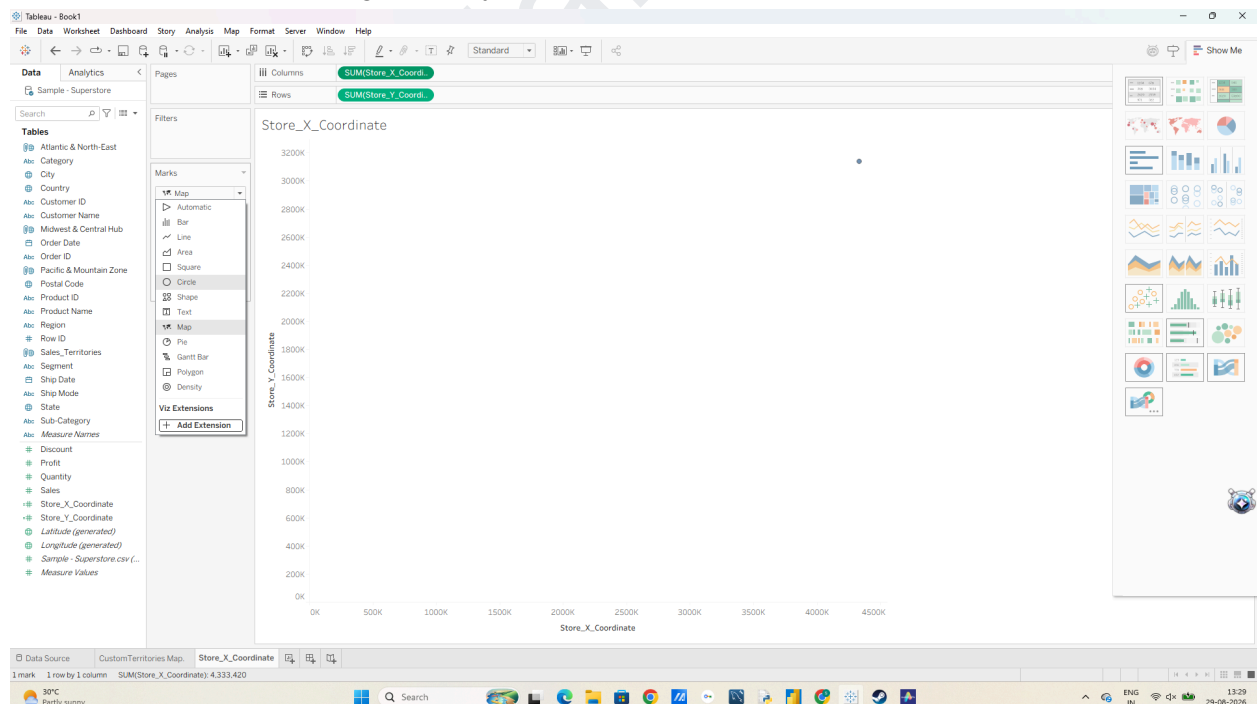


o Right-click SUM(Store_Y_Coordinate) on Rows --> select Dimension (or keep as AVG). 5.

Drag Sub-Category onto the Detail tile on the Marks card.

Step 4: Encode Metrics onto the Spatial Zones

1. On the Marks card, change mark type from Automatic to Circle (or Shape).



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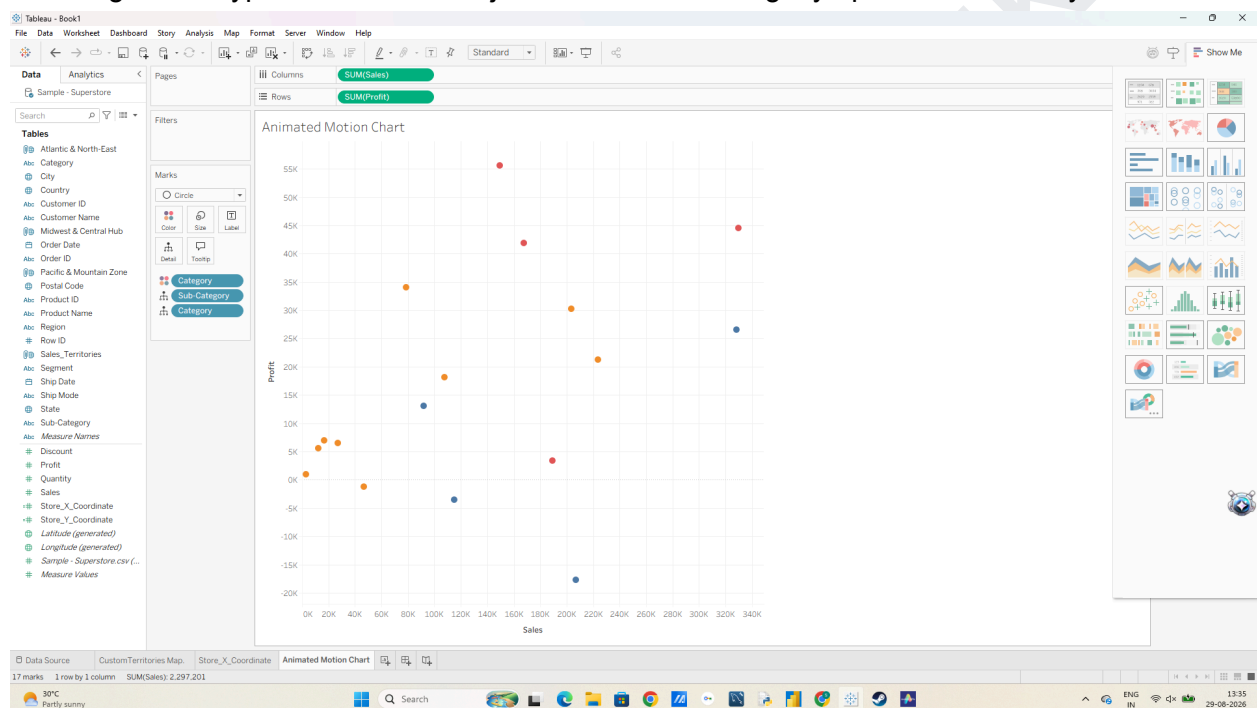
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2. Drag Sales onto the Size tile (so higher sales departments appear as larger footprint zones).
3. Drag Profit onto the Color tile: o Set color palette to Red-Green Diverging (Center = 0).

Task 1: Create an Animated Motion Chart (Pages Shelf)

Step 1: Set Up the Scatter Plot Worksheet

1. In Tableau Desktop, create a new worksheet named: Animated Motion Chart.
2. Drag Sales onto the Columns shelf (SUM(Sales)).
3. Drag Profit onto the Rows shelf (SUM(Profit)).
4. Drag Sub-Category onto the Detail tile and Color tile on the Marks card.
5. Drag Category onto the Shape tile (optional, for visual distinction).
6. Change mark type to Circle and adjust the Size slider slightly upward for visibility.



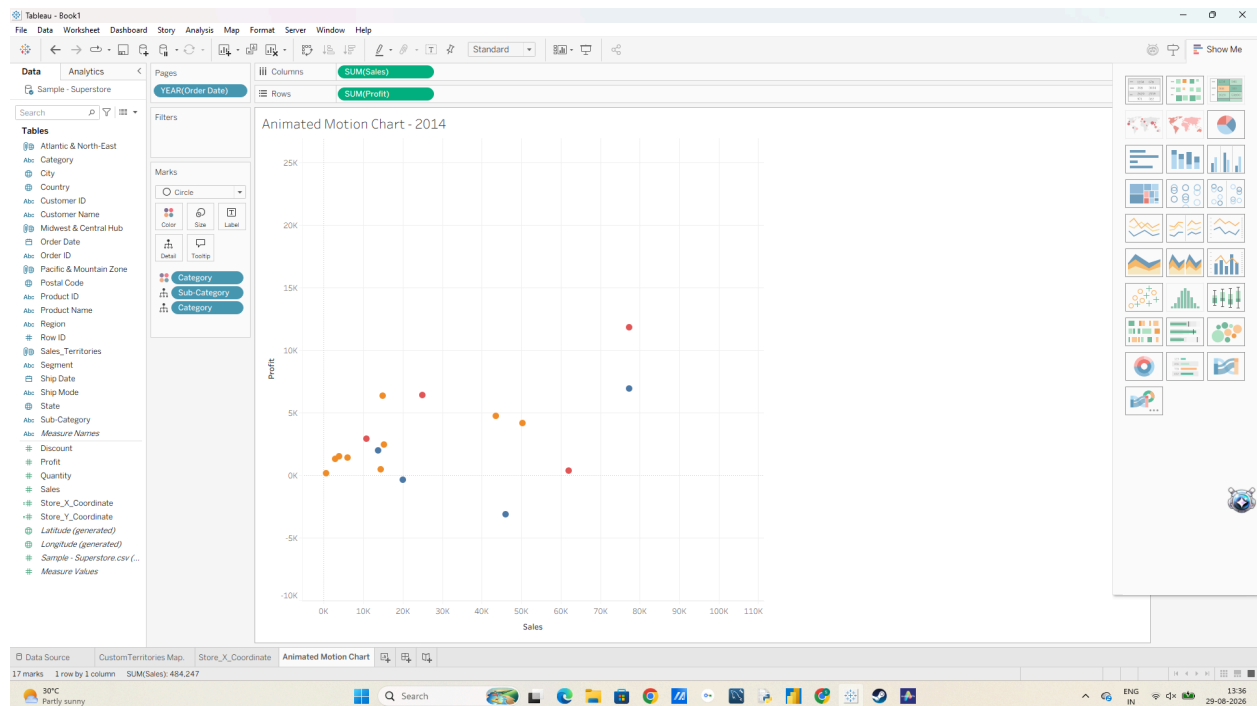
- Step 2: Add Time to the Pages Shelf 1. Drag Order Date onto the Pages shelf (located directly above the Filters shelf)

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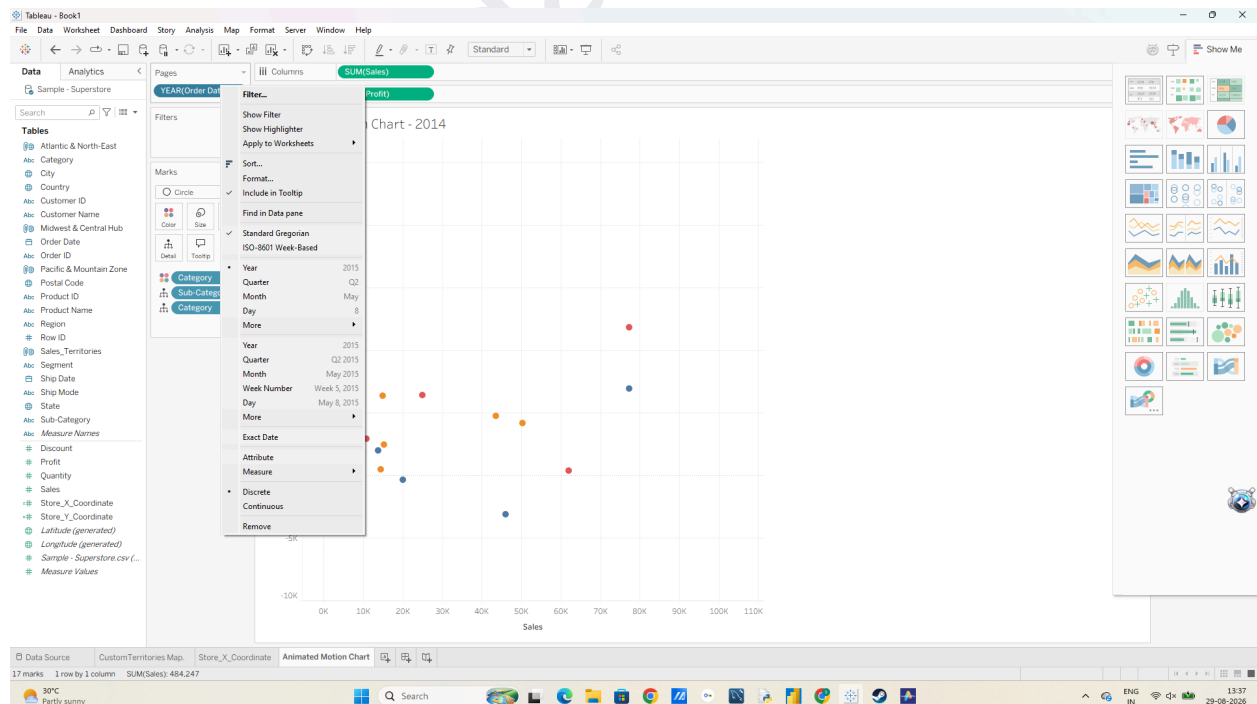
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2. Right-click the Order Date pill on the Pages shelf --> select Year (discrete YEAR(Order Date)). Tableau may have selected it by default



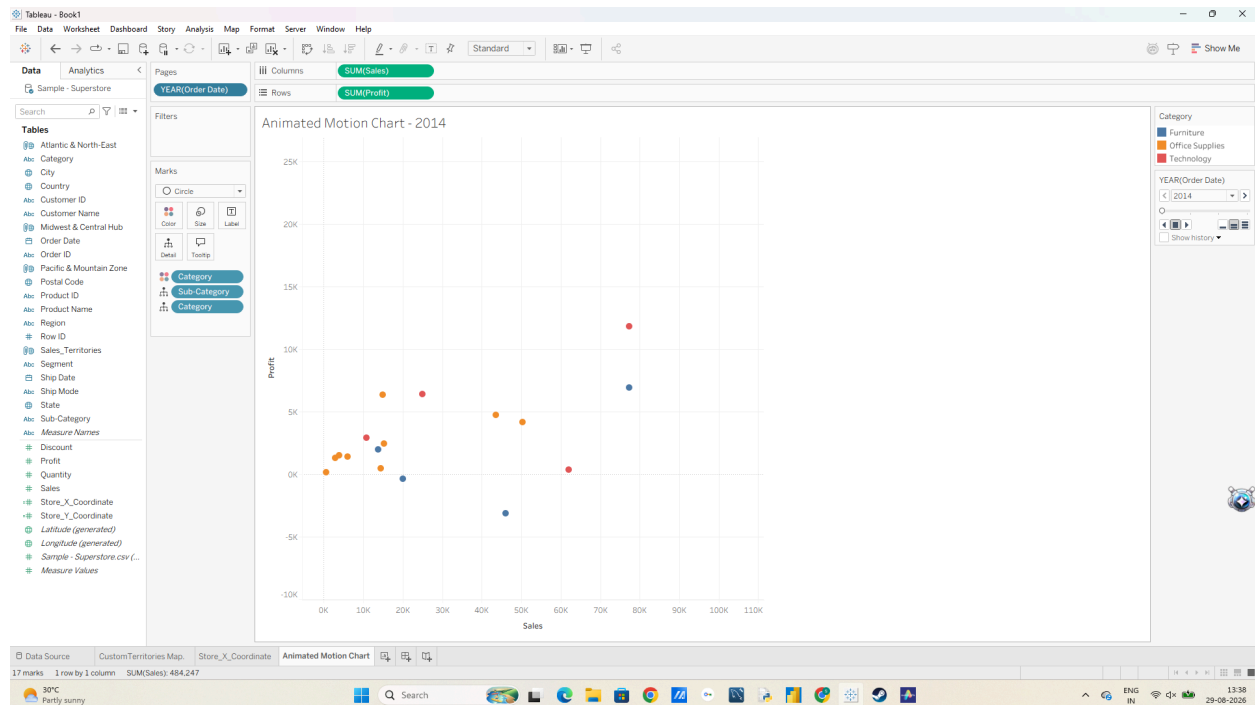
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3. A Pages Control Player will appear on the right side of the canvas with Play, Pause, and Speed controls.



Step 3: Enable Motion History and Trails

1. On the Pages Control panel on the right:

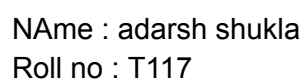
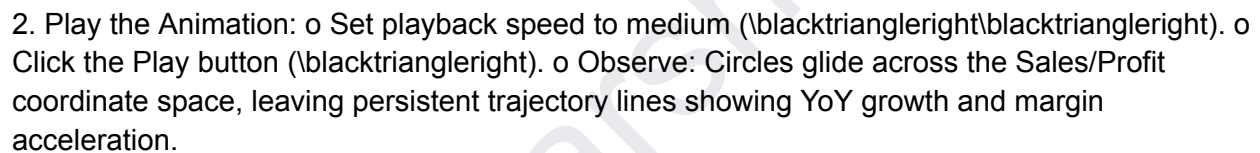
- o Check the box for Show History.

- o Click the dropdown arrow next to Show History to configure animation trail settings: Marks to show history for: Select All. Show: Select Both (Marks and Trails). Color: Choose a subtle faded trail color. Fade: Set to 50%.

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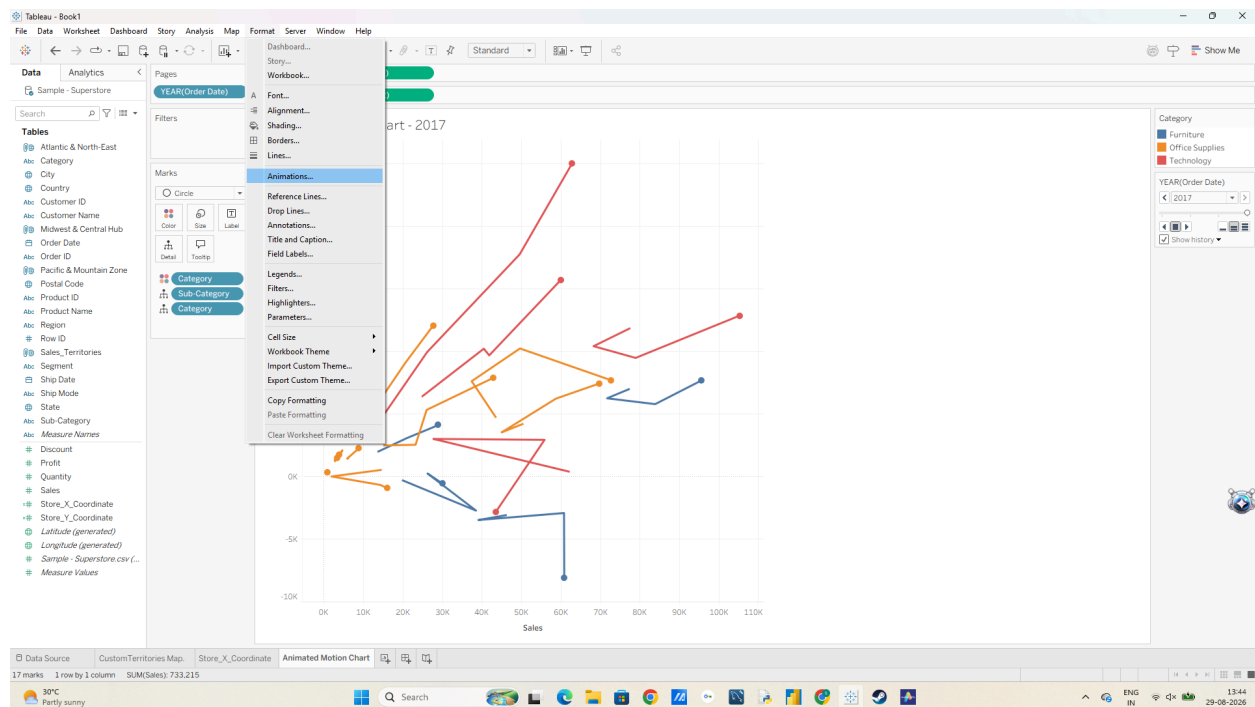
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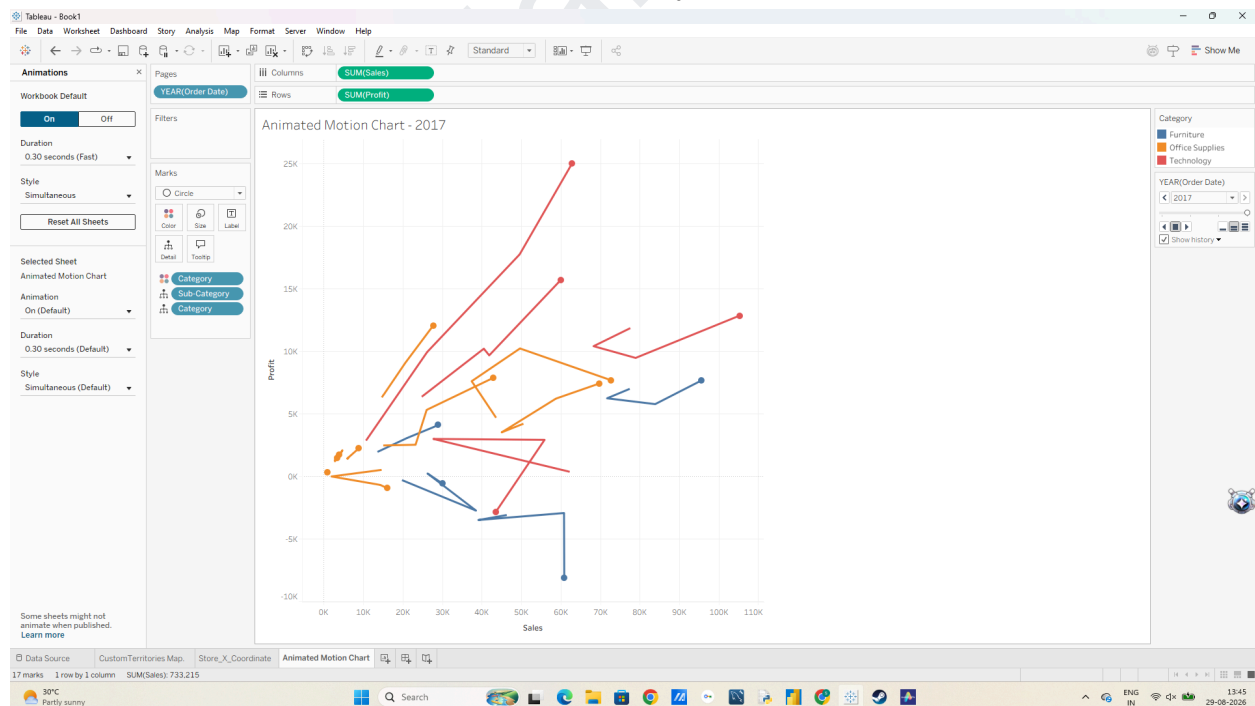
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Step 4: Enable Built-in Tableau Client Animations 1. From the top application menu bar, select Format --> Animations....



2. In the left-hand Animations pane: o Set Workbook Default to On. o Set Duration to 1.00 seconds (Slow) or 0.50 seconds (Medium). o Set Style to Simultaneous.



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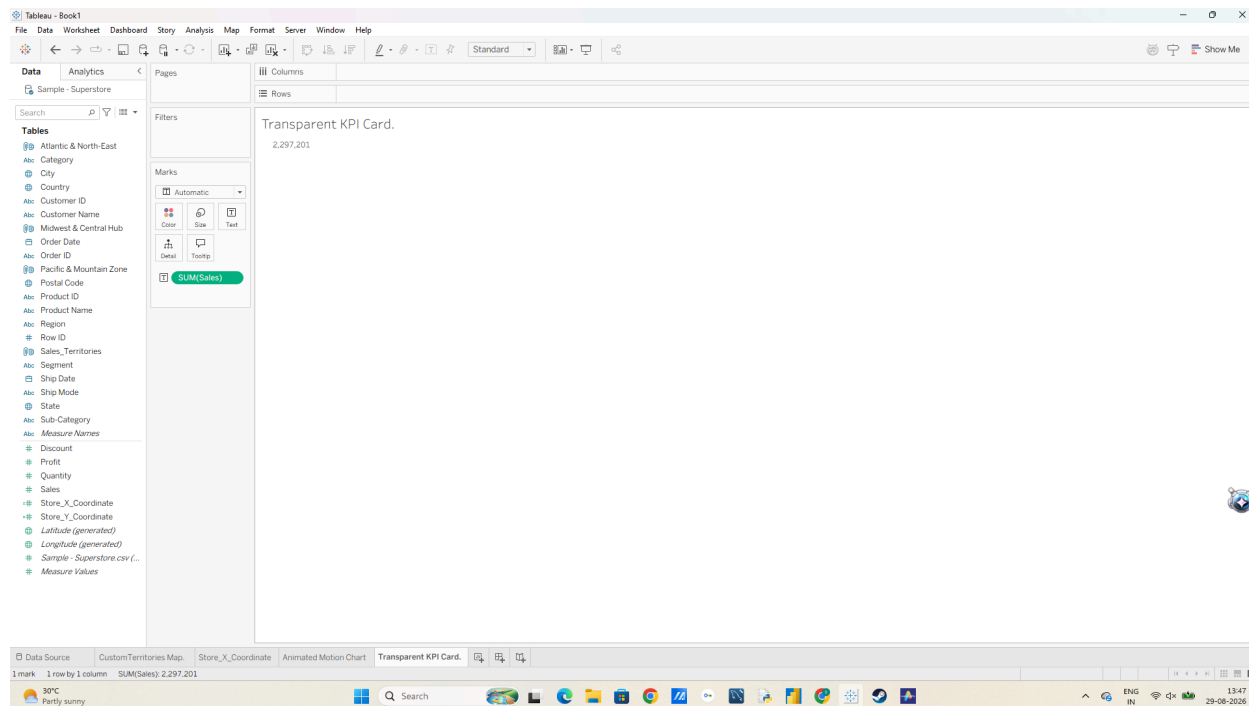
Task 2: Apply Sheet and Dashboard Transparency

Step 1: Build a Transparent KPI Card Worksheet 1. Create a new worksheet named:

Transparent KPI Card.

2. Drag Sales onto the Text tile on the Marks card.

3. Click Text --> increase font size to 28pt Bold and set font color to white (#FFFFFF).



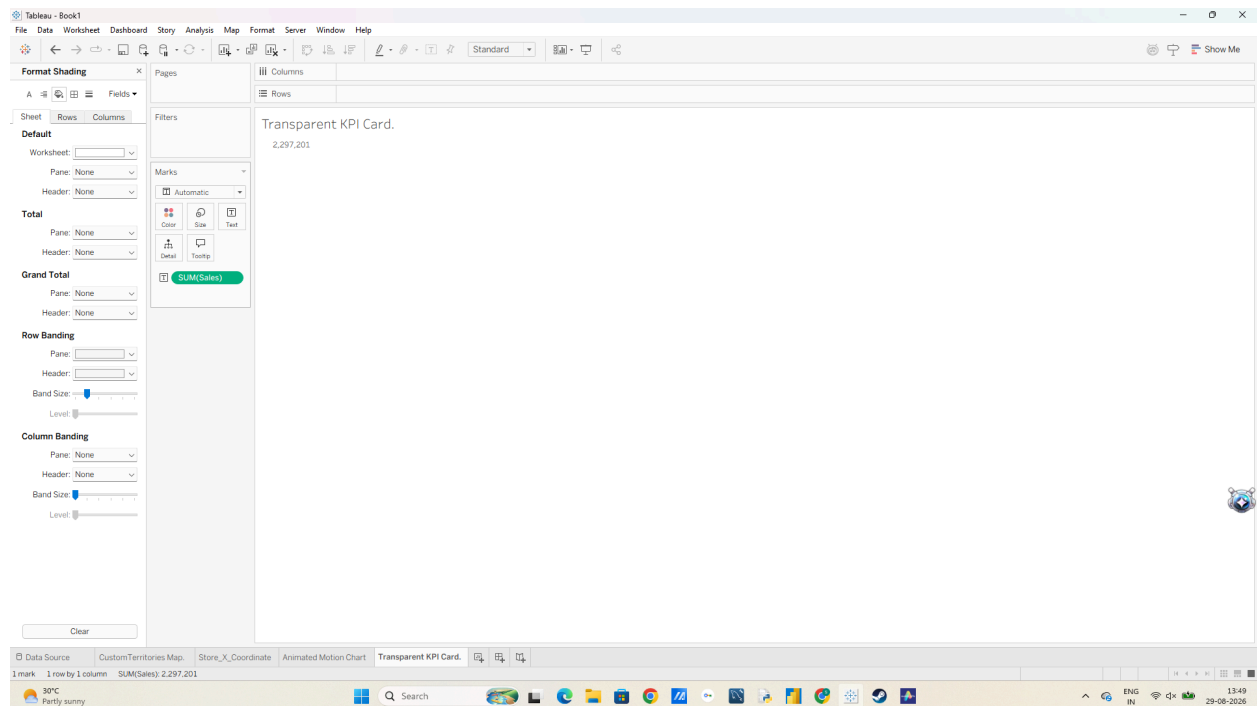
4. Set worksheet background to 100% Transparent: o From the top menu, select Format --> Shading.... o In the Format Shading pane on the left, under Worksheet, set the color dropdown to None (transparent). o Under Pane, set to None.

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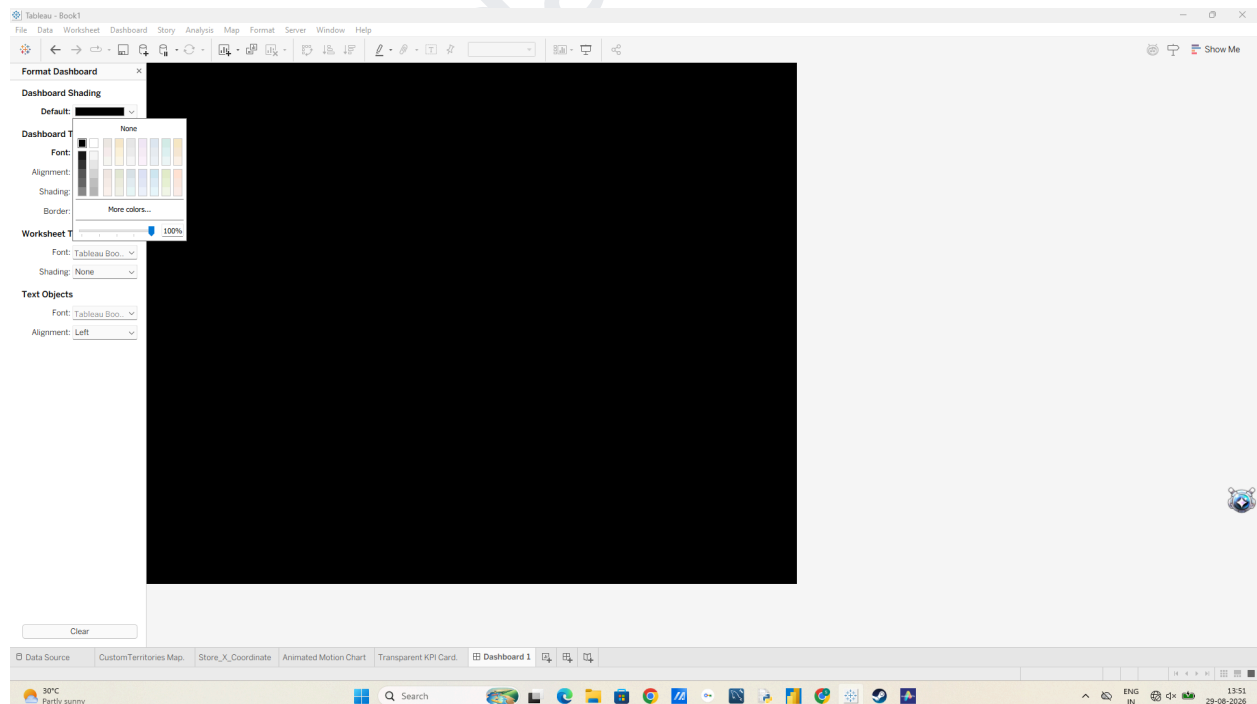
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Step 2: Assemble the Transparent Layered Dashboard 1. Create a new Dashboard and name it: Animated & Transparent Master View. 2. Under Dashboard properties, set the background color: o Select Format --> Dashboard... --> set Dashboard Shading to a deep navy or dark slate color (e.g., #1E293B).

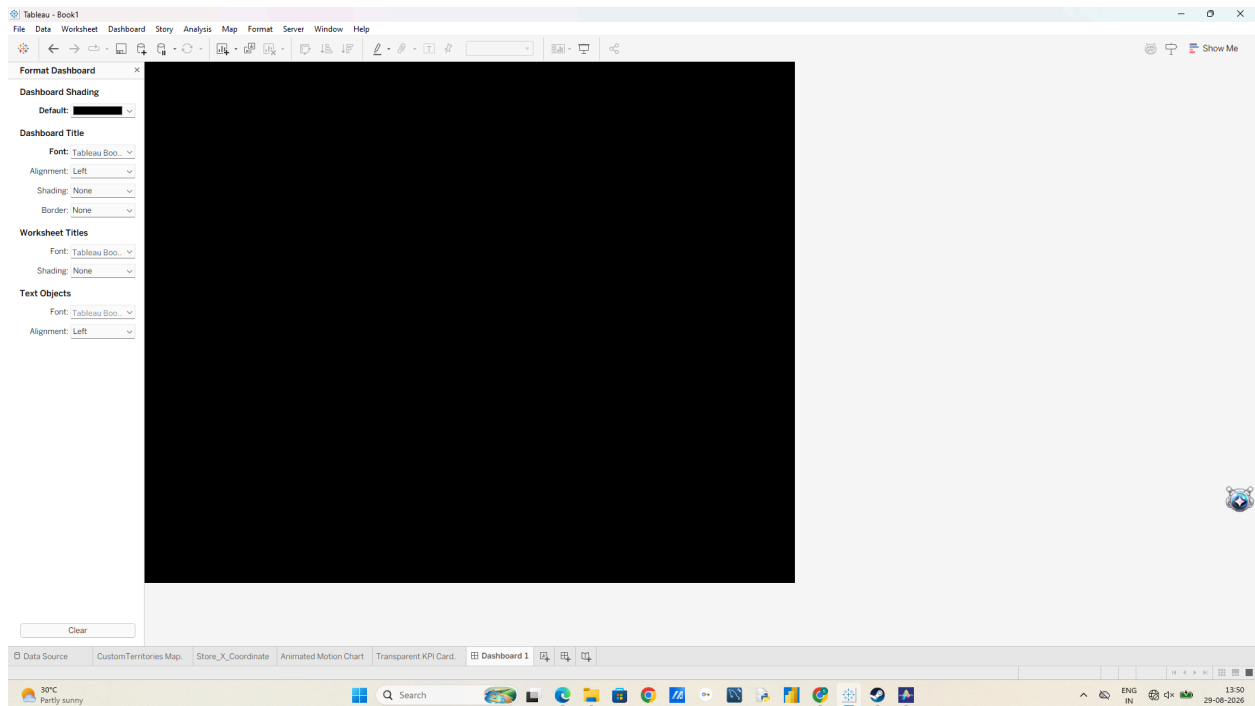


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3. In the Objects section on the left: o Select the Floating radio button (instead of Tiled).



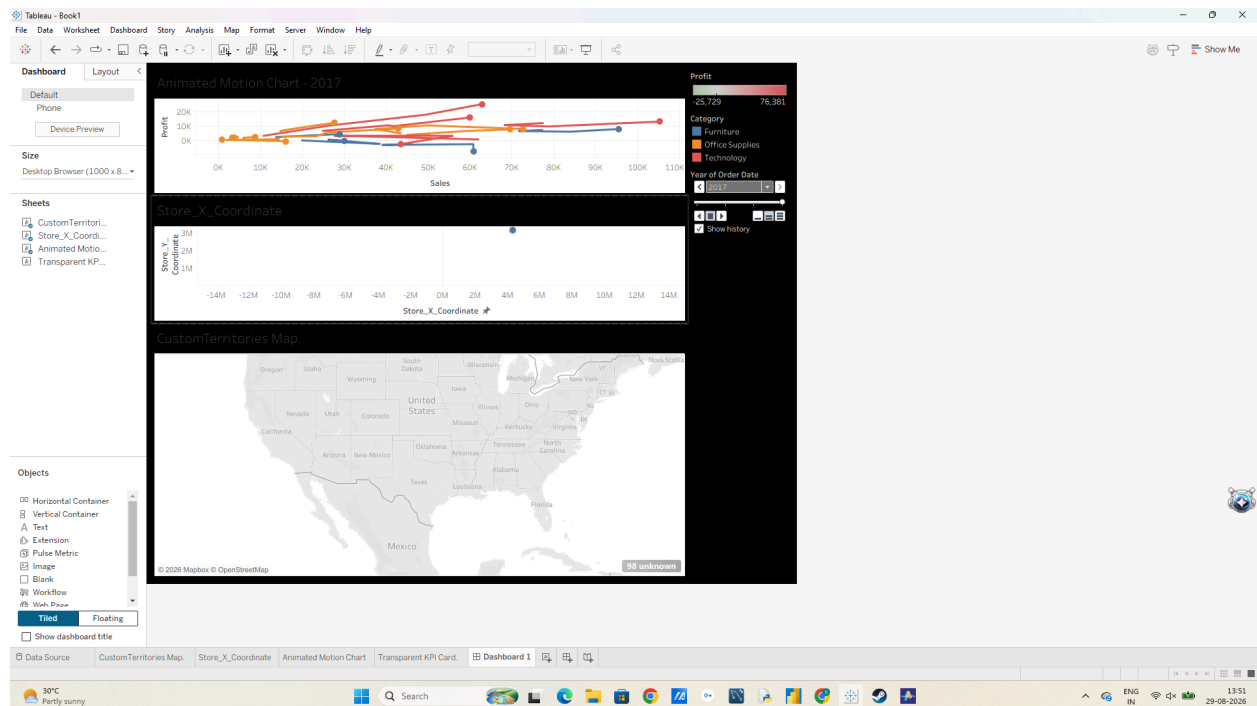
4. Drag your Store Floor Plan Heatmap (or Custom Territories Map) onto the dashboard canvas as the base background layer and resize it to fill the screen. 5. Drag your Transparent KPI Card onto the canvas as a Floating object directly over the map/image
6. Drag your Animated Motion Chart onto the dashboard as a floating panel to complete the interactive control deck.

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